SCALE-COVARIANT ATTENTION FOR TRAINING-FREE HIGH-RESOLUTION DIFFUSION TRANSFORMERS

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ABSTRACT

Diffusion transformers trained near one megapixel fail at higher resolutions: attention meets unseen token counts and position offsets, and its cost grows quadratically. Each head's balance between local detail and global layout, we find, is measurable at the training resolution and transfers to larger images. We introduce Scale-Covariant Attention (SCA), which calibrates once, fitting one weight per head and noise level in closed form against the model's full attention, and reuses that table at the target. Each head mixes a native-size sliding window with block-centre global attention, both in the trained position range, after a short global-only opening that sets the layout. For FLUX.1-dev at 4096^2, SCA beats every shared-sampler baseline on all per-image metrics, has the best FID and fine detail of eight methods, and runs $2.85\times$ faster than AReA. Dropped into HRDiT's progressive schedule, it matches HRDiT's preference scores in less than half the time. Gains hold at 6144^2, on HRGen-1K, with two more seeds, and on FLUX.2-dev; on FLUX.1-Krea-dev, SCA wins six metrics but trails AReA on ImageReward and patch FID. Code: github.com/shreshthsaini/scale-covariant-attention.

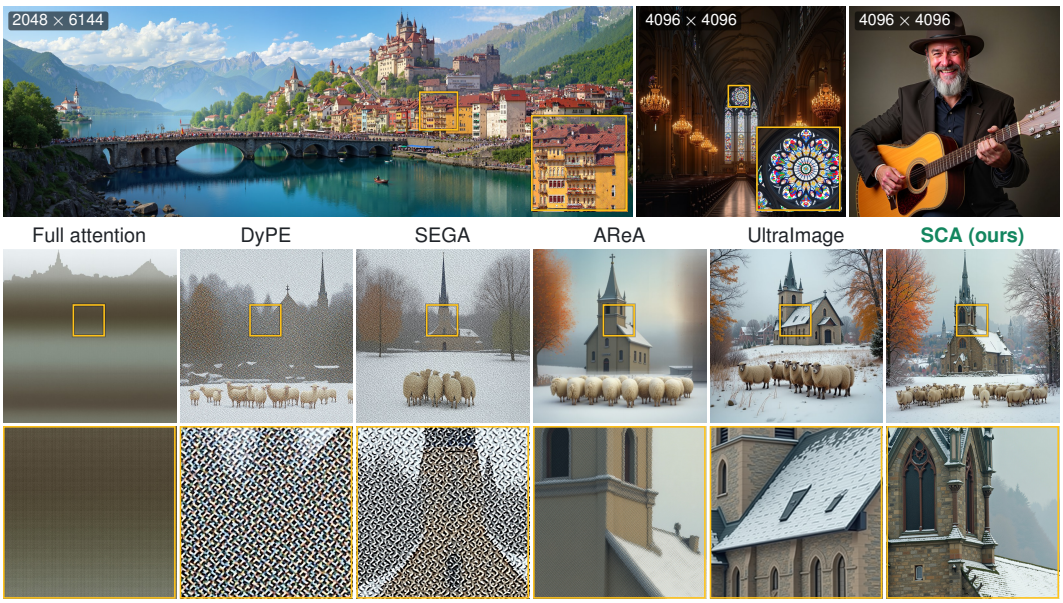

Figure 1: **SCA at 4K without training.** Top: curated SCA outputs of FLUX.1-dev at 2048×6144 and 4096^2, with insets at native pixels. Bottom: one curated 4096^2 prompt where no baseline scores unusually low, with 1:1 crops; full attention gives a flat field, DyPE and SEGA a lattice, and AReA, UltraImage, and SCA keep the church and flock.

1 INTRODUCTION

Diffusion transformers (Peebles & Xie, 2022) such as FLUX.1 (Black Forest Labs, 2024) and Stable Diffusion 3 (Esser et al., 2024) are trained near one megapixel. At 4096×4096 the image has 16 times as many tokens, full attention costs about 256 times as much, and every layer meets token

counts and position offsets it never saw in training. FLUX.1-dev with full attention then returns stripes or a flat field (Figure 1, bottom left), and such outputs show duplicated subjects, broken layouts, and blurred texture (He et al., 2023; Du et al., 2023; Issachar et al., 2025; Rajabi et al., 2026a). Restricting each query to a native-size window keeps the offsets in range, yet run from the first step it draws five Moai where the prompt asks for one (Figure 2a). We ask whether each head's balance between such a window and a coarse view of the whole image can be measured at the training resolution and reused at the target.

Training-free fixes either edit the positional encoding (Zhuo et al., 2024; Issachar et al., 2025; Zhao et al., 2025; Rajabi et al., 2026b) or restrict the keys each query sees (Zhang et al., 2023; Ma et al., 2025; Xue et al., 2026; Rajabi et al., 2026a). A native-size window cannot see an object larger than itself, so at high noise, when the layout forms, each window settles it alone. AReA (Rajabi et al., 2026a), the most recent key-restricting method, attends to keys subsampled onto a native-size grid for the first 8 of 28 steps, then to windows shrinking to native size, with one schedule for all heads. HRDiT and UltraImage give heads their own receptive fields, by cost budget or entropy (Xue et al., 2026; Zhao et al., 2025). None fits a head's balance at the training resolution, where full attention is trustworthy (Section 5).

(a) mix from step 1 (b) SCA

Figure 2: **Windows repeat the subject.** A development prompt asking for one Moai at 4096^2. (a) Mixing in native-size windows from step 1 draws five; (b) SCA's global-only opening keeps one large statue, plus a small one far off.

We answer by emulation. To mimic a target s times larger, we shrink both a native-size neighbourhood and a downsampled view of the whole image by s, so the image-to-window ratio matches the target. Full attention stays in distribution and serves as ground truth. A convex mix of the two views fits each head; its weight differs strongly between heads and is stable across prompt splits (correlation 0.99). It also transfers: the emulated scale-2 table explains scale 4 with 2.4% more output error than the scale-4 table, and a 1024^2 table explains 1280^2 with 0.8% more (Section 3). We call this measured transfer scale covariance: the local view follows a law fixed in tokens, the global view one fixed in image fractions, and their weight stays nearly constant over the tested scales.

We introduce Scale-Covariant Attention (SCA), which turns this measurement into an operator (Figure 3). Each head's calibrated weight mixes a global branch over block-centre keys on the native grid, with rescaled coordinates, and a local branch, an exact native-size sliding window. As the mix repeats the subject from the first step (Figure 2a), SCA runs the global branch alone for the first 8 of 28 steps, the opening phase, as a frequency argument suggests (Argument 1).

Against seven baselines with paired bootstrap intervals, SCA beats AReA on every per-image metric at 4096^2, at 71.1 s per image against 202.4 s (Section 4.1). In HRDiT's progressive schedule it matches HRDiT's preference scores in less than half the time. We highlight our contributions below.

- We fit each head's mix of local and global attention in closed form at the training resolution and show that the fit transfers across scales (Section 3).
- We propose SCA, a per-head mix of a native-size window and block-centre global keys whose image offsets stay in the trained range at a cost linear in the token count.
- We show that at 4096^2 on FLUX.1-dev, SCA beats AReA on 11 of 15 metrics with paired 95% intervals excluding zero, loses none, and runs $2.85\times$ faster.

2 Scale-Covariant Attention

2.1 Setting and notation

A multimodal diffusion transformer such as FLUX.1 concatenates T text tokens $\mathcal{T}$ with $N = HW$ image tokens on the grid $\mathcal{G} = [H] \times [W]$, where $[n] = \{0, \dots, n-1\}$. Each block applies joint attention (57 blocks of 24 heads in FLUX.1). It is trained at about 1024^2 pixels, a native grid of $n_0 \times n_0$ tokens with $n_0 = 64$. Rotary embeddings (Su et al., 2021) give token j a position $p_j \in \mathbb{Z}^2$

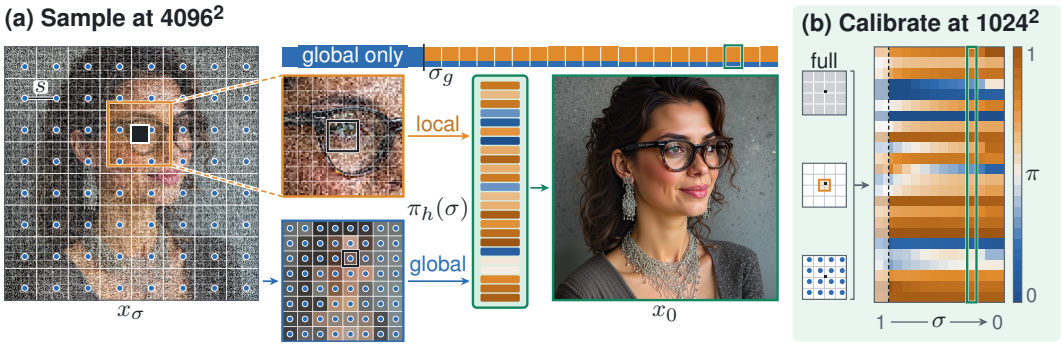

Figure 3: **SCA sampling and calibration.** (a) Each head mixes a native-size local window and block-centre global keys with weight $\pi_h(\sigma)$; steps with $\sigma > \sigma_g$ are global only (schematic grid, $s = 4$). (b) Calibration at 1024^2 fits π per head and noise bin to full attention, with both operators at an emulated scale; Figure D.1 zooms into each component.

(FLUX.1 puts all text tokens at 0) and a rotation R_p with $R_p^\top R_{p'} = R_{p'-p}$. For one head, a pre-rotation query q placed at u, and a set $\mathcal{K}$ of (key, value, key position) triples,

$$\mathcal{A}(q, u; \mathcal{K}) \;=\; \textstyle\sum_{(k,v,u')\in\mathcal{K}} \mathrm{softmax}_{\mathcal{K}}\big(q^\top R_{u'-u}\, k/\sqrt{d}\big)\; v, \tag{1}$$

where d is the head dimension. Full attention is $O_i^{\mathrm{full}} = \mathcal{A}\big(q_i, p_i; \{(k_j, v_j, p_j)\}_{j\in\mathcal{T}\cup\mathcal{G}}\big)$. The target grid is $H = s_H n_0$, $W = s_W n_0$ with integer factors (s when equal), so full attention meets offsets s times the trained range. The latent at noise level σ is $x_\sigma = (1-\sigma)x_0 + \sigma\epsilon$. Splitting the keys into a near and a far set gives exactly $O_i^{\mathrm{full}} = m_i O_i^{\mathrm{near}} + (1-m_i) O_i^{\mathrm{far}}$ (Lemma 1), where each O attends only to its own set and m_i is the softmax mass on the near set. Every head is thus a query-dependent mix of a near and a far field, which SCA approximates with two separately normalized operators and one weight per head. Both branches include text keys and their image key sets can overlap, so the identity does not decompose SCA exactly.

2.2 Two operators with native-range image geometry

Global operator. Let $\mathcal{G}_0 = [H/s_H] \times [W/s_W]$ be the $n_0 \times n_0$ grid of blocks. Block-centre subsampling keeps for each block b the token at $\kappa(b) = (s_H b_1 + \lfloor s_H/2 \rfloor,\, s_W b_2 + \lfloor s_W/2 \rfloor)$. The coarse map $c(p) = (\lfloor p_1/s_H \rfloor, \lfloor p_2/s_W \rfloor)$ sends a position to its block, so $c(\kappa(b)) = b$, and

$$\mathcal{K}^{\mathrm{glob}} = \{(k_j, v_j, 0)\}_{j\in\mathcal{T}} \cup \{(k_{\kappa(b)}, v_{\kappa(b)}, b)\}_{b\in\mathcal{G}_0}, \qquad O_i^{\mathrm{glob}} = \mathcal{A}\big(q_i,\, c(p_i);\, \mathcal{K}^{\mathrm{glob}}\big), \tag{2}$$

with $c(p_i)$ replaced by 0 for text queries. This is AReA's first phase with the pooling factor fixed at s. It models content whose extent is a fixed fraction of the image: layout, object count, identity.

Local operator. Image queries form tiles of $t \times t$ tokens ($t = 8$); query i lies in tile $\tau = \lfloor p_i/t \rfloor$. The window $\mathcal{W}(i)$ of tile τ has side $w = \min(n_0, H)$ and starts at $a(\tau) = \mathrm{clip}\,\big(t\tau + \lfloor t/2 \rfloor - \lfloor w/2 \rfloor, 0, H - w\big)$ per axis, centred on the tile and clamped to the grid:

$$\mathcal{K}_i^{\mathrm{loc}} = \{(k_j, v_j, 0)\}_{j\in\mathcal{T}} \cup \{(k_j, v_j, p_j)\}_{p_j\in\mathcal{W}(i)}, \qquad O_i^{\mathrm{loc}} = \mathcal{A}\big(q_i,\, p_i;\, \mathcal{K}_i^{\mathrm{loc}}\big), \tag{3}$$

an exact sliding window with the original positions. It models content whose extent is fixed in tokens: edges, texture, fine structure. Appendix A proves the two propositions below.

Proposition 1 (Trained position range)**.** *Assume exact integer positions and $t \le n_0 \le \min(H, W)$. Every query-key offset of the global operator and every image query-key offset of the local operator is at most $n_0 - 1$ per axis. Each image query sees the native $T + n_0^2$ keys in either operator.*

Proposition 2 (Cost)**.** *Per head and layer, full attention scores $(N+T)^2$ query-key pairs, the global operator $(N+T)(T+N_0)$ with $N_0 = |\mathcal{G}_0| = n_0^2$, and the local operator $N(T+w^2)$. For fixed n_0 and T the total is $O\big(N(T+w^2+N_0)\big)$, linear in N.*

Proposition 1 assumes exact coordinates; FLUX.1 rounds positions in bf16 beyond 256 tokens per axis (Appendix D). Native-range geometry also does not make hidden states in-distribution.

Algorithm 1: **Calibration**, once per model

```
# Input: prompts; full-attention sampler at 1024^2
# Output: pi*[s_c, l, h, beta]; full in App. E
def calibrate(q, k, v, l, sigma):  # every block
  O_full = attn(q, k, v)          # Eq. 1, all keys
  beta = min(int(14 * sigma), 13)    # noise bin
  for s_c in (2, 4):              # emulation, App. B
    f, w = s_c, n // s_c          # subsample, window
    t_c = round(t * w / n0)       # tile t n/(s_c n0)
    O_glob = global_branch(q, k, v, f)    # Eq. 2
    O_loc = local_branch(q, k, v, w, t_c) # Eq. 3
    dl, df = O_loc - O_glob, O_full - O_glob
    A[s_c, l, :, beta] += sum(dl * dl)    # Eq. 5
    B[s_c, l, :, beta] += sum(df * dl)  # over rows
    E[s_c, l, :, beta] += sum(df * df)
  return O_full                   # sampler unchanged
pi = clip(B / A, 0, 1); pi[A == 0] = 1  # Prop. 3
pi = fill_empty_bins(pi)       # nearest bin
```

Algorithm 2: **Sampling**, one SCA attention call

```
# Input: q, k, v (T text, then H x W image), l, σ
# Output: O for all T + HW queries; App. E
fh, fw = H // n0, W // n0       # subsampling s
kb = k_img[fh//2::fh, fw//2::fw]        # κ(b)
vb = v_img[fh//2::fh, fw//2::fw]
Q = cat(rope(q_txt, 0), rope(q_img, c(p)))
K = cat(rope(k_txt, 0), rope(kb, b))
O_glob = sdpa(Q, K, cat(v_txt, vb))     # Eq. 2
if sigma > sigma_g:                  # 0.89, Eq. 4
  return O_glob       # opening phase: no local
# keys: text + w x w window of the tile
mask = flex_layout(T, t=8, w=min(n0, H))  # App. D
O_loc = flex_attention(rope(q_img, p), rope(k, p),
    v, block_mask=mask)                 # Eq. 3
pi = interp(sigma, centres, table[l])   # per head
O_img = pi*O_loc + (1-pi)*O_glob[T:]    # Eq. 4
return cat(O_glob[:T], O_img)     # text: O_glob
```

Design choices. A block centre is a real token, so the global branch attends to keys and values the model produced; mean pooling instead lowers ImageReward by 1.52 (Section 4.3). The centre also lies within $\lfloor s/2 \rfloor$ tokens of its whole block per axis, against $s-1$ for a corner token. A window of side n_0 is the largest that keeps image offsets in the trained range and gives each query the native key count, so SCA keeps FLUX.1's RoPE and softmax scale. Text keys sit at position 0 in both branches. Text queries take O^{glob}, whose $T+n_0^2$ keys match the native size, and image queries keep offset p_i to text keys in the local branch, as in full attention (Remark 1).

Worked example. At 4096^2 the latent grid is 256×256 tokens ($s=4$, $N=65{,}536$, $T=512$). Each 8×8 query tile attends locally to the 64×64 window around it, 4,096 image keys at offsets of at most 63 per axis. The global branch keeps one centre per 4×4 block, again 4,096 keys on a 64×64 grid. With the 512 text keys a query sees 4,608 keys per branch, against 66,048 keys at offsets up to 255 in full attention. Per head and layer, SCA scores 6.06×10^8 query-key pairs against 4.36×10^9, 7.2 times fewer (8.4 with the opening phase), and one attention layer takes 18.7 ms instead of 87.3 ms (Table D.1).

2.3 Per-head mixing and the opening phase

For an image query of head h in block l at noise level σ,

$$O_i = O_i^{\text{glob}} \text{ if } \sigma > \sigma_g, \qquad O_i = \pi_{lh}(\sigma)\, O_i^{\text{loc}} + \big(1-\pi_{lh}(\sigma)\big)\, O_i^{\text{glob}} \text{ otherwise,} \tag{4}$$

where σ_g is the gate and text queries use O_i^{glob}. The weight $\pi_{lh}(\sigma)$, written $\pi_h(\sigma)$, stands in for the near-field mass m_i (Figure 4d); one weight per head is cheap and a few prompts determine it.

Argument 1 (A frequency-based explanation for the opening phase). At noise level σ a spatial frequency band has signal-to-noise ratio $\big((1-\sigma)/\sigma\big)^2 S(f)$ for image spectrum S. If latent power falls with frequency, only low frequencies carry signal at high noise, and a window of w tokens spans less than one period of any frequency below $1/w$. Independent windows may then repeat local structures; a global branch supplies context across them. As hidden states also carry earlier layers' information, this is a hypothesis, not an impossibility result (Appendix A).

Output matching cannot see this failure, since a tokenwise squared error can favour local detail over layout. We have not measured per-band signal-to-noise ratios of FLUX.1 latents, but as predicted, a 16-token window gives smaller, more numerous copies than a 64-token one (Figure A.1). We set $\sigma_g = 0.89$, the first 8 of 28 steps at maximum shift as in AReA; gating 13 steps gives a 4×4 block mosaic, as a block's queries share one coarse position.

2.4 Calibration by output matching

The weights must come from where the model is trustworthy, which the target is not. On an in-distribution $n \times n$ grid ($n=64$ at 1024^2), emulated scale s_c uses subsampling factor s_c, window side n/s_c, and tile side $\text{round}(tn/(s_c n_0))$. The ratio of grid to window is then s_c, as at a target of scale s_c. The sampler runs unmodified full attention (Algorithm 1), and at every block and step both

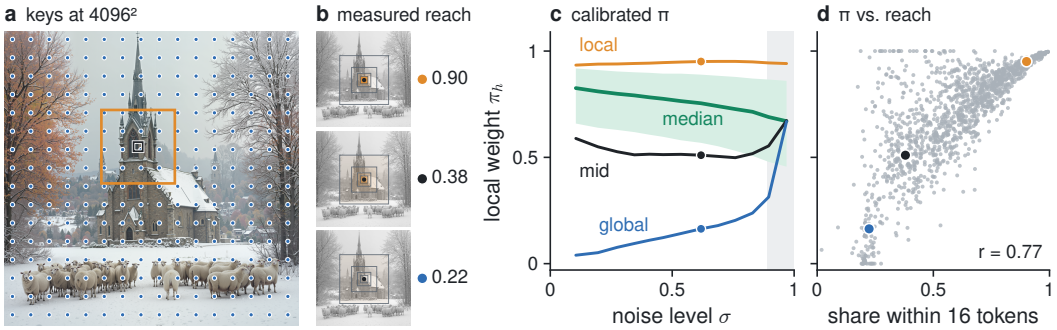

Figure 4: **Measured reach tracks calibrated π_h.** (a) Query tile, local window (amber), and every fourth block-centre key per axis (blue) on a 4096^2 SCA output; (b) full-attention mass of three heads at 1024^2, averaged over the 32 calibration prompts. (c) Their scale-2 $\pi_h(\sigma)$ with the median and interquartile range of all 1368 heads (grey: global only); (d) π_h against the mass share within 16 tokens, printed in (b), at $\sigma = 0.62$ ($r = 0.77$).

emulated operators see the same queries, keys, and values. For block l, head h, and noise bin β, sums over prompts, steps with $\sigma \in \beta$, image queries, and channels give

$$A = \sum \|O^{\text{loc}} - O^{\text{glob}}\|^2, \quad B = \sum \langle O^{\text{full}} - O^{\text{glob}},\, O^{\text{loc}} - O^{\text{glob}} \rangle, \quad E = \sum \|O^{\text{full}} - O^{\text{glob}}\|^2, \tag{5}$$

where $O^{\text{loc}}, O^{\text{glob}}$ are emulated. The mix with weight π has squared error $J(\pi) = E - 2\pi B + \pi^2 A$.

Proposition 3 (Closed-form calibration and transfer cost)**.** *In each cell (l, h, β), J is minimized on $[0, 1]$ by $\pi^\star = \text{clip}(B/A, 0, 1)$, with $\pi^\star = 1$ if $A = 0$. Any other weight $\pi' \in [0, 1]$ (fitted at another scale or resolution, per layer, shared, or a hard switch) has $J(\pi') - J(\pi^\star) \geq A(\pi' - \pi^\star)^2$, with equality when $0 < \pi^\star < 1$.*

So the per-head table has the lowest output error of any rule constant within a cell, and replacing it costs an A-weighted squared distance. This holds at the emulation scale; Section 3 measures transfer between scales and resolutions. The same sums score every alternative rule without another forward pass, and one pass collects emulated scales 2 and 4.

Inference. A target with factors (s_H, s_W) uses the table scale nearest $\max(s_H, s_W)$ in log space, here always 4. Each step interpolates a head's weight linearly in σ between the mean noise levels of occupied bins (14 of width $1/14$), constant beyond them. At 2048×4096, 2×4 blocks keep the global grid and the window at $n_0 \times n_0$; the scale-2 table would change metrics only slightly (Appendix N). The local operator is one exact FlexAttention call (Dong et al., 2024, Algorithm 2).

2.5 SCA INSIDE A PROGRESSIVE SCHEDULE

SCA replaces only the attention, so it fits inside a multi-stage sampler (Algorithm 5). HRDiT's released code (Xue et al., 2026) denoises at 1024^2 for 30 steps, then upsamples, re-noises, and repeats the last 17 and 10 steps of the same schedule at 2048^2 and 4096^2. There it changes attention in four ways: NTK-scaled RoPE, a proportional attention scale, per-head key pruning, and position alignment. Progressive SCA keeps HRDiT's schedule, transitions, guidance, and seeds and replaces all four with SCA (native RoPE, scale-2 and scale-4 tables). The stages re-enter at $\sigma = 0.81$ and 0.61, below σ_g, so the opening phase never runs and the 1024^2 stage sets the layout. All settings were fixed before any evaluation image was generated (Appendix I).

3 WHAT THE HEADS WANT

We calibrate FLUX.1-dev on 32 purpose-written prompts at 1024^2 and 16 at 1280^2, disjoint from the evaluation sets. Errors are relative, $\sum J / \sum \|O^{\text{full}}\|^2$, pooled over blocks, heads, and steps.

Heads differ. In the emulated scale-2 table, the standard deviation of $\pi_h(\sigma)$ over the 1368 heads is 0.24 to 0.26 at every noise level. Half of the (head, noise level) cells lie in $[0.2, 0.8]$, 11.7% exceed 0.95, and 2.9% are below 0.05 (Figures 4c, B.2a–c, and B.1). The mean weight is 0.64 to 0.74: the

Table 1: **Main comparison at 4096^2.** FLUX.1-dev, 195 Aesthetic-4K prompts, seed label 0; HPSv2.1 $\times 100$, MUSIQ_c on 1024^2 crops, FID_p patch FID, **best**, second best, ours shaded; seconds per image on one GH200 as a clean batch-1 median (Section 4.2) or a campaign mean on shared nodes (camp.), n/m not measured; Table J.1 adds all metrics. Key result: among single-pass methods, SCA is best on every metric shown and $2.85\times$ faster than AReA.

		Time (s)↓		Image↑	↑	Pick↑	↑	↑	↓	↓
Method	Steps	clean	camp.	Reward	HPSv2.1	Score	CLIP	MUSIQ_c	FID	FID_p
Single pass, shared sampler (28 steps, guidance 4.5)										
Full attention	28	186.5	196.2	−2.18	7.60	17.94	15.37	33.1	312.4	252.2
DyPE	28	n/m	195.9	−0.10	20.55	20.51	24.92	50.3	205.5	171.9
SEGA	28	n/m	190.4	0.56	24.34	21.64	26.95	50.3	164.9	125.8
AReA	28	202.4	264.4	0.92	27.16	22.36	27.69	50.4	145.8	102.9
SCA (ours)	28	**71.1**	**71.5**	1.23	29.67	23.09	**29.14**	**61.2**	**138.1**	**99.1**
Multi-stage, own recipe and budget										
UltraImage	30+30	n/m	308.5	1.28	30.77	23.33	28.84	41.7	142.0	109.5
HRDiT	30+17+10	108.6	110.4	**1.34**	**30.92**	**23.46**	29.12	53.5	138.4	103.0

typical head leans local but still uses the global branch. In the noise bin at $\sigma = 0.62$, the scale-2 π_h tracks each head's share of full-attention mass within 16 tokens of the query ($r = 0.77$; Figure 4d).

The table is stable across prompt splits. Tables fitted on disjoint prompt halves correlate at 0.989 over all cells (Figure B.2d; 0.986 at scale 4; 0.974 for 16 prompts at 1280^2). Each half's table, scored on the other half, gives a relative error of 0.0576 against 0.0574 in sample.

The weights transfer across tested scales and resolutions. Applied at scale 4, the scale-2 table gives a relative error of 0.205 against 0.200 for the scale-4 table, below every shared rule fitted at scale 4. The 1024^2 table scored at 1280^2 gives 0.0589 against 0.0585. Over all pairs of calibration targets, a transferred table adds at most 5.45% error, while each target's own per-layer fit adds 12 to 24%. As the two resolutions emulate scale 2 with windows of 32 and 40 tokens, the weights appear to depend mainly on the image-to-window ratio.

Per-head beats coarser rules; output error misses layout. Against the best hard switch, per-head mixing lowers the relative error from 0.092 to 0.057 at scale 2 and from 0.261 to 0.200 at scale 4. Per-layer tables reach 0.071 and 0.223, and one shared weight per noise level 0.077 and 0.230 (Figure B.3). In output energy the best hard switch never switches to global (Appendix B), so the opening phase deliberately overrides the fit (Argument 1).

4 Experiments

Data and seeds. The main set is the Aesthetic-4K evaluation split with 4K references (Zhang et al., 2025a): all 195 prompts; four of them (boats, Moai, woman and bear, underwater surfer) were used during development, which favours SCA slightly. A fixed 46-prompt subset, chosen before any ablation ran, serves the ablations, and we evaluate all 1000 HRGen-1K prompts (Rajabi et al., 2026a), which have no references. Sizes (height×width) are 4096^2, 2048×4096, and 2048×6144, plus 6144^2 (100 prompts) and 8192^2 (12 exploratory prompts). Seeds hash the prompt id and seed label 0 and are shared by all methods; labels 1 and 2 repeat all prompts.

Models and hyperparameters. The main model is FLUX.1-dev (Black Forest Labs, 2024); FLUX.2-dev (50 prompts, against full attention) and FLUX.1-Krea-dev (195 prompts, against AReA), each calibrated on 16 prompts at 1024^2, test other checkpoints (Appendices H and I). On FLUX.1-dev, SCA uses $t = 8$, $w = 64$, $\sigma_g = 0.89$, and the scale-4 table at every size (Table F.1), with the opening phase and window chosen on the development prompts only. Each process uses one NVIDIA GH200 120GB; worker logs give at least 123.3 GH200-hours (Appendix P).

Baselines. All baselines run on the main set. Full attention, AReA from its released code (Appendix H), DyPE (Issachar et al., 2025), SEGA (Rajabi et al., 2026b), and SigMa (Zhao et al.,

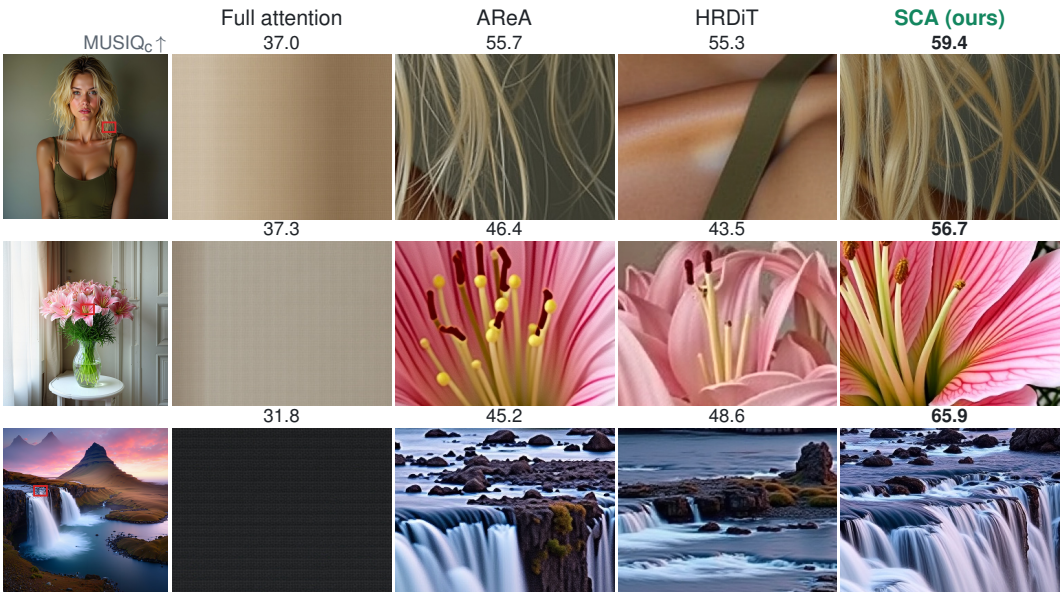

Figure 5: **SCA keeps native detail at 4096^2.** Three curated prompts (seed 0; curation favours SCA): SCA's image, then each method's red window at 1:1 under its MUSIQ_c (best bold). Full attention collapses to flat texture; Table 1 gives means and Figure R.2 uncurated samples.

2026) share AReA's sampler on FLUX.1-dev. It uses 28 Euler flow-matching steps, guidance 4.5, shift $\mu = 1.15$ at every size, bf16, and tiled VAE decoding. This fixes the sampler, not each baseline's tuned defaults (Appendix I). UltraImage (Zhao et al., 2025) runs its released guided two-stage recipe (30 steps per stage, guidance 3.5 then 7) and HRDiT (Xue et al., 2026) its progressive recipe (Section 2.5), each with its own budget.

Metrics. Preference and alignment scores are ImageReward, HPSv2, HPSv2.1, PickScore, CLIP score, and aesthetic score (Xu et al., 2023; Wu et al., 2023; Kirstain et al., 2023; Radford et al., 2021; Schuhmann, 2022). No-reference quality is MUSIQ on the full image and over four 1024^2 crops (MUSIQ_c), CLIP-IQA, and ARNIQA (Ke et al., 2021; Wang et al., 2022; Agnolucci et al., 2024). Distribution scores are FID and KID against all 195 official Aesthetic-4K reference rows (168 distinct images) and Inception score, all at 299^2, and patch FID and KID on the same crops (Heusel et al., 2017; Salimans et al., 2016). Crops depend only on the prompt id, so all methods share them. Patch FID is our own definition; AReA and SEGA do not define theirs, so we never compare it across papers. Table H.1 sets our numbers beside the published ones under AReA's protocol. Intervals are 95% paired percentile bootstraps over prompt ids (1000 or 2000 draws), which also resample reference images for the main FID and KID comparisons (Appendix G).

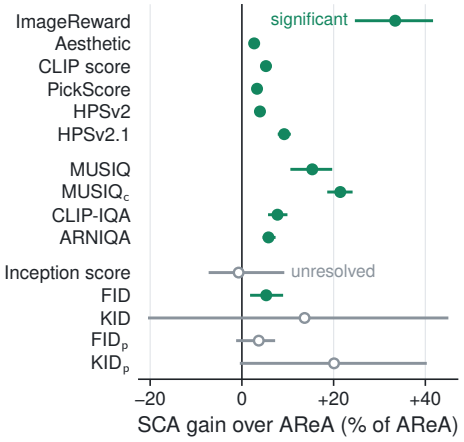

Figure 6: **Eleven of 15 metrics improve.** SCA minus AReA at 4096^2 as percent of AReA's mean, with 95% paired intervals. FID, KID, and their patch versions are negated so right is better; grey intervals span zero.

4.1 Main Results

Against AReA. SCA beats AReA on every per-image metric at $2.85\times$ the speed (Table 1; Figures 1 and 5). Eleven of 15 metrics improve significantly, including ImageReward ($+0.31$ $[0.23, 0.38]$) and FID (-7.7 $[-13.1, -2.6]$); four are unresolved (Figure 6; Table J.2). On FLUX.1-

Table 2: **Multi-stage recipes and sampling budgets.** 4096^2, 195 prompts, campaign times, columns as in Table 1 with the best in bold; against SCA at 28 steps, $^+$ is significantly better, $^\circ$ unresolved, and no mark significantly worse; 42 steps match HRDiT's time. Key result: SCA leads the single-pass methods, and progressive SCA (HRDiT's schedule, SCA attention) matches HRDiT's preference scores at 49.5 s against 110.4 s; Tables K.6, L.1, and L.2 add intervals.

		Time (s)↓		Image↑	↑	Pick↑	↑	↑	↓	↓
Method	Steps	clean	camp.	Reward	HPSv2.1	Score	CLIP	MUSIQ_c	FID	FID_p
Single pass, shared sampler (28 steps, guidance 4.5; other rows in Table 1)										
SigMa	28	n/m	192.5	−0.52	18.65	20.01	23.51	49.9	216.9	184.6
SCA (ours)	28	**71.1**	71.5	1.23	29.67	23.09	29.14	61.2	**138.1**	99.1
Multi-stage, own recipe and budget										
UltraImage	30+30	n/m	308.5	1.28°	30.77$^+$	23.33$^+$	28.84	41.7	142.0	109.5
HRDiT	30+17+10	108.6	110.4	**1.34**$^+$	30.92$^+$	**23.46**$^+$	29.12°	53.5	138.4°	103.0
SCA with other budgets or schedules										
56 steps	56	n/m	139.9	1.26$^+$	30.55$^+$	23.22$^+$	29.14°	**64.1**$^+$	139.8°	102.9
guidance 7	28	n/m	71.9	1.16	29.07	22.82	28.94°	61.4°	140.4°	**98.4**°
42 steps	42	n/m	106.8	1.26°	30.27$^+$	23.17$^+$	29.14°	63.1$^+$	140.5	100.6°
progressive	30+17+10	n/m	**49.5**	1.32$^+$	**30.99**$^+$	23.42$^+$	**29.36**°	54.2	141.1°	106.3

Krea-dev, which AReA's code loads and its published numbers use, our run reproduces its ImageReward, aesthetic score, and PickScore within 0.1 and its CLIP, HPS, and MUSIQ within 0.3 (Table H.1).

Other baselines. SCA also beats DyPE and SEGA on every per-image metric (Table K.2), and all five shared-sampler baselines, SigMa included (Table K.6). UltraImage, at over four times SCA's time, is significantly better on five per-image metrics, including HPSv2.1 and PickScore. It is worse on CLIP score and both MUSIQ scores, with ImageReward unresolved and higher FID (142.0 against 138.1; Tables K.2 and K.5). HRDiT's three-stage recipe leads single-pass SCA on learned preference scores at $1.5\times$ its time (Table 1).

HRDiT and progressive SCA. Progressive SCA matches HRDiT's preference scores in less than half the time, so HRDiT's lead comes from the schedule (Table 2; Figures L.1 and L.2). ImageReward differs by -0.02 $[-0.09, 0.03]$, HPSv2.1 by $+0.08$ $[-0.17, 0.32]$, and PickScore by -0.04 $[-0.13, 0.05]$. No per-image metric is significantly worse, HPSv2, MUSIQ, and ARNIQA are significantly better, and none of this establishes equivalence (Table L.2). Spending HRDiT's time on 42 single-pass steps does not close the gap. Against single-pass SCA, the parity costs fidelity: significantly higher patch FID, lower MUSIQ_c, and an unresolved rise in FID (Appendix L).

4.2 Other sizes and latency

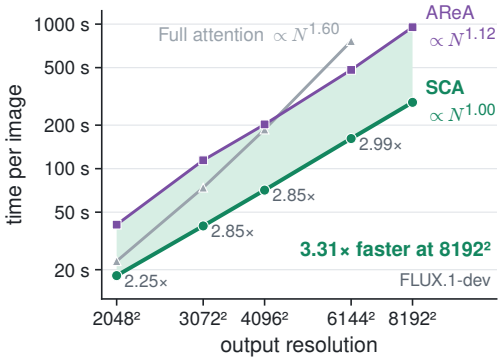

Figure 7: **Latency against size.** Batch 1, one GH200, median of three prompts; grey: speedup over AReA.

Coverage. SCA runs at every size and on every checkpoint we tried and is the fastest method we timed at each; quality depends on the aspect ratio (Tables J.3 and K.1). SCA beats AReA on 14 of 15 metrics at 6144^2 and on nine at 2048×6144, where it loses on aesthetic score and patch FID; results are mixed at 2048×4096. At 8192^2, 12 exploratory prompts favour SCA on preference and alignment scores, though grid artifacts remain (Appendix M).

Latency. Clean time (Table 1) is the median of three images after warm-up on an idle GPU, including text encoding and VAE decoding but not model loading; HRDiT's includes its cold first run. Figure 7 covers five sizes from 2048^2 to 8192^2. At 2048×4096 and 2048×6144, campaign times are 38.1 and 55.8 s for SCA against 112.4 and

Table 3: **Ablations.** 4096^2, fixed 46-prompt subset; the shaded default row gives absolute values and other rows the paired change from swapping one component, orange if significantly worse (none is better); IR, HPS, Pick, M_c: ImageReward, HPSv2.1 $\times 100$, PickScore, MUSIQ$_c$; Err.: relative output error at emulated scale 4; Table J.5 adds all metrics. Key result: every variant significantly lowers HPSv2.1 and PickScore.

Variant	Err.	IR	HPS	Pick	M_c
SCA (default)	0.200	1.12	29.98	22.96	62.5
Weight rule after the opening phase					
local only	0.261	−0.36	−2.86	−1.20	−0.3
shared $\pi(\sigma)$	0.230	−0.09	−1.83	−0.38	−10.4
per-layer $\pi_\ell(\sigma)$	0.223	−0.05	−0.70	−0.16	−2.4
Global branch and opening phase					
global only	0.627	−0.40	−5.66	−1.44	−11.7
mean pooling	n/a	−1.52	−9.75	−2.60	−4.2
no opening (v0)	0.200	−1.33	−4.07	−2.18	+2.5

Table 4: **Generality and sensitivity.** Paired differences at 4096^2 against each group's reference; default SCA: 32 calibration prompts, $\sigma_g = 0.89$; n: prompts; orange if significantly worse; Tables K.3, K.4, and K.1 add intervals. Key result: gains over AReA hold on HRGen-1K and with two more seeds.

Setting	n	ΔIR	ΔHPS	ΔM$_c$
Against AReA				
HRGen-1K	1000	+0.818	+4.93	+13.89
Seed 1	195	+0.274	+2.28	+11.51
Seed 2	195	+0.250	+2.50	+12.06
Against default SCA: calibration prompts, gate				
4 prompts	46	+0.012	−0.02	+0.09
8 prompts	46	+0.010	+0.01	−0.03
16 prompts	46	−0.007	−0.01	−0.11
$\sigma_g = 0.85$	46	−0.015	−0.85	−2.04
$\sigma_g = 0.93$	46	−0.004	−0.15	+0.65
Other checkpoints				
FLUX.2 vs. full	50	+0.456	+1.84	+21.44
Krea vs. AReA	195	−0.052	+0.63	+4.23

150.5 s for AReA. SCA's clean time grows nearly linearly with pixel count (Table D.2): fitting seconds $\propto N^b$ over the N image tokens gives $b = 1.00$ for SCA and 1.12 for AReA. At 6144^2 it takes 161.6 s, 4.7× and 3.0× faster than full attention and AReA. At 8192^2 it takes 287.6 s against AReA's 952.0 s, 3.31× faster; from 2048^2 to 8192^2, pixels grow 16× and SCA's time 15.8×.

4.3 Ablations

Table 3 changes one component at a time (examples in Figure R.7). **The opening phase is essential.** Without it (v0), ImageReward falls by 1.33 $[1.06, 1.60]$ and subjects repeat; CLIP-IQA and ARNIQA rise because each copy is sharp (Table J.5). **The per-head mix carries global context after the opening phase.** Local-only attention in the last 20 steps lowers ImageReward by 0.36 $[0.21, 0.52]$ and CLIP score by 1.93. **Per-head weights beat shared and per-layer weights, in the order output error predicts.** One shared weight per noise level (mix error 0.230) lowers ImageReward by 0.09 $[0.01, 0.17]$ and MUSIQ$_c$ by 10.4. Per-layer weights (0.223) still lose significantly on HPSv2, HPSv2.1, PickScore, ARNIQA, and MUSIQ$_c$. **Neither branch works alone, and block centres beat block means.** Global only loses on every metric; mean pooling blurs the global keys (ImageReward −1.52). Appendix O ablates a band guard and hard per-head routing.

4.4 Generality and sensitivity

Gains hold across seeds, prompt sets, and checkpoints. Gains over AReA hold with seeds 1 and 2 and on 1000 HRGen-1K prompts, improving all ten per-image metrics (Tables 4 and K.3). Calibrating on 4 to 16 prompts, not 32, moves ImageReward by ≤ 0.012; gates of 0.85 and 0.93 lose some metrics. On FLUX.2-dev it beats full attention on nine of ten per-image metrics in half the time (ImageReward 0.75 to 1.20). On FLUX.1-Krea-dev, over all 195 prompts, it wins six per-image metrics in a third of the time but loses 0.05 ImageReward, at the edge of significance, and patch FID (Table K.1). On all 1000 HRGen-1K prompts, Krea-calibrated SCA also exceeds every score AReA publishes there, though we did not rerun AReA on that checkpoint (Table H.1).

5 Related work

Training-free high-resolution sampling. Training-free methods alter convolutions, spectra, or attention temperature (He et al., 2023; Zhang et al., 2023; Huang et al., 2024; Jin et al., 2023), fuse global and patch paths (Du et al., 2023; Yang et al., 2024), by frequency in FreeScale (Qiu et al., 2024), add native-resolution guidance (Du et al., 2024; Bu et al., 2025; Koh et al., 2025), or adapt RoPE (Zhuo et al., 2024; Issachar et al., 2025; Zhao et al., 2026; Rajabi et al., 2026b; Liu et al., 2026). Of the closest methods (Table 5), AReA shares one schedule of pooled keys and

Table 5: **How training-free methods change attention.** Entries follow each paper's text or, where marked *, its mechanism; offsets: whether every image query-key offset stays in the trained range (text keys aside, Remark 1); cost: order in the image tokens N at a fixed native size. Key result: among single-pass methods, only SCA fits per-head weights in advance at the training resolution, and it keeps offsets in the trained range at linear cost.

Method	Local view	Global view	Per-head reach	Offsets	Cost	Passes
FreeScale	overlapping crops	full attention	×	n/a	$O(N^2)^*$	cascade
DyPE	×	full, scaled RoPE	×	×*	$O(N^2)^*$	single
SEGA	×	full, NTK, logit scale	×	×	$O(N^2)$	single
UltraImage	×	full, edited RoPE	entropy at run time	×*	$O(N^2)^*$	one or two
HRDiT	per-head scopes	×	budget, fitted per size	✓*	$O(N^2)^*$	cascade
ResDiT	native patches	full, interpolated	×	✓	$O(N^2)^*$	single
AReA	shrinking windows	pooled keys	×	×	$O(N^2)^*$	single
SCA (ours)	native window	block-centre keys	fitted at 1024^2	✓	$O(N)$	single

shrinking windows across heads (Rajabi et al., 2026a), and ResDiT fuses position-scaled global and native-size patch attention between global-only and local-only phases (Ma et al., 2025).

Head- and layer-specific attention. HRDiT picks per-head scopes under a cost budget (Xue et al., 2026); UltraImage concentrates attention per head by entropy and noted stable local and global head roles before us (Zhao et al., 2025). Sparse attention varies windows, spans, or key scales across layers or heads in diffusion transformers (Yuan et al., 2024; Xi et al., 2025; Zhang et al., 2025b), language models (Fu et al., 2024; Xiao et al., 2024), and vision transformers (Pan et al., 2022). Our local branch is neighbourhood attention (Hassani et al., 2022); ToDo attends to downsampled keys without rescaling coordinates (Smith et al., 2024). Head roles, two branches, and a global opening have precedents; SCA adds a closed-form per-head, per-noise-level mixture fitted by emulation at native resolution.

6 Limitations and Conclusion

Limitations. Late duplication is SCA's most visible failure. In 5 of the 11 uncurated FLUX.1-dev 4096^2 samples, it adds small people, animals, or objects in the last 20 steps, when the local branch dominates most heads. AReA's images of the same prompts blur or show lattice patterns instead (Appendix S). The opening phase prevents large repeats but not these small ones. A fix is in progress; no result here uses it. Multi-stage recipes keep higher learned preference scores than single-pass SCA, HRDiT at $1.5\times$ and UltraImage at over $4\times$ its time. Progressive SCA closes the gap to HRDiT at a cost in fidelity, with timings from different codebases. At 2048×4096 SCA has no resolved ImageReward or FID advantage over AReA and trails on aesthetic score, CLIP-IQA, and patch FID/KID. FLUX.2-dev covers 50 prompts, against full attention only. On FLUX.1-Krea-dev SCA trails AReA slightly on ImageReward and on patch FID. SigMa's dev-matched profile gives lattice-degraded images, possibly from our port or profile. We did not run ResDiT (no verified official code) or UltraImage's direct-extrapolation variant and claim no advantage over them. Grid artifacts at 8192^2 and one prompt's doubled limbs also remain. Metrics are automatic, low-sample FID is noisy, and intervals ignore multiple comparisons.

Conclusion. Each head's mix of a neighbourhood fixed in tokens and a view of the whole image can be measured once at the training resolution and reused at 4K. With that table and a global-only opening, SCA is the strongest and fastest single-pass training-free operator we tested for FLUX.1-dev at 4096^2, at about a third of AReA's time. In HRDiT's progressive schedule it matches HRDiT's preference scores in less than half the time, with no per-image metric significantly worse. Code: github.com/shreshthsaini/scale-covariant-attention.

REPRODUCIBILITY STATEMENT

Section 4 states the protocol, including timing (Section 4.2), and Appendix G defines every metric and interval. Appendix B gives the calibration geometry, statistics, and noise bins; Algorithms 1 and 2 give the procedure, and Appendix E gives it in full together with progressive SCA. Appendix F lists every hyperparameter, and Appendix D the kernel layout, numerical checks, and position rounding. Appendices H and I give each baseline's code revision, recipe, and seeds, and Appendix P the hardware and logged compute. Code, calibration table, per-image metrics, and the paired-difference files behind every table will be released in the code repository (link in the abstract).

REFERENCES

Lorenzo Agnolucci, Leonardo Galteri, Marco Bertini, and Alberto Del Bimbo. ARNIQA: Learning distortion manifold for image quality assessment. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2024.

Black Forest Labs. FLUX.1. `https://github.com/black-forest-labs/flux`, 2024.

Black Forest Labs. FLUX.2. `https://huggingface.co/black-forest-labs/FLUX.2-dev`, 2025.

Black Forest Labs and Krea AI. FLUX.1 Krea [dev]. `https://huggingface.co/black-forest-labs/FLUX.1-Krea-dev`, 2025.

Jiazi Bu, Pengyang Ling, Yujie Zhou, Pan Zhang, Tong Wu, Xiaoyi Dong, Yuhang Zang, Yuhang Cao, et al. HiFlow: Training-free high-resolution image generation with flow-aligned guidance. *arXiv preprint arXiv:2504.06232*, 2025.

Juechu Dong, Boyuan Feng, Driss Guessous, Yanbo Liang, and Horace He. Flex attention: A programming model for generating optimized attention kernels. *arXiv preprint arXiv:2412.05496*, 2024.

Ruoyi Du, Dongliang Chang, Timothy Hospedales, Yi-Zhe Song, and Zhanyu Ma. DemoFusion: Democratising high-resolution image generation with no $$$. *arXiv preprint arXiv:2311.16973*, 2023.

Ruoyi Du, Dongyang Liu, Le Zhuo, Qin Qi, Hongsheng Li, Zhanyu Ma, and Peng Gao. I-Max: Maximize the resolution potential of pre-trained rectified flow transformers with projected flow. *arXiv preprint arXiv:2410.07536*, 2024.

Patrick Esser, Sumith Kulal, Andreas Blattmann, Rahim Entezari, Jonas Müller, Harry Saini, Yam Levi, Dominik Lorenz, et al. Scaling rectified flow transformers for high-resolution image synthesis. *arXiv preprint arXiv:2403.03206*, 2024.

Tianyu Fu, Haofeng Huang, Xuefei Ning, Genghan Zhang, Boju Chen, Tianqi Wu, Hongyi Wang, Zixiao Huang, et al. Mixture of attention spans: Optimizing LLM inference efficiency with heterogeneous sliding-window lengths. *arXiv preprint arXiv:2406.14909*, 2024.

Ali Hassani, Steven Walton, Jiachen Li, Shen Li, and Humphrey Shi. Neighborhood attention transformer. *arXiv preprint arXiv:2204.07143*, 2022.

Yingqing He, Shaoshu Yang, Haoxin Chen, Xiaodong Cun, Menghan Xia, Yong Zhang, Xintao Wang, Ran He, et al. ScaleCrafter: Tuning-free higher-resolution visual generation with diffusion models. *arXiv preprint arXiv:2310.07702*, 2023.

Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. GANs trained by a two time-scale update rule converge to a local Nash equilibrium. *arXiv preprint arXiv:1706.08500*, 2017.

Linjiang Huang, Rongyao Fang, Aiping Zhang, Guanglu Song, Si Liu, Yu Liu, and Hongsheng Li. FouriScale: A frequency perspective on training-free high-resolution image synthesis. *arXiv preprint arXiv:2403.12963*, 2024.

Noam Issachar, Guy Yariv, Sagie Benaim, Yossi Adi, Dani Lischinski, and Raanan Fattal. DyPE: Dynamic position extrapolation for ultra high resolution diffusion. *arXiv preprint arXiv:2510.20766*, 2025.

Zhiyu Jin, Xuli Shen, Bin Li, and Xiangyang Xue. Training-free diffusion model adaptation for variable-sized text-to-image synthesis. *arXiv preprint arXiv:2306.08645*, 2023.

Junjie Ke, Qifei Wang, Yilin Wang, Peyman Milanfar, and Feng Yang. MUSIQ: Multi-scale image quality transformer. *arXiv preprint arXiv:2108.05997*, 2021.

Yuval Kirstain, Adam Polyak, Uriel Singer, Shahbuland Matiana, Joe Penna, and Omer Levy. Pick-a-Pic: An open dataset of user preferences for text-to-image generation. *arXiv preprint arXiv:2305.01569*, 2023.

Sungho Koh, SeungJu Cha, Hyunwoo Oh, Kwanyoung Lee, and Dong-Jin Kim. ScaleDiff: Higher-resolution image synthesis via efficient and model-agnostic diffusion. *arXiv preprint arXiv:2510.25818*, 2025.

Yihua Liu, Fanjiang Ye, Bowen Lin, Rongyu Fang, and Chengming Zhang. TIDE: Text-informed dynamic extrapolation with step-aware temperature control for diffusion transformers. *arXiv preprint arXiv:2603.08928*, 2026.

Yiyang Ma, Feng Zhou, Xuedan Yin, Pu Cao, Yonghao Dang, and Jianqin Yin. ResDiT: Evoking the intrinsic resolution scalability in diffusion transformers. *arXiv preprint arXiv:2512.01426*, 2025. URL https://arxiv.org/abs/2512.01426.

Zizheng Pan, Jianfei Cai, and Bohan Zhuang. Fast vision transformers with HiLo attention. *arXiv preprint arXiv:2205.13213*, 2022.

William Peebles and Saining Xie. Scalable diffusion models with transformers. *arXiv preprint arXiv:2212.09748*, 2022.

Haonan Qiu, Shiwei Zhang, Yujie Wei, Ruihang Chu, Hangjie Yuan, Xiang Wang, Yingya Zhang, and Ziwei Liu. FreeScale: Unleashing the resolution of diffusion models via tuning-free scale fusion. *arXiv preprint arXiv:2412.09626*, 2024.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, et al. Learning transferable visual models from natural language supervision. *arXiv preprint arXiv:2103.00020*, 2021.

Javad Rajabi, Koorosh Roohi, David B. Lindell, and Babak Taati. AReA: Adaptive receptive-field attention for efficient high-resolution diffusion transformers. *arXiv preprint arXiv:2610.17227*, 2026a. URL https://arxiv.org/abs/2610.17227.

Javad Rajabi, Kimia Shaban, Koorosh Roohi, David B. Lindell, and Babak Taati. SEGA: Spectral-energy guided attention for resolution extrapolation in diffusion transformers. *arXiv preprint arXiv:2605.22668*, 2026b.

Tim Salimans, Ian Goodfellow, Wojciech Zaremba, Vicki Cheung, Alec Radford, and Xi Chen. Improved techniques for training GANs. In *Advances in Neural Information Processing Systems*, 2016.

Christoph Schuhmann. LAION aesthetics predictor. https://github.com/LAION-AI/aesthetic-predictor, 2022.

Ethan Smith, Nayan Saxena, and Aninda Saha. ToDo: Token downsampling for efficient generation of high-resolution images. *arXiv preprint arXiv:2402.13573*, 2024.

Jianlin Su, Yu Lu, Shengfeng Pan, Ahmed Murtadha, Bo Wen, and Yunfeng Liu. RoFormer: Enhanced transformer with rotary position embedding. *arXiv preprint arXiv:2104.09864*, 2021.

Jianyi Wang, Kelvin C. K. Chan, and Chen Change Loy. Exploring CLIP for assessing the look and feel of images. *arXiv preprint arXiv:2207.12396*, 2022.

Xiaoshi Wu, Yiming Hao, Keqiang Sun, Yixiong Chen, Feng Zhu, Rui Zhao, and Hongsheng Li. Human preference score v2: A solid benchmark for evaluating human preferences of text-to-image synthesis. *arXiv preprint arXiv:2306.09341*, 2023.

Haocheng Xi, Shuo Yang, Yilong Zhao, Chenfeng Xu, Muyang Li, Xiuyu Li, Yujun Lin, Han Cai, et al. Sparse VideoGen: Accelerating video diffusion transformers with spatial-temporal sparsity. *arXiv preprint arXiv:2502.01776*, 2025.

Guangxuan Xiao, Jiaming Tang, Jingwei Zuo, Junxian Guo, Shang Yang, Haotian Tang, Yao Fu, and Song Han. DuoAttention: Efficient long-context LLM inference with retrieval and streaming heads. *arXiv preprint arXiv:2410.10819*, 2024.

Jiazheng Xu, Xiao Liu, Yuchen Wu, Yuxuan Tong, Qinkai Li, Ming Ding, Jie Tang, and Yuxiao Dong. ImageReward: Learning and evaluating human preferences for text-to-image generation. *arXiv preprint arXiv:2304.05977*, 2023.

Yu Xue, Haoxuan Qu, Zhuoling Li, Hongbin Xu, Jianxiong Yin, Simon See, Hossein Rahmani, and Jun Liu. HRDiT: Training-free high-resolution image generation with off-the-shelf diffusion transformer models. *arXiv preprint arXiv:2608.07003*, 2026. URL https://arxiv.org/abs/2608.07003.

Haosen Yang, Adrian Bulat, Isma Hadji, Hai X. Pham, Xiatian Zhu, Georgios Tzimiropoulos, and Brais Martinez. FAM diffusion: Frequency and attention modulation for high-resolution image generation with stable diffusion. *arXiv preprint arXiv:2411.18552*, 2024.

Zhihang Yuan, Hanling Zhang, Pu Lu, Xuefei Ning, Linfeng Zhang, Tianchen Zhao, Shengen Yan, Guohao Dai, et al. DiTFastAttn: Attention compression for diffusion transformer models. *arXiv preprint arXiv:2406.08552*, 2024.

Jinjin Zhang, Qiuyu Huang, Junjie Liu, Xiefan Guo, and Di Huang. Diffusion-4K: Ultra-high-resolution image synthesis with latent diffusion models. *arXiv preprint arXiv:2503.18352*, 2025a.

Peiyuan Zhang, Yongqi Chen, Runlong Su, Hangliang Ding, Ion Stoica, Zhengzhong Liu, and Hao Zhang. Fast video generation with sliding tile attention. *arXiv preprint arXiv:2502.04507*, 2025b.

Shen Zhang, Zhaowei Chen, Zhenyu Zhao, Yuhao Chen, Yao Tang, and Jiajun Liang. HiDiffusion: Unlocking higher-resolution creativity and efficiency in pretrained diffusion models. *arXiv preprint arXiv:2311.17528*, 2023.

Bingxuan Zhao, Qing Zhou, Yu Wang, Chuang Yang, and Qi Wang. σ: Sigmoid modulation for ultra high resolution diffusion. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pp. 162993–163008. PMLR, 2026. URL https://proceedings.mlr.press/v306/zhao26cc.html.

Min Zhao, Bokai Yan, Xue Yang, Hongzhou Zhu, Jintao Zhang, Shilong Liu, Chongxuan Li, and Jun Zhu. UltraImage: Rethinking resolution extrapolation in image diffusion transformers. *arXiv preprint arXiv:2512.04504*, 2025. URL https://arxiv.org/abs/2512.04504.

Le Zhuo, Ruoyi Du, Han Xiao, Yangguang Li, Dongyang Liu, Rongjie Huang, Wenze Liu, Lirui Zhao, et al. Lumina-Next: Making Lumina-T2X stronger and faster with Next-DiT. *arXiv preprint arXiv:2406.18583*, 2024.

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A PROOFS

Lemma 1 (Softmax split)**.** *Let the keys of a query be partitioned into a near set $\mathcal{K}^{\text{near}}$ and a far set $\mathcal{K}^{\text{far}}$, with logits ℓ_j. Then $O^{\text{full}} = m\,O^{\text{near}} + (1-m)\,O^{\text{far}}$ with $m = \sum_{j\in\mathcal{K}^{\text{near}}} e^{\ell_j} / \sum_j e^{\ell_j}$ and $O^{\text{near}}, O^{\text{far}}$ the softmax outputs restricted to $\mathcal{K}^{\text{near}}$ and $\mathcal{K}^{\text{far}}$.*

Proof. Write $Z^{\text{near}} = \sum_{j\in\mathcal{K}^{\text{near}}} e^{\ell_j}$, so that $Z^{\text{near}}O^{\text{near}} = \sum_{j\in\mathcal{K}^{\text{near}}} e^{\ell_j} v_j$, and likewise for the far set. Then $O^{\text{full}} = (Z^{\text{near}}O^{\text{near}} + Z^{\text{far}}O^{\text{far}})/(Z^{\text{near}} + Z^{\text{far}})$, and $Z^{\text{near}}/(Z^{\text{near}} + Z^{\text{far}}) = m$. □

Proof of Proposition 1. Global operator. Image queries sit at $c(p_i) \in \mathcal{G}_0 = [n_0]^2$ and text queries at $0 \in \mathcal{G}_0$; subsampled keys sit at $b \in \mathcal{G}_0$ and text keys at 0. Any two points of $[n_0]^2$ differ by at most $n_0 - 1$ per axis. The key count is $T + |\mathcal{G}_0| = T + n_0^2$.

Local operator. Fix one axis of length $H \geq w$ and a tile $[t\tau,\, t\tau + t)$ with $t \leq w$. The unclamped start $a' = t\tau + \lfloor t/2 \rfloor - \lfloor w/2 \rfloor$ satisfies $a' \leq t\tau$ because $\lfloor t/2 \rfloor \leq \lfloor w/2 \rfloor$, and $a' + w \geq t\tau + t$ because $w - \lfloor w/2 \rfloor = \lceil w/2 \rceil \geq \lceil t/2 \rceil = t - \lfloor t/2 \rfloor$. If $a' < 0$ the window is $[0, w)$, and $w \geq a' + w \geq t\tau + t$. If $a' > H - w$ the window is $[H - w, H)$, with $H - w < a' \leq t\tau$ and $H \geq t\tau + t$. In every case the window contains the tile, so it contains each query of the tile, and two positions in a window of side $w \leq n_0$ differ by at most $n_0 - 1$. The key count is $T + w^2 = T + n_0^2$ when $n_0 \leq \min(H, W)$. □

Remark 1 (Text-image offsets). FLUX.1 places all text tokens at position 0, so the offset between an image query at p_i and a text key is p_i. In the global operator the query sits at $c(p_i) \in [n_0]^2$, inside the trained range. In the local operator it keeps p_i, up to $sn_0 - 1$ per axis, exactly as in full attention and AReA's windows at the target. Mapping text offsets through $c(\cdot)$ in the local operator is a possible refinement that we have not tested.

Proof of Proposition 2. Full attention scores every query against every key, $(N + T)^2$ pairs. The global operator scores $N + T$ queries against $T + |\mathcal{G}_0|$ keys with $|\mathcal{G}_0| = N/(s_H s_W) = n_0^2$. The local operator scores N image queries against T text keys and w^2 window keys. The sum is at most $(2N + T)(T + w^2 + N_0)$, which is $O(N(T + w^2 + N_0))$ for fixed n_0, T. Value aggregation has the same pair counts. In the opening phase the local term is absent. The FlexAttention kernel visits whole blocks and masks the extra pairs: a query block of 8×16 tokens spans two tiles whose windows are 8 columns apart, so it visits a 64×72 key region, 12.5% more than one window, plus partial blocks at clamped borders. This changes the constant, not the order. □

Proof of Proposition 3. $J(\pi) = E - 2\pi B + \pi^2 A$ with $A \geq 0$. If $A = 0$ then $O^{\text{loc}} = O^{\text{glob}}$ on every summand, so $B = 0$ and J is constant; any π, in particular $\pi^\star = 1$, is optimal and the bound holds trivially. If $A > 0$, J is a strictly convex quadratic with unconstrained minimizer B/A, and its minimizer on $[0, 1]$ is the projection $\text{clip}(B/A, 0, 1)$. For any π',

$$J(\pi') - J(\pi^\star) = A(\pi' - \pi^\star)^2 + 2(\pi' - \pi^\star)(A\pi^\star - B).$$

If $0 < \pi^\star < 1$ then $A\pi^\star = B$ and the second term vanishes. If $\pi^\star = 0$ then $B \le 0$, so $A\pi^\star - B \ge 0$ and $\pi' - \pi^\star \ge 0$. If $\pi^\star = 1$ then $B \ge A$, so $A\pi^\star - B \le 0$ and $\pi' - \pi^\star \le 0$. In both clipped cases the second term is nonnegative.

Any rule that assigns one weight per cell, such as a per-layer table, a weight shared per noise level, a hard switch, or a table fitted elsewhere, is a feasible π' in every cell, so its total error is at least the per-head table's, by $\sum_{\text{cells}} A(\pi' - \pi^\star)^2$ or more. A table interpolated in σ between the mean sampled noise levels of occupied bins is not constant within a bin; the statement applies to the bin-constant table used to score rules in Section 3. □

The frequency argument in more detail. Argument 1 rests on two facts and one assumption. The first fact is that the noisy latent is a sum of signal and independent white noise, so a band's signal-to-noise ratio scales as $((1-\sigma)/\sigma)^2 S(f)$; at $\sigma = 0.89$ the factor is 0.015, and at $\sigma = 0.5$ it is 1. The second is that a window of w tokens cannot distinguish a component of period longer than w from a smooth trend within the window, so windows cannot coordinate a layout at that scale except through what earlier layers mixed into their keys. The assumption is that natural-image latent spectra fall with frequency, which we have not measured for FLUX.1 latents. Together they predict that windowed attention fails first and worst at high noise and for objects larger than the window, which matches the duplication we observe (Figure A.1) and the effect of a 16-token window.

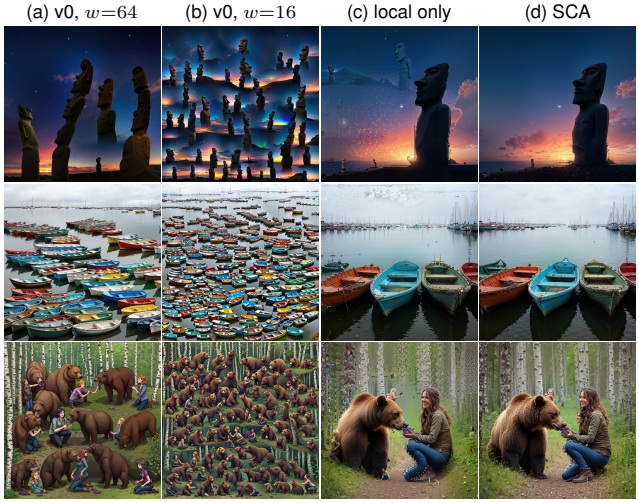

Figure A.1: **Windows at high noise duplicate subjects.** 4096^2, three development prompts. (a) v0, the calibrated mix from the first step: each 64-token window settles the layout on its own. (b) A 16-token window gives smaller, more numerous copies. (c) With the opening phase but local-only attention afterwards ($\pi = 1$), ghost copies and grid speckle remain. (d) SCA keeps one main subject per row, though a small second Moai stands on the far shore and extra faces appear around the woman and the bear.

B Calibration details and statistics

This section expands Section 3 and Figures 3b and 4. Figure B.1 shows the calibrated tables, Figure B.2 every head's $\pi_h(\sigma)$ and the split-half agreement, Figures B.3 and B.4 the output errors per rule and per noise level, and Figure B.5 every head's weight per block.

Statistics. For every emulated scale, block, head, and noise bin, the pass stores A, B, E, the error of the local branch alone $\sum \|O^{\text{full}} - O^{\text{loc}}\|^2$, $\sum \|O^{\text{full}}\|^2$, and the query count, in float64, summed over image queries and channels (`sca/calib.py`). The sampler's trajectory is that of unmodified full attention; the emulated operators only observe. Statistics are written per prompt and merged by summation, so prompts can be added later and split-half fits need no extra pass. Text-query outputs are not in the objective, because SCA sends text queries through the global branch only.

Geometry at emulated scale. At 1024^2 the grid is 64×64. Emulated scale 2 uses a 32×32 window, 2×2 subsampling onto a 32×32 grid, and 4×4 query tiles; emulated scale 4 uses a 16×16 window, 4×4 subsampling onto a 16×16 grid, and 2×2 tiles. At 1280^2 the grid is 80×80, and emulated scale 2 uses a 40×40 window, 2×2 subsampling onto a 40×40 grid, and 5×5 tiles.

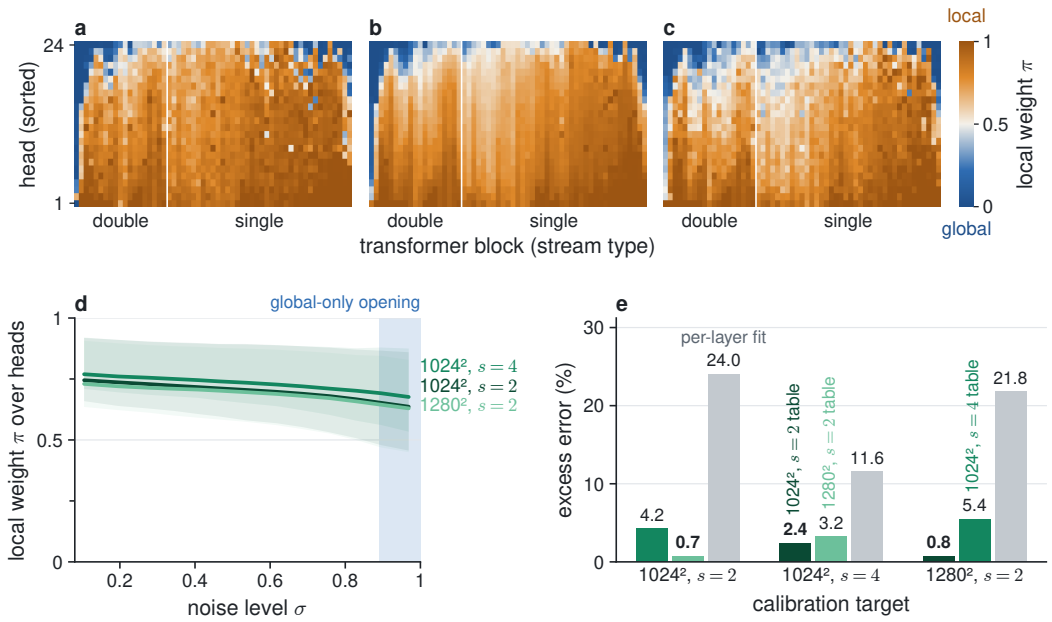

Figure B.1: **Heads differ widely, yet a calibrated table transfers across scale and resolution.** (a to c) $\pi_h(\sigma)$ at $\sigma = 0.10, 0.46, 0.83$ ($\sigma = 1$: pure noise); (d) mean and interquartile range across heads, shaded: global-only opening. (e) Excess error over each target's table: at most 5.4% transferred, 12 to 24% per-layer.

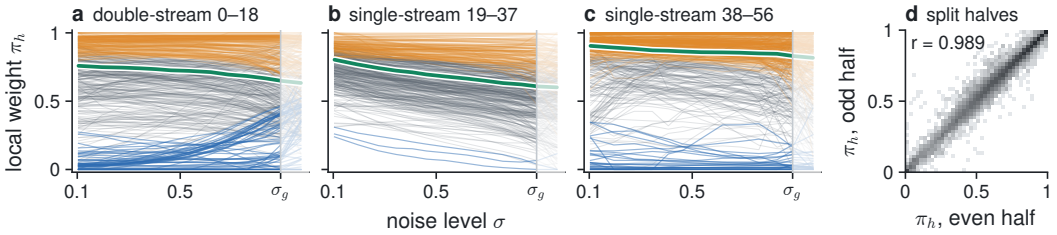

Figure B.2: **Heads want different mixes.** (a–c) $\pi_h(\sigma)$ of all 1368 heads at emulated scale 2; amber: mean above 0.8, blue: below 0.2, green: median; $\sigma > \sigma_g$ veiled. (d) Tables from disjoint prompt halves agree ($r = 0.989$).

Noise bins. With 28 steps and $\mu = 1.15$, σ runs from 1.0 to 0.105, so the lowest of the 14 bins is empty and copies its neighbour; 13 bins are populated. Under this schedule 8 steps have $\sigma > 0.89$ (the ninth has $\sigma = 0.888$) and 13 have $\sigma > 0.80$.

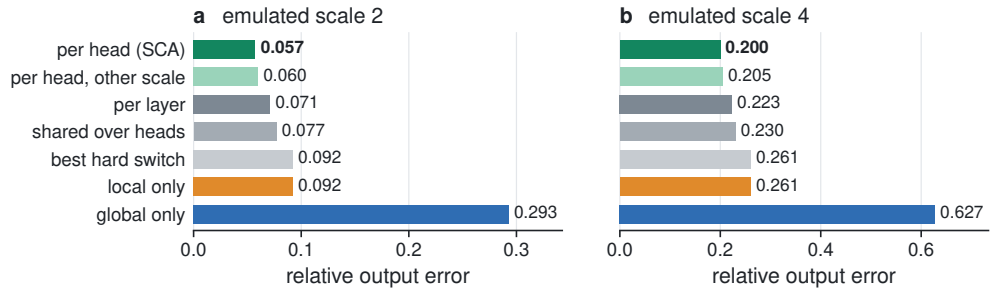

Figure B.3: **Per-head weights explain full attention best, and transfer across scales.** Relative output error per mixing rule at emulated scales 2 (a) and 4 (b), 32 prompts; "other scale" swaps the scale-2 and scale-4 tables; the best hard switch ($\sigma^\star = 1$) equals local only. In-sample and held-out errors agree within 0.0003.

No global phase in output energy. The best single shared weight falls from 0.88 at $\sigma = 0.10$ to 0.72 at $\sigma = 0.97$, and the mean per-head weight from 0.74 to 0.64 (Figure B.4); within a block, the

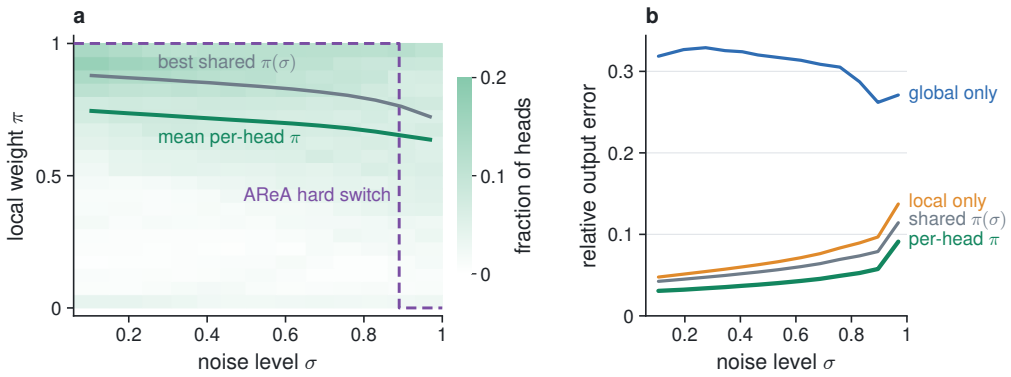

Figure B.4: **In output energy, no noise level prefers the global branch.** (a) Fraction of heads per π and noise bin, with mean $\pi_h(\sigma)$, best shared $\pi(\sigma)$ and AReA's switch; scale 2, 32 prompts, $\sigma = 1$: pure noise. (b) Error per noise bin: local beats global everywhere; per-head mixing beats both.

per-head weights span a median range of 0.66 to 0.84 at every noise level (Figure B.5). Among hard switches from global to local, the best threshold is $\sigma^\star = 1$, which equals local only. A switch at $\sigma^\star = 0.89$ between our two branches raises the error from 0.092 to 0.152 at scale 2 and from 0.261 to 0.376 at scale 4. In the noisiest bin, the global branch alone has relative error 0.27 at scale 2, the local branch alone 0.14, and the per-head mix 0.09. The generation results of Section 4.3 show the opposite preference for the opening steps, which is the gap between squared output error and layout.

Attention mass. For 256 random image queries per block and step, the pass also records the softmax mass on text keys and on image keys within Chebyshev distance 2, 4, 8, and 16. At 1024^2, averaged over heads and queries, text receives 15 to 19% of the mass; image keys within distance 16 receive 52% at $\sigma = 0.10$ and 40% at $\sigma = 0.97$, and keys outside that 33×33 neighbourhood still receive 29% and 41%. A purely local receptive field cannot account for any step, even at native resolution. Figure 4b and d use the per-head shares in the noise bin at $\sigma = 0.62$; panel b caps the opacity at 12 times the uniform density, which clips only the local head's core.

Calibration budget. We merge the stored sufficient statistics from the first 4, 8, or 16 calibration prompts and rerun the 46 ablation prompts with each resulting table. Table K.4 reports generation-level differences from the 32-prompt table. The test reuses calibration observations but uses separate prompts for generation evaluation; it is not a held-out output-error benchmark.

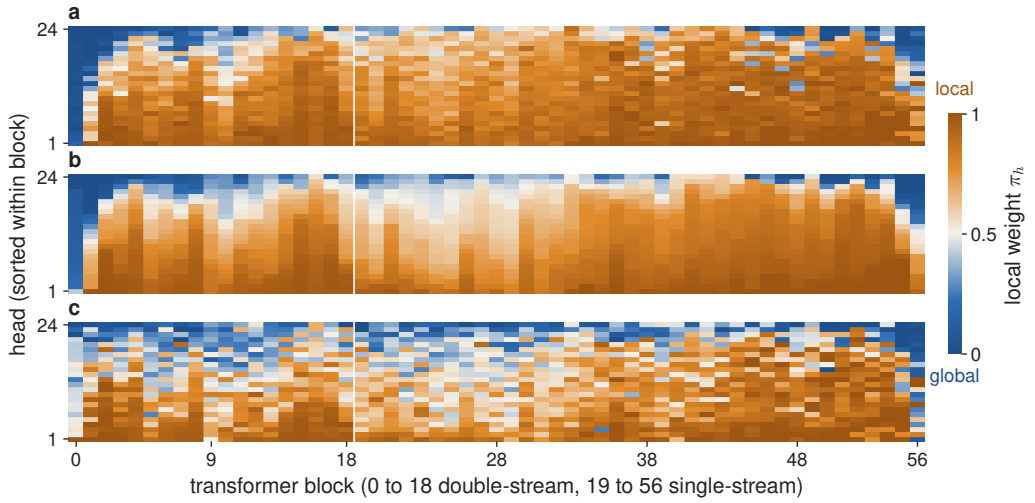

Figure B.5: **Heads in the same block differ widely at every noise level.** $\pi_h(\sigma)$ of the scale-2 table at $\sigma = 0.10$ (a), 0.54 (b), 0.97 (c), 57 blocks by 24 heads, 32 prompts, heads sorted per block by (b). Median within-block range 0.66 to 0.84; Figure B.1 shows the scale-4 inference table.

C CALIBRATION PROMPTS

This section lists the prompts behind the calibration of Section 3 and Algorithm 1 (`scripts/calib_prompts.jsonl`). The 1024^2 table uses prompts 1 to 32 and the 1280^2 table prompts 1 to 16. None of them is an evaluation prompt.

1. A candid street portrait of an elderly fisherman with a weathered face and a knitted wool cap, harbor in the background, overcast light
2. Two children laughing while running through a lawn sprinkler on a summer afternoon, backlit by the sun
3. A professional headshot of a young woman with curly red hair and freckles, soft studio lighting, neutral grey backdrop
4. A jazz trio performing in a dim basement club, saxophone player in the foreground, warm stage lights and haze
5. A chef plating a dessert in a busy restaurant kitchen, steam rising, stainless steel counters
6. A ballet dancer mid-leap in an empty rehearsal studio with tall windows and wooden floors
7. A red fox standing in fresh snow at the edge of a birch forest, early morning light
8. A macro photograph of a honeybee covered in pollen on a purple lavender flower
9. A herd of elephants crossing a dusty savanna river at sunset, long shadows
10. A tabby cat sleeping curled up on a stack of old books next to a window
11. A school of tropical fish swimming around a coral reef, sun rays through clear blue water
12. A bald eagle perched on a dead pine branch above a misty lake
13. A neon sign reading "OPEN 24 HOURS" in the window of a rainy diner at night, reflections on the wet sidewalk
14. A hand-painted wooden sign that says "Fresh Eggs" leaning against a farm fence, chickens nearby
15. A vintage movie theater marquee with the words "NOW SHOWING" lit by hundreds of bulbs
16. A chalkboard menu in a cafe listing coffee drinks and prices in neat handwriting
17. A highway road sign reading "Exit 42 Pine Valley" against a clear blue sky
18. A bookstore storefront with gold lettering "Second Chapter Books" on the glass, autumn leaves on the ground
19. A wide alpine valley with a winding river, snow-capped peaks and wildflower meadows, midsummer
20. Sand dunes in the desert at golden hour with sharp ridges and wind ripples
21. A rugged coastline with sea stacks and crashing waves under a stormy sky
22. A terraced rice field on a hillside in morning fog, farmers in conical hats
23. The northern lights over a frozen lake with a small wooden cabin, stars visible
24. A dense tropical rainforest with a waterfall and hanging vines, light filtering through the canopy
25. A cozy Scandinavian living room with a fireplace, sheepskin rug, and large windows showing snow outside
26. A modern open-plan kitchen with a marble island, pendant lights, and potted herbs
27. A cluttered artist studio with canvases, paint tubes, brushes in jars, and afternoon sunlight
28. A grand library reading room with long oak tables, green lamps, and arched ceilings
29. A small Japanese tatami room with a low table, sliding paper doors, and a garden view
30. A product photograph of a matte black wristwatch on a slate surface, dramatic side light
31. A pair of white running shoes floating against a pastel gradient background, studio product shot
32. A glass bottle of olive oil with a sprig of rosemary on a rustic wooden table, soft daylight
33. A flat lay of a leather notebook, fountain pen, glasses, and a cup of espresso on a desk
34. A ceramic teapot with a blue glaze and matching cups arranged on a linen tablecloth
35. A crowded night market in Taipei with food stalls, lanterns, and hundreds of people
36. A packed football stadium during a night match, floodlights and cheering fans
37. Commuters crossing a huge intersection in Tokyo during rush hour, seen from above
38. A music festival crowd at sunset with raised hands and confetti in the air
39. A busy train station concourse with travelers, departure boards, and a glass roof
40. A gothic cathedral facade with intricate stone carvings and a rose window, low angle
41. A row of colorful Victorian townhouses on a hilly street with a cable car
42. A brutalist concrete apartment block with repeating balconies under a pale sky

43. A glass skyscraper reflecting clouds, shot looking straight up from the street
44. A traditional Chinese pagoda beside a lake with cherry blossoms in full bloom
45. An old stone bridge over a canal in Venice with gondolas and laundry lines
46. An astronaut riding a horse across a red desert planet with two moons, cinematic
47. A detailed watercolor illustration of a lighthouse on a rocky island with seagulls
48. A steampunk airship floating above a Victorian city at dusk, intricate brass details

D IMPLEMENTATION DETAILS AND COST

This section expands Section 2.2 and Figure 3. Figure D.1 draws the two operators, the gate, and the mix at component level, together with the calibrated table for all 57 blocks. Table D.1 gives attention cost, and Table D.2 and Figure D.2 end-to-end latency; Appendix E writes out the corresponding algorithms.

FlexAttention window. Image tokens are reordered tile-major, so that a sparse block is a small spatial tile. Query blocks cover 8×16 tokens (128 queries) and key blocks 8×8 (64 keys), shifted so that unclamped windows start on a block edge; text keys come first. The mask function states window membership per query-key pair, so the result does not depend on the block layout, only the amount of skipped work does. On GH200, 8×16 query blocks with 8×8 key blocks were fastest at 2048^2, 4096^2, and 2048×6144. RoPE, subsampling, both branches, and the mix compile into one graph per layer; the mix is in fp32.

Numerical checks. The 18 CPU tests in `tests/test_sca.py` use float32 to check full-mode agreement with stock FLUX attention, both branches against brute-force Equations 2 and 3, full-grid and factor-1 identities, calibration when the local branch is exact, the gate, and FlexAttention against an explicit gather reference. A separate GPU check in `scripts/bench_attn.py` compares the gather and FlexAttention implementations in bf16: the recorded maximum absolute difference is 0.015625, and both have the same maximum error against the fp32 reference on the tested grids. This is an observed absolute-error check; no per-element ULP bound was measured.

Position ids in bf16. The FLUX.1 pipeline builds image position ids in bf16, which cannot represent every integer above 256. Our frozen runs retain these rounded ids for RoPE. Passing the true grid from the sampler fixes window membership and subsampling geometry, but does not restore exact RoPE coordinates: the coarse-query positions are also computed from the rounded ids. At a 384-token axis, index 383 rounds to 384 and its scale-6 coarse coordinate becomes 64; at a 512-token axis, index 511 rounds to 512 and its scale-8 coarse coordinate also becomes 64. A local 64-token window can likewise have a rounded endpoint difference of 64 instead of the ideal bound 63. Proposition 1 therefore applies exactly to the mathematical operator, not to these finite-precision positions at 2048×6144, 6144^2, or 8192^2. The 4096^2 grid uses indices 0–255, which bf16 represents exactly. FLUX.2 uses integer position ids. Rebuilding FLUX.1 positions in higher precision would require a separate controlled experiment; the reported runs are unchanged.

Non-divisible grids. When H/n_0 is not an integer, the subsampling factor per axis is the divisor of H nearest to H/n_0, and the tile side the divisor nearest to t; neither case arises at the sizes we evaluate.

End-to-end latency. Table D.2 and Figure D.2 come from one matched protocol on one GH200 (`results/latency_curve_batch1.csv`, which also lists the excluded warm-up images; `results/latency_curve_summary.csv` summarizes all cells but full attention at 6144^2). At each size, each method generates one excluded warm-up image and then three timed images in the same process, at batch 1 and 28 steps; we report the median. At 4096^2 the medians are 186.5 s for full attention (186.0, 186.6, 186.5), 202.4 s for AReA (192.7, 202.4, 204.3), and 71.1 s for SCA (71.0, 71.1, 71.1). At 2048^2 and 3072^2 they are 22.9 and 73.9 s for full attention, 41.0 and 114.3 s for AReA, and 18.2 and 40.2 s for SCA. At 6144^2 and 8192^2, AReA takes 482.3 and 952.0 s and SCA 161.6 and 287.6 s, so SCA reduces measured time by 66.5% and 69.8%. At 6144^2 full attention takes 762.7 s (763.5, 762.7, 762.5), $4.7\times$ SCA's time. An earlier 4096^2 benchmark, kept as a separate repeat (`results/latency_4096_batch1.csv`), gave 184.6 s for full attention (184.1, 184.7, 184.6), 212.6 s for AReA (207.3, 225.8, 212.6), 88.1 s for SCA + band guard (88.1, 98.7, 88.1; single-pass compiled band), and 71.0 s for SCA final (71.0, 83.1, 70.9). On the shared

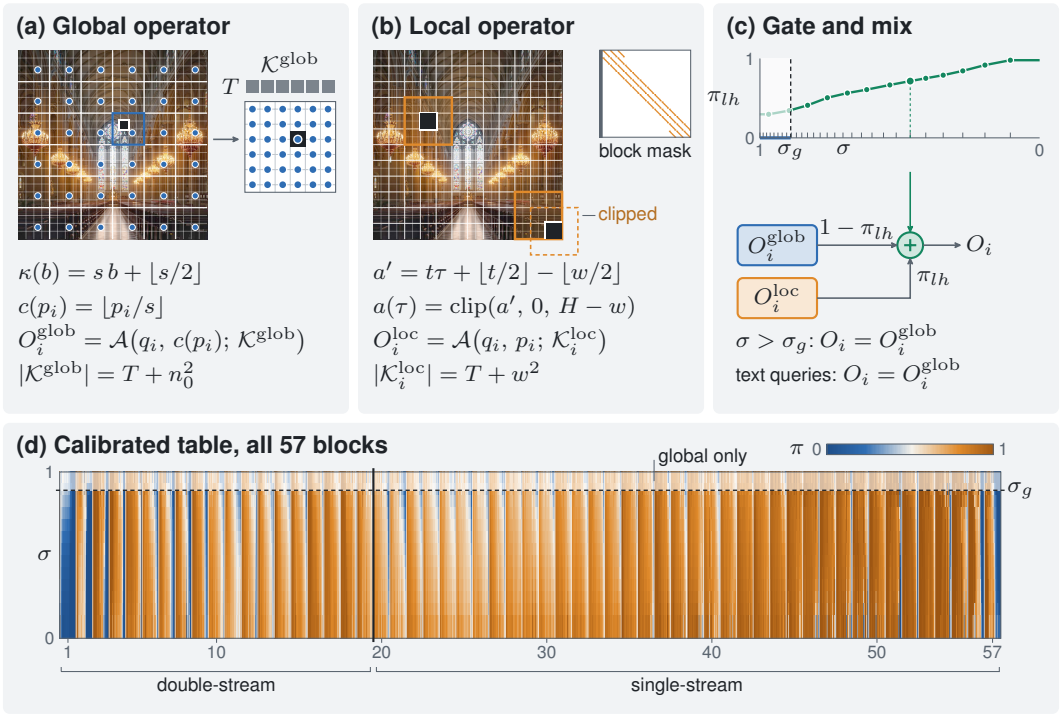

Figure D.1: **The operators at component level.** (a) The global operator keeps the centre token $\kappa(b)$ of each $s \times s$ block as a key at coarse position b and places query i at $c(p_i)$, so each softmax sees $T + n_0^2$ keys (Equation 2). (b) The local operator lets each $t \times t$ query tile attend to a $w \times w$ window at original positions, clipped at the border (dashed: unclipped a'); one FlexAttention call applies the exact block mask (inset; grey: text keys; Equation 3). (c) Each head's weight $\pi_{lh}(\sigma)$ (block 29, head 18) is interpolated linearly between bin centres; for $\sigma > \sigma_g$, the first 8 of 28 steps, every head returns O_i^{glob} and skips the local branch (Equation 4). (d) The scale-4 table used at inference covers all 57×24 heads, sorted by mean π within each block, and 14 noise bins; over the populated bins, π averages 0.67 in the double-stream and 0.76 in the single-stream blocks. Panels (a) and (b) use a toy grid with $H = 24$, $s = 4$, $n_0 = w = 6$, and $t = 2$.

Table D.1: **Cost.** Top: analytic attention FLOPs per image (57 blocks, 24 heads, head dimension 128, 28 steps, $T = 512$); "gated" skips the local branch in the first 8 steps; ratio is full over gated SCA. Bottom: measured time of one attention layer in bf16 on GH200, minimum over interleaved runs; SCA is both branches and the mix.

Size ($H \times W$)	Grid	(s_H, s_W)	Full	SCA, no gate	SCA, gated	Ratio
2048×4096	128×256	(2, 4)	2.17×10^{16}	5.97×10^{15}	5.12×10^{15}	4.2
4096×4096	256×256	(4, 4)	8.56×10^{16}	1.19×10^{16}	1.02×10^{16}	8.4
2048×6144	128×384	(2, 6)	4.84×10^{16}	8.93×10^{15}	7.66×10^{15}	6.3

Token grid	Pixels	Full (SDPA, ms)	SCA (ms)	Speedup
128×128	2048^2	7.1	5.2	1.4
256×256	4096^2	87.3	18.7	4.7
128×384	2048×6144	49.6	14.0	3.5

Table D.2: **Seconds per image**, batch 1, idle GH200, 28 steps with text encoding and VAE decoding, median of three timed images after an excluded warm-up; full attention not run at 8192^2.

Size (N tokens)	Full attention	AReA	SCA
2048^2 (16,384)	22.9	41.0	**18.2**
3072^2 (36,864)	73.9	114.3	**40.2**
4096^2 (65,536)	186.5	202.4	**71.1**
6144^2 (147,456)	762.7	482.3	**161.6**
8192^2 (262,144)	not run	952.0	**287.6**

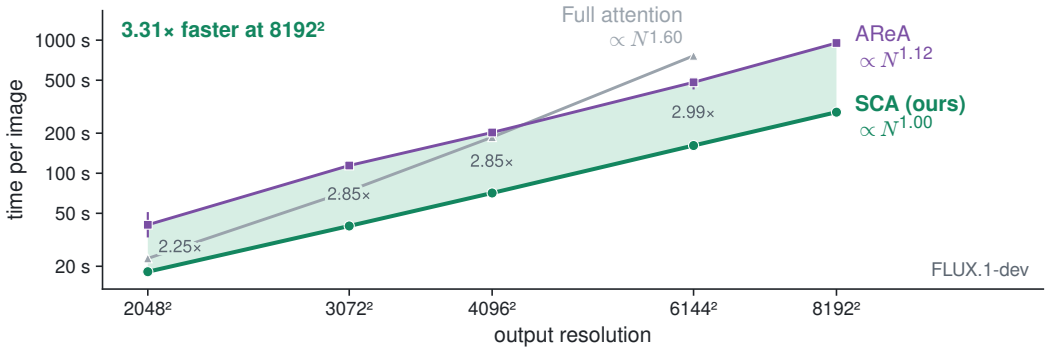

Figure D.2: **Latency at every size.** Figure 7 at full width: batch 1 on one GH200, median of three prompts, whiskers min to max; full attention was not run at 8192^2. Grey ratios: speedup over AReA per size; exponents fit seconds $\propto N^b$ (N image tokens).

campaign nodes, in the 191-prompt run that also timed the band guard, per-image times at 4096^2 were 71 s for SCA final, 120 s for SCA + band guard, and 262 s for AReA; the campaign code used an unfused band, so dropping it saved more there (49 s) than in the clean measurement (17 s). At the non-square sizes we have only campaign timings, with both methods on the same pool of shared nodes (mean over 195 prompts): SCA takes 38.1 s and AReA 112.4 s at 2048×4096, and 55.8 s and 150.5 s at 2048×6144.

E ALGORITHMS IN FULL

Algorithms 3 and 4 expand Algorithms 1 and 2 to every step the code performs, and Algorithm 5 gives progressive SCA (Section 2.5). Each step follows `sca/calib.py`, `sca/attn.py`, or `scripts/sample_progressive.py` in the code repository (link in the abstract), and each comment names the equation or setting that a line implements.

Algorithm 3: **Calibration**, in full

```
# Input: 32 prompts; FLUX.1-dev at 1024^2 (n = 64),
#   28 steps, guidance 4.5, full attention
# Output: pi[s_c, l, h, beta], centres[beta]
def bin(sigma): return min(int(14 * sigma), 13)
def on_step(sigma):           # once per step
  S[bin(sigma)] += sigma; C[bin(sigma)] += 1
def calibrate(q, k, v, l, sigma):  # every block
  O_full = attn(q, k, v)      # Eq. 1, all keys
  beta = bin(sigma)           # 14 bins, width 1/14
  for s_c in (2, 4):          # emulation, App. B
    f, w = s_c, n // s_c      # subsample, window
    t_c = round(t * w / n0)   # tile sides 4 and 2
    O_glob = global_branch(q, k, v, f)    # Eq. 2
    O_loc = local_branch(q, k, v, w, t_c) # Eq. 3
    dl, df = O_loc - O_glob, O_full - O_glob
    # Eq. 5: per head, over image rows, channels
    A[s_c, l, :, beta] += sum(dl * dl)
    B[s_c, l, :, beta] += sum(df * dl)
    E[s_c, l, :, beta] += sum(df * df)
    F[s_c, l, :, beta] += sum(O_full * O_full)
  return O_full               # sampler unchanged
sample(prompts, on_step, calibrate)
pop = C > 0                   # 13 of 14 bins
centres = where(pop, S / C, (arange(14)+0.5)/14)
pi = clip(B / A, 0, 1); pi[A == 0] = 1  # Prop. 3
pi = fill_empty_bins(pi, pop)  # nearest bin
J = E - 2 * pi * B + pi**2 * A # error of the mix
rel_err = J.sum() / F.sum()    # Section 3
```

Algorithm 4: **Sampling**, in full

```
# Input: q, k, v before RoPE (T text, then H x W
#   image), block l, noise level sigma
# Output: O for all T + HW queries; sca/attn.py
# Per image: s_H, s_W = H // n0, W // n0, and the
#   table entry s_c nearest max(s_H, s_W) in log
# Per step, all 57 x 24 heads, ends held:
pi = interp(sigma, centres, table[s_c])
kb = k_img[s_H//2::s_H, s_W//2::s_W]  # κ(b)
vb = v_img[s_H//2::s_H, s_W//2::s_W]
cp = p // (s_H, s_W)          # c(p_i); text at 0
Q = cat(rope(q_txt, 0), rope(q_img, cp))
K = cat(rope(k_txt, 0), rope(kb, b))
O_glob = sdpa(Q, K, cat(v_txt, vb))   # Eq. 2
if sigma > sigma_g:                   # 0.89, Eq. 4
  return O_glob               # opening phase
w = (min(n0, H), min(n0, W))  # window sides
a = clip(t*(p//t) + t//2 - w//2, 0, (H, W) - w)
keep(i, j) = j < T or a_i <= p_j < a_i + w  # Eq. 3
perm = tile_major(H, W)       # 8 x 16 query blocks
mask = block_mask(keep, perm) # 8 x 8 key blocks
O_loc = flex_attention(rope(q_img, p)[perm],
    rope(k, p), v, mask)[inverse(perm)]
O_img = pi[l] * O_loc + (1 - pi[l]) * O_glob[T:]
return cat(O_glob[:T], O_img) # fp32 mix, Eq. 4
```

Algorithm 5: **Progressive SCA**, HRDiT's schedule with SCA attention (Section 2.5)

```
# Input: prompt, seed; FLUX.1-dev in bf16; SCA tables at scales 2 and 4
# Output: 4096^2 image; scripts/sample_progressive.py
sig = shift(linspace(1, 1/30, 30), mu=1.15)     # one 30-step schedule serves all stages, as in HRDiT
x = noise(1024^2, seed)                         # stage 1: stock dense attention, native RoPE
for i in range(30):                             # Euler; ref keeps each step's x0 estimate
  v = flux(x, sig[i], g=3.5); ref[i] = x - sig[i] * v; x = x + (sig[i+1] - sig[i]) * v
for res, k, g, s_c, a in ((2048, 13, 4.5, 2, 1.0), (4096, 20, 6.0, 4, 0.25)):
  img = sharpen(bicubic(decode(x), res))        # 2 img - blur(img), blur width ~ U(0.1, 2.0)
  x = sig[k] * noise + (1 - sig[k]) * encode(img)   # re-enter at sigma 0.81, then 0.61
  use_sca(table[s_c], n0=64, t=8, sigma_g=0.89) # replaces NTK RoPE, attention scale, pruning, alignment
  N = 30 - k                                    # 17, then 10 steps; sig[k] < sigma_g: no opening phase
  for n, i in enumerate(range(k, 30)):
    v = flux(x, sig[i], g); x0 = x - sig[i] * v; r, ref[i] = ref[i], x0    # SCA in every block
    x0 = x0 + a * (N - n) / N * (lowpass(up(r)) - lowpass(x0))  # Butterworth, order 4, cut-off 0.2
    u = momentum((x - x0) / sig[i], 0.5 * (N - n) / N)          # HRDiT's velocity momentum
    x = x + (sig[i+1] - sig[i]) * u
return decode(x)
```

F HYPERPARAMETERS

Table F.1 expands the Models and hyperparameters paragraph of Section 4 and lists every setting of the main runs.

Table F.1: **All hyperparameters** of the main runs (code commit `2d11740`), produced by `scripts/sample.py --method sca --pi-table calib/k1_1024.npz --gate-sigma 0.89`. Appendix I gives the settings that differ for UltraImage, HRDiT, SigMa, and FLUX.2-dev.

Group	Setting	Value
Model	checkpoint, precision	FLUX.1-dev, bf16 weights and activations
	blocks, heads, head dim	19 double-stream + 38 single-stream, 24, 128
	native grid n_0, text tokens T	64 (1024^2 pixels, 16 pixels per token), 512
Sampler	steps, solver	28, flow-matching Euler
	guidance, shift	4.5 (distilled guidance), $\mu = 1.15$ at every size
	seed, decoding	seed 0 per prompt, shared across methods; tiled VAE decoding
Local branch	tile t, window w	8, 64 (clamped at borders), original positions
	kernel	FlexAttention; query blocks 8×16, key blocks 8×8, tile-major order
Global branch	subsampling	block centre, factor (s_H, s_W) = target grid / n_0
	positions	queries at $c(p)$, keys at block index, text at 0
Mixing	table	per (block, head, σ bin), $57 \times 24 \times 14$, scale-4 entry
	interpolation	linear in σ between the mean sampled noise levels of occupied bins, constant outside
	gate	$\sigma_g = 0.89$ (8 of 28 steps global only)
Calibration	prompts, resolution	32 at 1024^2 (16 more at 1280^2 for transfer)
	emulated scales	2 and 4, one pass; 14 bins of width $1/14$
	hardware	one GH200; complete calibration runtime was not retained
Band guard	(not in final method)	average-pool $p = 16$, bilinear upsampling

G Evaluation protocol and metrics

This section expands the Data and Metrics paragraphs of Section 4: Table G.1 summarizes the protocol, and the paragraphs below define each metric and interval.

Table G.1: **Evaluation protocol.** The FLUX.1-dev attention comparisons share sampler settings and prompt seeds. UltraImage and HRDiT use their own multi-stage protocols; FLUX.2-dev uses its own native-resolution schedule (Appendix I). Prompt counts include the four development prompts. Table F.1 lists the hyperparameters; the paragraphs below define the metrics.

Models	FLUX.1-dev (Black Forest Labs, 2024) (main results); FLUX.1-Krea-dev (Black Forest Labs & Krea AI, 2025) (195 prompts); FLUX.2-dev (Black Forest Labs, 2025) (50 prompts); bf16.
Sampler	28 Euler flow-matching steps, guidance 4.5, shift $\mu = 1.15$ at every size, seed label 0 per prompt, tiled VAE decoding; UltraImage, HRDiT, and FLUX.2-dev exceptions are detailed in Appendix I.
Sizes	4096^2, 2048×4096, 2048×6144 (195 prompts each); 6144^2 (100-prompt subset); 8192^2 (12 exploratory paired prompts, metrics and qualitative examples).
Prompts	Aesthetic-4K evaluation set with 4K references (Zhang et al., 2025a), 195 prompts, fixed 46-prompt ablation subset; HRGen-1K (Rajabi et al., 2026a), all 1000 prompts.
Baselines	Full attention; AReA, released code (Rajabi et al., 2026a); DyPE (Issachar et al., 2025), SEGA (Rajabi et al., 2026b), UltraImage (two-stage defaults; Zhao et al., 2025), HRDiT (released progressive protocol; Xue et al., 2026); SigMa (Zhao et al., 2026) on 195 prompts.
Metrics	ImageReward, HPSv2/2.1, PickScore, CLIP, aesthetic; FID, KID, IS, patch FID/KID; MUSIQ (full, 1024^2 crops), CLIP-IQA, ARNIQA; 95% paired bootstrap intervals over prompts.
Hardware	One NVIDIA GH200 per process; verified timing logs identify the GH200 120 GB model. Clean latency uses batch 1; campaign timings are marked separately.

Prompts and references. The Aesthetic-4K evaluation split (Zhang et al., 2025a) has 195 prompts with 4K reference images. Four of them (boats, Moai, woman and bear, underwater surfer) were used to choose the opening phase and window; they stay in the evaluation set for every method, which favours SCA slightly. The 195 reference rows hold 168 distinct images (27 duplicate rows, each with its own caption); distribution metrics keep the official rows and weights, as Diffusion-4K's statistics do. The 46-prompt ablation subset was fixed before any ablation was run. HRGen-

1K (Rajabi et al., 2026a) has 1000 prompts and no references; we use all of them. The 48 calibration prompts were written for this purpose and share no prompt with Aesthetic-4K (the largest word-set Jaccard overlap with any evaluation prompt is 0.26).

Implementations. torch-fidelity 0.3.0, pyiqa 0.1.16 (MUSIQ) and 0.1.15 (CLIP-IQA, ARNIQA), hpsv2 1.2.0, and image-reward 1.5.

Preference and alignment metrics. ImageReward v1.0 (Xu et al., 2023) (bicubic resize to 224, centre crop). HPSv2 and HPSv2.1 (Wu et al., 2023): cosine of normalized image and text features with the official validation transform (longest side 224, padded). PickScore (Kirstain et al., 2023): $\exp(\text{logit scale})$ times the cosine. CLIP score (Radford et al., 2021): $100\max(\cos, 0)$ with ViT-L/14. Aesthetic: the LAION improved aesthetic predictor (sac+logos+ava1, linear MSE) on the L2-normalized CLIP ViT-L/14 image embedding (Schuhmann, 2022).

No-reference quality. MUSIQ (Ke et al., 2021) (pyiqa defaults) on the full image at native resolution, and MUSIQ_c, its mean over four fixed random 1024^2 crops per image; crop positions depend only on the prompt id, so they are identical across methods. CLIP-IQA (Wang et al., 2022) and ARNIQA (Agnolucci et al., 2024) (pyiqa defaults) on the image downscaled with antialiased bicubic resampling to a long side of 1024 pixels, the scale both were trained on.

Distribution metrics and reference correction. FID (Heusel et al., 2017) and KID compare the generated images (195 on the main set, 195 on the panel) with all 195 official reference rows, using Inception-v3 features (torch-fidelity) after bilinear resizing to 299^2. The reference rows contain 168 distinct image contents; repeated rows retain their official weights. Patch FID and KID are our own definition: the same Inception features on four fixed random 1024^2 crops per generated image and per reference row (780 generated and 780 reference crops on the main set). AReA and SEGA report a patch FID without defining its crops, so patch FID is never compared across papers. An audit found that the original feature-bank assembly keyed reference crops by resolved image path, so repeated images reused the last row's crop features. We corrected the lookup to retain each row's intended cached crops. Whole-image FID/KID are unchanged; all patch point estimates reported here use the corrected assembly. Original CSVs remain archived for provenance. Inception score uses full images at 299^2, with 10 splits.

Intervals. Per-image intervals are 95% paired percentile bootstraps over prompt ids: original summaries use 1000 draws; separately recomputed comparisons use 2000 draws with seed 0. For the main SCA–AReA FID/KID and patch-FID/KID comparison, 1000 draws resample generated prompt ids jointly across methods and resample the 168 distinct reference-image identities, keeping all rows and crops from each sampled identity together. This retains the 195-row weighted point estimate while accounting for repeated reference images. The resulting FID difference is -7.74 $[-13.11, -2.61]$; patch FID is -3.79 $[-7.46, 1.29]$ and patch KID ($\times 10^3$) is -2.29 $[-4.60, 0.05]$. An earlier version of this analysis reported a significant patch-KID improvement; under the corrected evaluation its interval includes zero. The whole-image and patch FID/KID comparisons at both non-square sizes, against every added baseline, and for the sampling-budget and progressive runs also use reference-image clusters. Other whole-image FID/KID intervals are labelled historical reference-row bootstraps; they have not been recomputed with image-identity clusters. Historical patch intervals are not attached to corrected estimates and are omitted where recomputation is incomplete. FID's finite-sample bias can place point estimates outside percentile intervals. Intervals are unadjusted for multiple comparisons and measure prompt/reference sampling uncertainty, not variability over independent model training runs.

H Reproducing AReA

This section expands the AReA comparison of Section 4.1: our protocol for AReA and our runs beside its published numbers at 4096^2 (Table H.1) and on HRGen-1K (Table H.2). We run AReA's released code with its FLUX.1 defaults (block-centre pooling, first phase for 8 of 28 steps, halo width 8, bridge ratio 0.5), the FLUX.1-dev checkpoint, guidance 4.5, 28 steps, μ at its maximum, and the seeds shared with the other methods. AReA's published numbers match FLUX.1-Krea-dev, the default of its released code: rerunning that code on Krea-dev reproduces its published

Table H.1: **Published numbers at 4096^2 next to our runs.** Aesthetic-4K, 195 prompts: AReA's Table 1 (HPS and MUSIQ from its Table 6; "FLUX.1"), our runs with FLUX.1-Krea-dev, the checkpoint AReA's released code uses (28 steps, guidance 4.5), and our FLUX.1-dev protocol. Last block: HRGen-1K (all rows in Table H.2). HPS: HPSv2 $\times 100$; MUSIQ on the full image; FID at 299^2 against the 195 reference rows; patch FID is omitted because AReA does not state its crops. †Fewer than all prompts; not ranked. **Bold** / underline: best / second best among ranked rows (HRGen-1K: bold only); ours shaded. Key result: Our run of AReA's code on FLUX.1-Krea-dev reproduces its published ImageReward, Aesthetic, and PickScore within 0.1 and its CLIP, HPS, and MUSIQ within 0.28; against that run SCA is higher on HPS and MUSIQ, within 0.1 on ImageReward, Aesthetic, CLIP, and PickScore, and worse on FID; on HRGen-1K, where we did not rerun AReA on FLUX.1-Krea-dev, SCA on that checkpoint exceeds every published AReA score.

Method	n	Image-Reward↑	Aes-thetic↑	CLIP↑	Pick-Score↑	HPS↑	MUSIQ↑	FID↓
Published by AReA, single pass (FLUX.1)								
Full attention	195	−0.72	5.80	25.72	20.34	–	–	179.8
UltraImage	195	−0.14	6.11	26.77	20.85	–	–	169.4
DyPE	195	1.11	6.32	28.98	22.79	–	–	155.2
SigMa	195	1.01	6.16	28.36	22.47	–	–	156.8
SEGA	195	1.27	6.31	29.16	23.16	–	–	150.6
AReA	195	**1.32**	<u>6.37</u>	**29.71**	**23.31**	<u>29.69</u>	<u>45.2</u>	148.5
Our runs, FLUX.1-Krea-dev, same prompts and metrics								
AReA	195	<u>1.28</u>	6.32	<u>29.49</u>	<u>23.21</u>	29.41	45.0	**142.6**
DyPE	100†	0.96	6.41	28.73	22.50	28.39	42.7	171.6
SEGA	100†	1.13	6.31	29.19	23.01	29.15	46.0	164.2
SCA (ours)	195	1.23	**6.39**	29.41	23.20	**29.70**	**46.3**	<u>144.1</u>
Our runs, FLUX.1-dev, shared sampler								
AReA	195	0.92	6.27	27.69	22.36	28.62	39.4	145.8
SCA (ours)	195	1.23	6.44	29.14	23.09	29.75	45.4	138.1
HRGen-1K, 1000 prompts without references: AReA's Table 2 and our FLUX.1-Krea-dev run								
AReA, published (FLUX.1)	1000	0.60	5.87	29.89	21.44	27.50	–	–
SCA (ours), FLUX.1-Krea-dev	1000	**0.78**	**6.03**	**30.10**	**21.64**	**28.06**	43.7	–

ImageReward, aesthetic score, and PickScore within 0.1 and its CLIP, HPS, and MUSIQ within 0.3 (Table H.1), while on FLUX.1-dev the same code gives ImageReward 0.92 against the published 1.32. Our direct SCA–AReA comparison holds the FLUX.1-dev settings fixed. UltraImage and FLUX.2-dev use separate protocols (Appendix I). AReA's paper reports 151 s for AReA and 314 s for full attention at 4096^2 on an H100; the matched sweep measures 202.4 s and 186.5 s on one GH200 (medians of three images). We use commit `9e38ec2` of the released repository, which includes a cost cap on halo and bridge tokens and a jagged halo edge; neither appears in AReA's paper, which does not say whether its Table 1 used them. Recomputed from the code with 512 text tokens, the first-phase pooling factors are 4, 4, 4, 4, 2, 2, 2, 2 at 4096×4096 and 2048×4096, and 8, 8, 8, 4, 4, 4, 2, 2 at 2048×6144. The second-phase windows shrink over 20 steps from 180, 151, and 192 tokens respectively to 64. At 2048×6144 no window is cheaper than full attention once the text tokens are counted, so the first window falls back to 192 tokens and the cap drops the bridge and halo on the first three second-phase steps. Other differences from the release defaults are the FLUX.1-dev checkpoint instead of FLUX.1-Krea-dev, and one seed per prompt shared across methods instead of one seed for all prompts. On FLUX.1-Krea-dev, AReA's default checkpoint, we rerun AReA and recalibrate SCA (16 prompts at 1024^2, scale-4 table, other settings unchanged) with the same 28-step, guidance-4.5 sampler on all 195 prompts; DyPE and SEGA run the Krea transformer through the frozen baseline harness on the 100 prompts of rows 4 to 103. There SCA wins six metrics and loses patch FID and, at the edge of significance, ImageReward, at a third of AReA's time (Table K.1). Table H.2 expands the HRGen-1K block of Table H.1 to every row of AReA's Table 2.

Table H.2: **HRGen-1K: AReA's published Table 2 next to our runs.** All 1000 prompts at 4096^2, no reference images. Top: AReA's Table 2 ("FLUX.1", which its released code runs as FLUX.1-Krea-dev). Bottom: SCA with a Krea-calibrated table on FLUX.1-Krea-dev (28 steps, guidance 4.5), and our paired FLUX.1-dev runs of AReA and SCA. HPS: HPSv2 $\times 100$; IS: Inception score; MUSIQ on the full image; †fewer than 1000 prompts. **Bold** / underline: best / second best among the published rows and our FLUX.1-Krea-dev row. Table H.1 holds the Aesthetic-4K comparison. Key result: SCA on FLUX.1-Krea-dev exceeds AReA's published ImageReward, Aesthetic, CLIP, PickScore, HPS, and IS.

Method	n	Image-Reward↑	Aes-thetic↑	CLIP↑	Pick-Score↑	HPS↑	IS↑	MUSIQ↑
Published by AReA (FLUX.1)								
Full attention	1000	−1.46	5.12	23.36	19.04	23.20	7.2	–
UltraImage	1000	−0.03	5.77	28.26	20.46	26.90	10.2	–
DyPE	1000	0.33	5.83	29.35	20.94	26.60	10.9	–
SigMa	1000	0.06	5.63	27.91	20.54	25.80	10.7	–
SEGA	1000	0.44	5.82	29.63	21.18	27.00	12.0	–
AReA	1000	<u>0.60</u>	<u>5.87</u>	<u>29.89</u>	<u>21.44</u>	<u>27.50</u>	<u>13.9</u>	–
Our runs								
SCA (ours), FLUX.1-Krea-dev	1000	**0.78**	**6.03**	**30.10**	**21.64**	**28.06**	**15.7**	43.7
AReA, our run, FLUX.1-dev	1000	−0.03	5.92	26.08	20.43	26.16	12.7	37.4
SCA (ours), FLUX.1-dev	1000	0.79	6.25	28.73	21.61	28.11	15.3	45.1

I Other baselines and FLUX.2-dev

DyPE and SEGA. We use the existing baseline harness frozen at commit `c05ebdc`, with FLUX.1-dev, 28 steps, guidance 4.5, maximum shift, tiled VAE decoding, and the same prompt-derived seeds as SCA. These settings isolate the attention/position treatment on the checkpoint used in our main comparison. They do not reproduce the methods' reported FLUX.1-Krea-dev results; on Krea weights both trail AReA on ImageReward (Table K.1). Their low FLUX.1-dev scores should therefore not be read as a ranking under each method's preferred checkpoint.

UltraImage. We run a guided two-stage configuration from the FLUX branch of the released implementation through `scripts/run_ultraimage.py`. It uses a 1024^2 first stage and a 4096^2 second stage, 30 steps per stage, guidance 3.5 then 7, the `super_imax` guidance schedule, shifted-window patchification, proportional attention, RIFLEx positions, and attention decay. Each prompt also needs a 2+2-step block-average preparation pass. The reported campaign time includes preparation when not cached and the two-stage call; main-stage and preparation times are retained separately in sidecars. Each call reseeds both the generator and the global RNG, including the VAE sampling path, from the prompt id and seed label. Thus seed labels are paired, but the two-stage method does not share an identical latent trajectory with the one-stage methods. The wrapper preserves an upstream scheduler behaviour: assigning `scheduler.config.shift` does not update the scheduler's internal shift in the installed diffusers version, so both stages retain shift 3.0. This is a reproduction caveat, not a newly tuned setting. We report this system-level comparison separately from the common-sampler attention comparisons.

HRDiT. We run the released progressive protocol of the official repository (commit `f3be935`) on FLUX.1-dev in fp16: 30 steps at 1024^2 with guidance 3.5, then 17 steps at 2048^2 with guidance 4.5 and 10 steps at 4096^2 with guidance 6, 57 steps in all, with latent upsampling and the shipped per-head scope plan and attention settings. Its sampling budget therefore differs from the shared protocol, and we report it with the multi-stage recipes. HRDiT's global Torch random state is reset per prompt because its Gaussian blur draws a random width. Its clean time, 108.6 s, is the median of three batch-1 runs on the latency prompts (the first, 144.4 s, includes warm-up). The 195-prompt HRDiT evaluation used the original crop assembly, under which patch FID is 109.9 for HRDiT and 105.9 for SCA. Recomputed from the cached crop features with the corrected assembly of Table 1 (`repro/evaluate_corrected.py`), HRDiT's patch FID is 103.0, against 99.1 for SCA and 102.9 for AReA.

Progressive SCA. `scripts/sample_progressive.py` reimplements HRDiT's released schedule with the stock diffusers FLUX components and changes only the attention (Section 2.5).

One 30-step schedule with dynamic shift $\mu = 1.15$ (from the 1024^2 sequence length) serves all stages, and the recorded HRDiT noise levels are reproduced to 10^{-6}. The 2048^2 stage repeats steps 14 to 30 and the 4096^2 stage steps 21 to 30. Each transition decodes the latent, upsamples it bicubically, applies HRDiT's unsharp mask with a blur width drawn from the per-prompt global seed, re-encodes it, and re-noises it with the per-prompt generator; guidance is 3.5, 4.5, and 6.0, and HRDiT's low-band structure guidance and velocity momentum are kept. At 1024^2 both methods use dense attention with native RoPE. At 2048^2 and 4096^2 we remove NTK RoPE, the proportional scale, head pruning, and position alignment, and use SCA with the FLUX.1-dev table at scales 2 and 4, $n_0 = 64$, $t = 8$, and $\sigma_g = 0.89$, which no high-resolution step exceeds. Deliberate differences from HRDiT's runner are bf16 instead of fp16 and a finite-value check at the end of each stage instead of after every step; with bf16 and stock components the 1024^2 images, and hence the layouts, differ from HRDiT's for the same seed (Figure L.2). Reported time is CUDA-synchronized and covers text encoding, all stages, transitions, and the final decode, excluding model loading; per-stage means are 5.1, 12.3, and 26.1 s, peak memory 37.8 GiB. The gate and every other setting were fixed before generation; two development prompts served as a smoke test.

SigMa. The dev-matched profile runs the released code (commit `fbe3a86`) on FLUX.1-dev under the shared sampler: 28 steps, guidance 4.5, bf16, YaRN positions, and the shift capped at its maximum from the target sequence length. SigMa's paper states FLUX.1-dev with guidance 3.5, while its released command line hardcodes FLUX.1-Krea-dev with guidance 4.5; we ran only the profile that matches our protocol. The outputs are finite, fully decoded 4096^2 images, but the images we inspected show a regular lattice of dots, as DyPE's do at 4096^2 on FLUX.1-dev. This may reflect our port or profile rather than the method, so the SigMa row should not be read as the method's best result.

Adapter for HRDiT and SigMa. SigMa runs on 195 prompts and HRDiT on all 195, both through one adapter with campaign per-prompt CPU generators, resident models, and tiled VAE decoding. These differ from the upstream command-line seed, offload, and decode choices, so the runs are controlled adaptations, not reproductions of the authors' tables. Campaign time includes text encoding, decoding, and finite-value checks after every step, which add synchronization overhead; it excludes model loading and is not a clean latency benchmark.

FLUX.2-dev. The model has 56 attention blocks and 48 heads per block. We reuse the two branches and calibration rule with a native grid of 64^2 tokens, local window 64, and query tile 8. The four-axis positions preserve the time and text axes while coarsening the two spatial axes. Calibration uses 16 prompts at 1024^2, 14 noise bins, and emulated scales 2 and 4; sampling uses the scale-4 table. Both full attention and SCA use 28 steps, guidance 4.0, seed label 0, and a shift fixed to the native sequence length, $\mu = 2.1514$. The pipeline's default extrapolated shift at 4096^2 would be about 11.55; we therefore do not interpret this comparison as a default-pipeline benchmark. The gate is 0.959 so that the first 8 steps use only global attention. The 50 evaluation prompts are Aesthetic-4K rows 4–53, a 50-prompt subset. Prompt embeddings are cached for both methods; reported times include denoising and tiled VAE decoding, and exclude text encoding and model loading. At emulated scale 4, fitted per-head relative output error is 0.1790, versus 0.1889 for per-layer weights and 0.1924 for local-only attention; split-half correlation is 0.950 and cross-half error 0.1794. These are calibration diagnostics, not generation ablations of the individual components on FLUX.2.

J Extended results with intervals

This section expands the main tables to all metrics and sizes. Table J.1 expands Table 1, Table J.2 gives its paired intervals against AReA, Table J.3 covers the non-square sizes, and Table J.4 checks the calibration table there. Table J.5 expands Table 3; Tables J.6 and J.7 cover the band guard.

Appendix K pairs the added baselines against SCA. All these tables are populated from archived evaluation CSV files. Per-image means were checked against the per-image records; distribution-level estimates use cached features and the corrected crop assembly (Appendix G). Set-level metrics (FID, KID, their patch versions, Inception score) use the whole set. FID/KID and patch-FID/KID intervals on the 195-prompt sets resample reference-image clusters; other FID/KID intervals retain

Table J.1: **At 4096^2, SCA beats AReA on every per-image metric at about a third of its latency.** FLUX.1-dev, 195 prompts (100 at 6144^2). HPS, CLIP-IQA, and ARNIQA are $\times 100$; higher is better except latency and FID. Bold: best per size. * / †: significantly better / worse than AReA (95% paired interval; none for full attention's FID and FID_p). SCA + band guard: the 191-prompt run without the development prompts. Latency at 4096^2: batch-1 seconds per image on an idle GH200, median of three. ‡Campaign mean on shared nodes, not clean latency. $^\circ$UltraImage uses its two-stage defaults.

Method	Lat. (s)↓	Image Reward	Aesth.	CLIP	Pick Score	HPSv2	HPSv2.1	FID↓	FID_p↓	MUSIQ	MUSIQ_c	CLIP-IQA	ARNIQA
4096×4096, scale factors (4, 4)													
Full attention	186.5	$-2.18^\dagger$	$4.50^\dagger$	$15.37^\dagger$	$17.94^\dagger$	$20.75^\dagger$	$7.60^\dagger$	312.4	252.2	$22.9^\dagger$	$33.1^\dagger$	$49.0^\dagger$	$54.4^\dagger$
DyPE	$195.9^\ddagger$	-0.10	5.72	24.92	20.51	25.53	20.55	205.5	171.9	36.3	50.3	59.6	60.2
SEGA	$190.4^\ddagger$	0.56	6.02	26.95	21.64	27.13	24.34	164.9	125.8	37.3	50.3	66.7	64.8
UltraImage$^\circ$	$308.5^\ddagger$	**1.28**	**6.50**	28.84	**23.33**	**29.96**	**30.77**	142.0	109.5	33.6	41.7	72.8	**72.3**
AReA (released code)	202.4	0.92	6.27	27.69	22.36	28.62	27.16	145.8	102.9	39.4	50.4	67.7	67.4
SCA + band guard	88.1	1.20^*	6.46^*	28.87^*	22.90^*	29.40^*	28.95^*	142.3	**98.8**	41.4^*	61.0^*	70.9^*	70.8^*
SCA (final)	**71.1**	1.23^*	6.44^*	**29.14***	23.09^*	29.75^*	29.67^*	**138.1***	99.1	**45.4***	**61.2***	**73.0***	71.3^*
2048×4096, scale factors (2, 4)													
AReA (released code)	$112.4^\ddagger$	**1.13**	**6.41**	**28.59**	**22.66**	28.36	**27.64**	144.6	**111.7**	51.9	61.6	**74.9**	**73.0**
SCA (final)	**38.1**‡	**1.13**	$6.21^\dagger$	28.24	22.65	**28.54***	27.46	**143.4**	$117.0^\dagger$	**56.1***	**65.9***	$73.6^\dagger$	72.9
2048×6144, scale factors (2, 6)													
AReA (released code)	$150.5^\ddagger$	0.39	**6.14**	26.52	21.51	26.41	23.36	163.0	**105.7**	42.8	53.0	67.5	69.3
SCA (final)	**55.8**‡	**0.67***	$5.99^\dagger$	**27.07**	**21.74***	**26.53**	**23.46**	**143.3***	$109.3^\dagger$	**48.5***	**60.6***	**69.1***	**70.2***
6144×6144, scale factors (6, 6), 100 prompts													
AReA (released code)	$507.9^\ddagger$	-0.42	5.80	23.84	20.30	24.95	18.78	216.7	188.6	28.3	44.5	53.8	47.8
SCA (final)	**164.7**‡	**0.84***	**6.24***	**27.12***	**21.89***	**27.48***	**25.16***	**169.9***	**122.4***	**37.5***	**48.3***	**60.1***	**62.6***

the historical reference-row bootstrap unless a corrected image-cluster interval exists, and Inception-score intervals resample only generated images.

Table J.2: **All metrics at 4096^2**, 195 prompts, paired against AReA. Δ is method minus AReA with its 95% paired bootstrap interval; **blue** / *orange*: significantly better / worse than AReA. Intervals use 1000 prompt draws for SCA final (FID, KID, and patch versions also resample reference-image clusters) and 2000 for full attention's per-image metrics; full attention's set-level differences and campaign time are points without marks. SCA + band: the 191-prompt run without the development prompts, paired on those prompts; its corrected patch values have no intervals. ‡Campaign mean on shared nodes (clean latency: Table D.2). Sources: `results/a4k195_main/`, `final4k_*`.

Metric	AReA	SCA final	Δ [95% CI]	SCA + band	Δ [95% CI]	Full attention	Δ [95% CI]
ImageReward	0.921	1.228	**+0.308** [+0.227, +0.384]	1.198	**+0.281** [+0.208, +0.358]	−2.180	−3.101 [−3.218, −2.984]
Aesthetic	6.27	6.44	**+0.17** [+0.12, +0.22]	6.46	**+0.18** [+0.12, +0.24]	4.50	−1.77 [−1.87, −1.67]
CLIP score	27.69	29.14	**+1.45** [+1.12, +1.80]	28.87	**+1.20** [+0.79, +1.61]	15.37	−12.32 [−13.12, −11.54]
PickScore	22.36	23.09	**+0.74** [+0.61, +0.86]	22.90	**+0.54** [+0.40, +0.71]	17.94	−4.42 [−4.63, −4.22]
HPSv2 (×100)	28.62	29.75	**+1.13** [+0.95, +1.31]	29.40	**+0.75** [+0.53, +0.99]	20.75	−7.87 [−8.24, −7.50]
HPSv2.1 (×100)	27.16	29.67	**+2.51** [+2.12, +2.90]	28.95	**+1.75** [+1.31, +2.22]	7.60	−19.56 [−20.41, −18.72]
MUSIQ	39.4	45.4	**+6.0** [+4.2, +7.8]	41.4	**+2.0** [+0.0, +3.9]	22.9	−16.5 [−18.0, −14.9]
MUSIQ_c	50.4	61.2	**+10.8** [+9.4, +12.2]	61.0	**+10.6** [+9.2, +12.3]	33.1	−17.3 [−19.1, −15.5]
CLIP-IQA (×100)	67.7	73.0	**+5.3** [+3.9, +6.7]	70.9	**+3.3** [+1.8, +4.9]	49.0	−18.7 [−20.5, −16.9]
ARNIQA (×100)	67.4	71.3	**+3.9** [+3.1, +4.9]	70.8	**+3.5** [+2.4, +4.6]	54.4	−12.9 [−14.5, −11.3]
Inception score	6.66	6.62	−0.05 [−0.48, +0.62]	6.10	−0.33 [−0.88, +0.18]	3.82	−2.84
FID↓	145.8	138.1	**−7.7** [−13.1, −2.6]	142.3	−3.7 [−9.6, +1.0]	312.4	+166.6
KID↓ ($\times 10^3$)	7.17	6.19	−0.98 [−3.23, +1.47]	5.48	−1.63 [−3.84, +0.51]	164.15	+156.98
FID_p ↓	102.9	99.1	−3.8 [−7.5, +1.3]	98.8	−4.7	252.2	+149.3
KID_p ↓ ($\times 10^3$)	11.40	9.11	−2.29 [−4.60, +0.05]	9.67	−1.77	171.01	+159.61
Seconds per image‡ ↓	264.4	71.5	**−192.9** [−206.4, −180.3]	120.0	**−141.7** [−155.8, −127.1]	196.2	−68.1

Table J.3: **All metrics at the non-square sizes**, 195 prompts each, SCA final paired against AReA (released code). Conventions as in Table J.2: evaluator intervals over 1000 prompt draws, and for FID, KID, and their patch versions 1000 draws that also resample reference-image clusters. Sources: `results/a4k195_2048x4096/`, `a4k195_2048x6144/`.

	2048 × 4096, factors (2, 4)			2048 × 6144, factors (2, 6)		
Metric	AReA	SCA	Δ [95% CI]	AReA	SCA	Δ [95% CI]
ImageReward	1.132	1.126	−0.006 [−0.087, +0.062]	0.391	0.670	**+0.279** [+0.157, +0.399]
Aesthetic	6.41	6.21	−0.20 [−0.26, −0.14]	6.14	5.99	−0.15 [−0.26, −0.04]
CLIP score	28.59	28.24	−0.35 [−0.76, +0.01]	26.52	27.07	+0.55 [−0.01, +1.07]
PickScore	22.66	22.65	−0.02 [−0.12, +0.10]	21.51	21.74	**+0.23** [+0.05, +0.41]
HPSv2 (×100)	28.36	28.54	**+0.18** [+0.07, +0.29]	26.41	26.53	+0.12 [−0.01, +0.27]
HPSv2.1 (×100)	27.64	27.46	−0.18 [−0.47, +0.08]	23.36	23.46	+0.10 [−0.29, +0.53]
MUSIQ	51.9	56.1	**+4.2** [+2.7, +5.7]	42.8	48.5	**+5.7** [+3.8, +7.7]
MUSIQ_c	61.6	65.9	**+4.3** [+3.1, +5.4]	53.0	60.6	**+7.6** [+6.2, +9.2]
CLIP-IQA (×100)	74.9	73.6	−1.3 [−2.2, −0.4]	67.5	69.1	**+1.6** [+0.2, +3.1]
ARNIQA (×100)	73.0	72.9	−0.2 [−0.9, +0.5]	69.3	70.2	**+0.9** [+0.1, +1.7]
Inception score	6.27	6.67	+0.41 [−0.14, +0.97]	5.48	6.22	**+0.74** [+0.13, +1.23]
FID↓	144.6	143.4	−1.2 [−6.3, +4.0]	163.0	143.3	**−19.7** [−24.4, −11.4]
KID↓ ($\times 10^3$)	7.81	7.74	−0.07 [−2.73, +3.12]	17.92	8.36	**−9.57** [−13.66, −5.35]
FID_p ↓	111.7	117.0	+5.3 [+1.6, +9.4]	105.7	109.3	+3.6 [+0.2, +11.1]
KID_p ↓ ($\times 10^3$)	14.26	16.14	+1.88 [+0.39, +3.64]	13.56	15.14	+1.58 [−2.38, +6.01]
Seconds per image‡ ↓	112.4	38.1	**−74.3** [−76.3, −72.2]	150.5	55.8	**−94.7** [−97.1, −92.4]

Table J.4: **Calibration table at 2048 × 4096**, 46 ablation prompts, paired against SCA final with the scale-4 table; **blue** / *orange*: significantly better / worse than this reference. Sources: `results/ps2_*`.

Metric	SCA, scale-2 table	AReA (released code)
ImageReward	+0.012 [−0.025, +0.050]	+0.027 [−0.141, +0.192]
Aesthetic	−0.00 [−0.03, +0.02]	**+0.29** [+0.16, +0.43]
CLIP score	−0.02 [−0.22, +0.17]	+0.19 [−0.53, +0.91]
PickScore	**+0.05** [+0.00, +0.09]	+0.14 [−0.09, +0.39]
HPSv2 (×100)	**+0.09** [+0.04, +0.13]	−0.19 [−0.44, +0.06]
HPSv2.1 (×100)	**+0.17** [+0.08, +0.26]	+0.39 [−0.21, +1.01]
MUSIQ	+0.0 [−0.5, +0.5]	−3.1 [−7.5, +1.2]
MUSIQ_c	−0.0 [−0.4, +0.2]	−4.1 [−6.7, −1.3]
CLIP-IQA (×100)	−0.6 [−1.2, +0.0]	+1.3 [−0.8, +3.4]
ARNIQA (×100)	−0.2 [−0.6, +0.1]	+0.4 [−1.5, +2.8]

Table J.5: **Ablation intervals for Table 3**, fixed 46-prompt ablation subset at 4096^2, paired against SCA final (2000 bootstrap draws), whose column gives absolute values; **blue** / *orange*: significantly better / worse. Panels group the variants as in Table 3; each panel's last row gives patch FID (set-level, no paired interval). Sources: `results/ablnb_*`.

Weight rule after the opening phase				
Metric	SCA final	$\pi = 1$	shared $\pi(\sigma)$	per-layer
ImageReward	1.124	-0.357 $[-0.524, -0.207]$	-0.090 $[-0.173, -0.012]$	-0.054 $[-0.159, +0.030]$
Aesthetic	6.45	-0.21 $[-0.35, -0.07]$	-0.09 $[-0.18, -0.02]$	$+0.01$ $[-0.06, +0.07]$
CLIP score	29.41	-1.93 $[-2.68, -1.20]$	-0.38 $[-0.78, +0.08]$	-0.13 $[-0.54, +0.28]$
PickScore	22.96	-1.20 $[-1.44, -0.99]$	-0.38 $[-0.51, -0.26]$	-0.16 $[-0.28, -0.05]$
HPSv2 ($\times 100$)	29.71	-1.78 $[-2.07, -1.50]$	-0.96 $[-1.19, -0.73]$	-0.50 $[-0.63, -0.37]$
HPSv2.1 ($\times 100$)	29.98	-2.86 $[-3.35, -2.36]$	-1.83 $[-2.24, -1.43]$	-0.70 $[-0.98, -0.41]$
MUSIQ	46.1	-3.5 $[-6.5, -0.7]$	-4.9 $[-7.9, -1.9]$	$+0.7$ $[-1.6, +3.0]$
MUSIQ_c	62.5	-0.3 $[-2.5, +1.7]$	-10.4 $[-12.8, -8.3]$	-2.4 $[-3.6, -1.1]$
CLIP-IQA ($\times 100$)	71.3	-4.6 $[-6.3, -3.0]$	-7.7 $[-9.4, -6.0]$	-0.9 $[-2.0, +0.1]$
ARNIQA ($\times 100$)	70.5	-1.0 $[-2.0, +0.0]$	-6.5 $[-8.5, -4.8]$	-1.5 $[-2.3, -0.9]$
$\text{FID}_p \downarrow$	168.0	192.2	166.7	167.7

Global branch and opening phase				
Metric	SCA final	$\pi = 0$	mean pool	v0, no gate
ImageReward	1.124	-0.395 $[-0.507, -0.295]$	-1.518 $[-1.847, -1.193]$	-1.328 $[-1.595, -1.064]$
Aesthetic	6.45	-0.76 $[-0.87, -0.64]$	-0.61 $[-0.77, -0.44]$	-0.08 $[-0.17, +0.01]$
CLIP score	29.41	-3.48 $[-4.43, -2.59]$	-3.59 $[-4.76, -2.49]$	-1.65 $[-2.58, -0.84]$
PickScore	22.96	-1.44 $[-1.66, -1.22]$	-2.60 $[-3.00, -2.17]$	-2.18 $[-2.51, -1.85]$
HPSv2 ($\times 100$)	29.71	-3.32 $[-3.72, -2.94]$	-4.04 $[-4.66, -3.43]$	-1.89 $[-2.34, -1.45]$
HPSv2.1 ($\times 100$)	29.98	-5.66 $[-6.35, -4.98]$	-9.75 $[-11.30, -8.17]$	-4.07 $[-5.11, -3.04]$
MUSIQ	46.1	-3.5 $[-6.7, -0.2]$	-1.8 $[-5.3, +1.9]$	-5.6 $[-9.0, -2.1]$
MUSIQ_c	62.5	-11.7 $[-13.6, -9.8]$	-4.2 $[-8.4, -0.5]$	$+2.5$ $[-0.4, +5.5]$
CLIP-IQA ($\times 100$)	71.3	-27.0 $[-30.2, -23.8]$	-10.1 $[-13.0, -7.1]$	$\mathbf{+2.9}$ $[+1.0, +4.9]$
ARNIQA ($\times 100$)	70.5	-45.1 $[-47.6, -42.5]$	-4.8 $[-6.7, -2.6]$	$\mathbf{+4.3}$ $[+2.7, +6.1]$
$\text{FID}_p \downarrow$	168.0	237.0	184.6	202.6

Table J.6: **Intervals for the band-guard ablations** (Table O.1 and further variants), 46 ablation prompts at 4096^2, paired against SCA + band guard, whose absolute values form the first column; **blue** / *orange*: significantly better / worse than this reference. Corrected patch FID is shown without intervals. Sources: `results/abl_*`, `route_*`.

Metric	Reference	no band	no gate	shared $\pi(\sigma)$	per-layer	$\pi = 1$
ImageReward	1.165	−0.041 [−0.109, +0.021]	*−0.357* [−0.511, −0.222]	−0.032 [−0.084, +0.015]	−0.024 [−0.055, +0.004]	−0.015 [−0.075, +0.040]
Aesthetic	6.43	+0.01 [−0.05, +0.08]	*−0.23* [−0.35, −0.11]	−0.06 [−0.13, +0.00]	+0.03 [−0.01, +0.06]	+0.03 [−0.02, +0.10]
CLIP score	29.01	+0.40 [−0.04, +0.92]	*−1.15* [−1.82, −0.47]	−0.09 [−0.53, +0.31]	−0.03 [−0.28, +0.23]	+0.16 [−0.19, +0.53]
PickScore	22.71	**+0.25** [+0.09, +0.41]	*−0.86* [−1.13, −0.59]	+0.02 [−0.10, +0.14]	+0.01 [−0.06, +0.08]	**+0.12** [+0.03, +0.22]
HPSv2 (×100)	29.18	**+0.54** [+0.31, +0.75]	*−0.49* [−0.87, −0.14]	−0.20 [−0.42, +0.01]	−0.08 [−0.19, +0.04]	**+0.16** [+0.04, +0.29]
HPSv2.1 (×100)	28.81	**+1.17** [+0.71, +1.64]	*−1.39* [−2.18, −0.63]	−0.33 [−0.68, +0.01]	−0.01 [−0.19, +0.19]	**+0.55** [+0.30, +0.79]
MUSIQ	43.1	**+2.9** [+0.7, +5.7]	−0.7 [−3.6, +2.4]	+0.5 [−2.5, +4.0]	−0.3 [−1.9, +1.8]	+1.0 [−0.3, +2.6]
MUSIQ_c	62.1	+0.4 [−0.9, +1.6]	−1.0 [−3.4, +1.7]	*−5.7* [−7.2, −4.5]	*−1.5* [−2.4, −0.7]	**+0.9** [+0.3, +1.4]
CLIP-IQA (×100)	69.6	**+1.7** [+0.3, +3.2]	−0.6 [−2.7, +1.5]	*−6.5* [−8.4, −4.7]	*−1.2* [−2.1, −0.5]	+0.5 [−0.9, +1.9]
ARNIQA (×100)	69.6	+0.9 [−0.6, +3.1]	+1.6 [−0.4, +4.5]	*−4.6* [−7.1, −2.9]	*−1.2* [−2.0, −0.6]	+0.6 [−0.1, +1.4]
$\text{FID}_p \downarrow$	166.7	+1.3	+8.9	−4.6	−1.3	+1.6

Table J.7: **Band-guard ablations, continued.** Conventions as in Table J.6.

Metric	Reference	route 0.5	route 0.3	mean pool	$\pi = 0$	v0
ImageReward	1.165	*−0.051* [−0.095, −0.009]	−0.026 [−0.082, +0.024]	*−0.848* [−1.083, −0.615]	*−0.436* [−0.537, −0.336]	*−1.369* [−1.633, −1.113]
Aesthetic	6.43	−0.02 [−0.08, +0.04]	+0.02 [−0.03, +0.07]	*−0.39* [−0.54, −0.25]	*−0.75* [−0.86, −0.63]	−0.07 [−0.15, +0.03]
CLIP score	29.01	−0.00 [−0.31, +0.28]	+0.13 [−0.23, +0.51]	*−2.96* [−4.18, −1.67]	*−3.08* [−3.98, −2.16]	*−1.25* [−2.02, −0.40]
PickScore	22.71	+0.04 [−0.03, +0.12]	**+0.09** [+0.01, +0.17]	*−1.80* [−2.10, −1.49]	*−1.19* [−1.41, −0.96]	*−1.94* [−2.23, −1.65]
HPSv2 (×100)	29.18	+0.02 [−0.11, +0.15]	+0.11 [−0.01, +0.24]	*−2.35* [−2.81, −1.94]	*−2.79* [−3.18, −2.45]	*−1.35* [−1.74, −0.95]
HPSv2.1 (×100)	28.81	+0.18 [−0.04, +0.40]	**+0.34** [+0.12, +0.58]	*−5.81* [−6.96, −4.74]	*−4.48* [−5.17, −3.80]	*−2.90* [−3.82, −2.01]
MUSIQ	43.1	+1.2 [−0.2, +3.0]	+0.8 [−0.4, +2.5]	−1.8 [−5.5, +1.7]	−0.6 [−4.4, +3.4]	−2.7 [−6.1, +0.7]
MUSIQ_c	62.1	+0.5 [−0.0, +1.1]	**+0.8** [+0.2, +1.4]	−0.8 [−5.0, +3.6]	*−11.3* [−13.3, −9.6]	+2.9 [−0.3, +6.2]
CLIP-IQA (×100)	69.6	+0.3 [−1.3, +1.5]	+0.6 [−0.7, +1.7]	*−5.4* [−9.3, −1.6]	*−25.2* [−29.1, −21.4]	**+4.6** [+2.1, +7.3]
ARNIQA (×100)	69.6	+0.1 [−0.6, +0.8]	+0.6 [−0.1, +1.3]	−1.1 [−3.6, +1.9]	*−44.3* [−47.4, −40.7]	**+5.1** [+2.8, +8.1]
$\text{FID}_p \downarrow$	166.7	+1.5	+1.1	+18.7	+70.3	+35.9

K ADDITIONAL PAIRED INTERVALS

This section expands Table 4 and the baseline rows of Tables 1 and 2 with per-metric intervals. Tables K.2 and K.5 cover the added baselines, Table K.3 the prompt-set, seed, and model checks, Table K.4 the calibration-budget and gate checks, Table K.6 every baseline of Table 2, and Table K.1 the 6144^2 and FLUX.1-Krea-dev runs. Figure K.1 plots four of these metrics for five of the seven baselines run on all 195 prompts (HRDiT and SigMa are in Table K.6), signed as SCA minus baseline. Every difference in the tables is method minus the named reference. Positive values favour the method for the ten per-image metrics, and **blue** / *orange* mark intervals that exclude zero in favour of / against the method. These comparisons resample matched prompt ids, with 2000 draws and seed 0; they do not treat multiple crops as independent images. The HRGen-1K sample has 1000 prompts, the two extra seeds 195, the sensitivity experiments 46, FLUX.2-dev 50, and the main baseline comparison 195. Tables use the separately recomputed per-image CSVs; these intervals differ slightly from the original 1000-draw evaluator summaries. No multiplicity correction is applied.

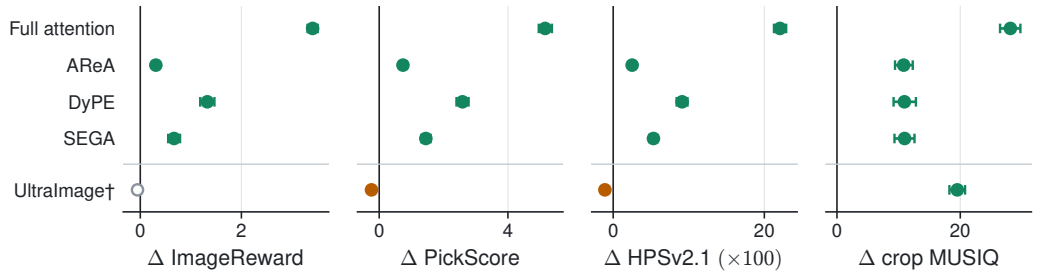

Figure K.1: **SCA beats every shared-protocol baseline on all four metrics; against UltraImage the result is mixed.** SCA minus baseline, mean over the same 195 prompts at 4096^2, whiskers 95% prompt-bootstrap intervals (2000 draws, one seed); open markers: interval contains zero; orange: significant loss. †Own two-stage recipe, not an equal-cost comparison.

6144^2 and FLUX.1-Krea-dev. Table K.1 gives every metric with its paired interval for the 100-prompt 6144^2 comparison and for FLUX.1-Krea-dev, where DyPE and SEGA also trail AReA on ImageReward. The Krea table follows the FLUX.1-dev calibration procedure with 16 prompts at 1024^2 (Appendix H).

Table K.1: **At 6144^2 SCA beats AReA on 14 of 15 metrics; on FLUX.1-Krea-dev it wins six metrics and loses ImageReward and patch FID, in a third of the time.** Method minus AReA with 95% paired intervals over 100 prompts at 6144^2 and for DyPE and SEGA, which ran only those, and all 195 for SCA on FLUX.1-Krea-dev; first column FLUX.1-dev at 6144^2, others FLUX.1-Krea-dev at 4096^2 with a Krea-calibrated SCA table. Per-image intervals use 1000 prompt draws; FID, KID, and their patch versions resample prompts and reference-image clusters. Lower is better for FID, KID, their patch versions, and time. **Blue** / *orange*: significantly better / worse than AReA. Time is the campaign mean on shared GH200 nodes.

	6144^2	FLUX.1-Krea-dev, 4096^2		
Metric	SCA	SCA	DyPE	SEGA
ImageReward	**+1.262** [+1.051, +1.450]	−0.052 [−0.103, −0.001]	−0.282 [−0.424, −0.173]	−0.110 [−0.189, −0.030]
Aesthetic	**+0.44** [+0.31, +0.55]	**+0.07** [+0.04, +0.10]	+0.05 [−0.02, +0.12]	−0.05 [−0.11, +0.02]
CLIP score	**+3.28** [+2.32, +4.18]	−0.08 [−0.33, +0.15]	−0.95 [−1.47, −0.49]	−0.49 [−0.91, −0.07]
PickScore	**+1.58** [+1.31, +1.88]	−0.01 [−0.10, +0.07]	−0.73 [−0.93, −0.55]	−0.23 [−0.36, −0.10]
HPSv2 (×100)	**+2.53** [+2.12, +2.94]	**+0.29** [+0.17, +0.42]	−1.02 [−1.24, −0.80]	−0.25 [−0.43, −0.09]
HPSv2.1 (×100)	**+6.37** [+5.39, +7.38]	**+0.63** [+0.37, +0.91]	−1.24 [−1.76, −0.76]	−0.25 [−0.61, +0.09]
MUSIQ	**+9.2** [+6.9, +11.7]	**+1.3** [+0.3, +2.3]	−2.6 [−4.6, −0.4]	+0.7 [−1.2, +2.8]
MUSIQ$_c$	**+3.8** [+1.4, +6.2]	**+4.2** [+3.4, +5.0]	**+3.2** [+1.8, +4.6]	+1.3 [−0.1, +2.6]
CLIP-IQA (×100)	**+6.3** [+4.3, +8.4]	+0.2 [−0.6, +1.2]	+0.9 [−0.7, +2.3]	**+1.8** [+0.4, +3.2]
ARNIQA (×100)	**+14.9** [+13.0, +17.0]	**+1.4** [+1.0, +1.8]	−1.6 [−2.6, −0.7]	−0.4 [−1.3, +0.4]
FID↓	**−46.8** [−58.6, −35.3]	+1.4 [−3.0, +7.0]	+5.0 [−2.8, +16.3]	−2.3 [−10.0, +7.4]
KID↓ (×10^3)	**−23.20** [−31.77, −16.84]	−0.12 [−2.15, +1.79]	−0.81 [−4.63, +3.26]	−0.85 [−4.00, +1.53]
FID$_p$ ↓	**−66.2** [−70.2, −48.9]	+8.1 [+5.5, +12.8]	+7.7 [+5.0, +18.5]	+0.1 [−3.1, +6.7]
KID$_p$ ↓ (×10^3)	**−56.88** [−68.46, −47.50]	+1.79 [−0.26, +4.00]	+1.32 [−1.44, +4.94]	−0.81 [−2.48, +0.90]
Inception score	+0.45 [−0.06, +0.84]	+0.32 [−0.31, +0.77]	+0.25 [−0.22, +0.69]	+0.09 [−0.43, +0.56]
Time (s)↓	**−343.1** [−349.3, −337.4]	**−165.8** [−177.9, −154.4]	**−24.3** [−28.9, −19.6]	**−20.1** [−25.0, −15.5]

Table K.2: **DyPE and SEGA trail SCA on every per-image metric; UltraImage and HRDiT are mixed.** Baseline minus SCA at 4096^2, 195 prompts; **blue** / *orange*: baseline significantly better / worse than SCA. UltraImage and HRDiT use their own multi-stage recipes; HRDiT's set-level metrics are in Table 1.

Metric	DyPE	SEGA	UltraImage	HRDiT
ImageReward	−1.326 [−1.466, −1.185]	−0.668 [−0.786, −0.551]	+0.055 [−0.011, +0.120]	**+0.114** [+0.049, +0.181]
Aesthetic	−0.72 [−0.81, −0.62]	−0.42 [−0.50, −0.35]	**+0.06** [+0.01, +0.10]	**+0.09** [+0.04, +0.13]
CLIP score	−4.22 [−4.85, −3.62]	−2.19 [−2.66, −1.76]	−0.31 [−0.61, −0.01]	−0.02 [−0.32, +0.27]
PickScore	−2.59 [−2.78, −2.40]	−1.45 [−1.60, −1.31]	**+0.24** [+0.14, +0.33]	**+0.37** [+0.27, +0.46]
HPSv2 (×100)	−4.22 [−4.53, −3.95]	−2.61 [−2.84, −2.38]	**+0.21** [+0.06, +0.37]	+0.13 [−0.02, +0.28]
HPSv2.1 (×100)	−9.12 [−9.83, −8.39]	−5.33 [−5.84, −4.78]	**+1.10** [+0.81, +1.39]	**+1.25** [+0.97, +1.53]
MUSIQ	−9.1 [−11.2, −7.0]	−8.1 [−9.9, −6.3]	−11.8 [−13.7, −9.8]	−5.1 [−6.7, −3.4]
MUSIQ_c	−10.9 [−12.8, −9.2]	−11.0 [−12.6, −9.3]	−19.5 [−20.8, −18.2]	−7.7 [−9.0, −6.4]
CLIP-IQA (×100)	−13.4 [−15.2, −11.7]	−6.3 [−8.0, −4.8]	−0.2 [−1.2, +0.9]	**+2.8** [+1.9, +3.8]
ARNIQA (×100)	−11.1 [−12.4, −9.7]	−6.5 [−7.6, −5.4]	**+1.0** [+0.3, +1.8]	+0.3 [−0.3, +0.9]

Table K.3: **The gains hold on HRGen-1K, two more seeds, and FLUX.2-dev.** SCA minus AReA for HRGen-1K and seeds 1 and 2; SCA minus full attention for FLUX.2-dev. All at 4096^2; **blue**: significantly better than the reference.

Metric	HRGen-1K	Seed 1	Seed 2	FLUX.2-dev
ImageReward	**+0.818** [+0.768, +0.870]	**+0.274** [+0.204, +0.349]	**+0.250** [+0.183, +0.320]	**+0.456** [+0.320, +0.598]
Aesthetic	**+0.33** [+0.30, +0.36]	**+0.14** [+0.08, +0.18]	**+0.12** [+0.06, +0.17]	**+0.30** [+0.18, +0.42]
CLIP score	**+2.65** [+2.40, +2.89]	**+1.33** [+1.00, +1.63]	**+1.45** [+1.15, +1.74]	+0.38 [−0.38, +1.12]
PickScore	**+1.19** [+1.13, +1.24]	**+0.72** [+0.60, +0.83]	**+0.72** [+0.59, +0.84]	**+0.56** [+0.29, +0.86]
HPSv2 (×100)	**+1.95** [+1.86, +2.05]	**+1.10** [+0.94, +1.28]	**+1.18** [+0.99, +1.37]	**+0.85** [+0.50, +1.21]
HPSv2.1 (×100)	**+4.93** [+4.70, +5.17]	**+2.28** [+1.94, +2.62]	**+2.50** [+2.10, +2.89]	**+1.84** [+1.10, +2.55]
MUSIQ	**+7.7** [+6.8, +8.5]	**+5.6** [+3.8, +7.4]	**+5.9** [+4.0, +7.6]	**+16.0** [+11.9, +19.9]
MUSIQ_c	**+13.9** [+13.2, +14.6]	**+11.5** [+10.2, +12.9]	**+12.1** [+10.5, +13.6]	**+21.4** [+18.2, +24.8]
CLIP-IQA (×100)	**+9.0** [+8.3, +9.8]	**+4.9** [+3.5, +6.3]	**+5.0** [+3.8, +6.2]	**+6.6** [+4.7, +8.6]
ARNIQA (×100)	**+7.2** [+6.7, +7.7]	**+4.0** [+3.1, +5.0]	**+4.0** [+3.1, +4.8]	**+7.1** [+5.4, +8.8]

Table K.4: **The calibration budget barely matters; a gate at $\sigma_g = 0.85$ lowers seven metrics.** Variant minus default SCA, 46 ablation prompts; **blue** / *orange*: variant significantly better / worse than default SCA. Budgets use the first 4, 8, or 16 of the 32 calibration prompts; the default gate is $\sigma_g = 0.89$.

Metric	4 prompts	8 prompts	16 prompts	$\sigma_g = 0.85$	$\sigma_g = 0.93$
ImageReward	+0.012 [−0.007, +0.029]	+0.010 [−0.004, +0.024]	−0.007 [−0.020, +0.005]	−0.015 [−0.098, +0.064]	−0.004 [−0.097, +0.069]
Aesthetic	+0.01 [−0.00, +0.02]	+0.01 [−0.00, +0.02]	+0.00 [−0.01, +0.01]	−0.19 [−0.26, −0.12]	+0.00 [−0.05, +0.06]
CLIP score	+0.08 [−0.03, +0.19]	+0.05 [−0.04, +0.15]	+0.04 [−0.04, +0.13]	−0.27 [−0.82, +0.26]	−0.64 [−1.03, −0.29]
PickScore	−0.01 [−0.03, +0.01]	+0.00 [−0.01, +0.02]	+0.01 [−0.01, +0.03]	−0.12 [−0.27, +0.02]	−0.22 [−0.38, −0.07]
HPSv2 (×100)	+0.02 [−0.02, +0.05]	+0.02 [−0.00, +0.05]	+0.00 [−0.02, +0.03]	−0.48 [−0.69, −0.28]	+0.02 [−0.16, +0.20]
HPSv2.1 (×100)	−0.02 [−0.07, +0.04]	+0.01 [−0.03, +0.05]	−0.01 [−0.05, +0.04]	−0.85 [−1.22, −0.48]	−0.15 [−0.56, +0.24]
MUSIQ	−0.1 [−0.4, +0.3]	−0.1 [−0.5, +0.2]	**+0.1** [+0.0, +0.3]	−1.9 [−3.5, −0.2]	+0.4 [−1.0, +2.0]
MUSIQ_c	+0.1 [−0.1, +0.3]	−0.0 [−0.2, +0.1]	−0.1 [−0.2, +0.0]	−2.0 [−3.0, −1.1]	+0.6 [−0.4, +1.9]
CLIP-IQA (×100)	+0.2 [−0.1, +0.4]	+0.2 [−0.1, +0.4]	−0.1 [−0.4, +0.1]	−3.2 [−4.8, −1.7]	+0.0 [−1.2, +1.1]
ARNIQA (×100)	+0.1 [−0.1, +0.3]	+0.0 [−0.1, +0.2]	−0.0 [−0.1, +0.1]	−4.0 [−5.9, −2.3]	**+1.8** [+0.7, +3.1]

Table K.5: **All three added baselines have significantly higher FID, patch FID, and patch KID than SCA.** Method minus SCA on 195 prompts (negative favours the method); *orange*: significantly worse than SCA. Patch estimates use corrected crops; all four intervals use 1000 draws that resample generated prompts and reference-image clusters.

Metric	DyPE	SEGA	UltraImage
FID↓	+67.4 [+56.2, +80.3]	+26.9 [+21.1, +39.1]	+4.0 [+1.1, +10.2]
KID↓ ($\times 10^3$)	+40.39 [+30.63, +52.87]	+4.70 [+0.14, +10.48]	+0.19 [−1.83, +2.18]
FID_p ↓	+72.8 [+58.7, +81.6]	+26.7 [+18.8, +35.3]	+10.4 [+6.2, +13.9]
KID_p ↓ ($\times 10^3$)	+49.34 [+40.09, +59.95]	+13.97 [+9.33, +20.00]	+7.31 [+5.10, +9.47]

195-prompt panel. Table K.6 pairs each baseline of Table 2 with SCA on the same 195 prompts and seeds. Per-image intervals use 2000 prompt draws with seed 0; FID, KID, and their patch versions use 1000 draws that resample generated prompts and reference-image clusters. Table L.1 gives the matching intervals for the 56-step and guidance-7 SCA runs.

Table K.6: **On the 195-prompt panel, every shared-protocol baseline trails SCA on all ten per-image metrics, while HRDiT and UltraImage lead on several preference metrics.** Baseline minus SCA (28 steps) at 4096^2 with 95% paired intervals; HPSv2, HPSv2.1, CLIP-IQA, and ARNIQA are $\times 100$, KID and KID_p $\times 10^3$. For FID, KID, their patch versions, and campaign time (seconds per image), lower is better, so negative favours the baseline. **Blue** / *orange*: baseline significantly better / worse than SCA. SigMa uses its dev-matched profile; UltraImage and HRDiT use their own recipes (Appendix I).

	Single pass, shared sampler				
Metric	Full attn.	DyPE	SigMa	SEGA	AReA
ImageReward	−3.409 [−3.509, −3.308]	−1.326 [−1.466, −1.185]	−1.751 [−1.898, −1.605]	−0.668 [−0.786, −0.551]	−0.308 [−0.381, −0.234]
Aesthetic	−1.94 [−2.03, −1.85]	−0.72 [−0.81, −0.62]	−0.84 [−0.94, −0.75]	−0.42 [−0.50, −0.35]	−0.17 [−0.22, −0.12]
CLIP score	−13.77 [−14.59, −12.98]	−4.22 [−4.85, −3.62]	−5.63 [−6.31, −4.95]	−2.19 [−2.66, −1.76]	−1.45 [−1.81, −1.11]
PickScore	−5.15 [−5.37, −4.96]	−2.59 [−2.78, −2.40]	−3.08 [−3.29, −2.87]	−1.45 [−1.60, −1.31]	−0.74 [−0.86, −0.63]
HPSv2	−9.00 [−9.33, −8.66]	−4.22 [−4.53, −3.95]	−4.97 [−5.29, −4.69]	−2.61 [−2.84, −2.38]	−1.13 [−1.31, −0.95]
HPSv2.1	−22.07 [−22.89, −21.27]	−9.12 [−9.83, −8.39]	−11.02 [−11.78, −10.29]	−5.33 [−5.84, −4.78]	−2.51 [−2.87, −2.13]
MUSIQ	−22.5 [−24.2, −20.8]	−9.1 [−11.2, −7.0]	−9.9 [−12.0, −7.8]	−8.1 [−9.9, −6.3]	−6.0 [−7.8, −4.4]
MUSIQ_c	−28.1 [−29.8, −26.5]	−10.9 [−12.8, −9.2]	−11.4 [−13.2, −9.6]	−11.0 [−12.6, −9.3]	−10.8 [−12.3, −9.4]
CLIP-IQA	−24.0 [−25.5, −22.5]	−13.4 [−15.2, −11.7]	−15.0 [−16.9, −13.1]	−6.3 [−8.0, −4.8]	−5.3 [−6.7, −3.9]
ARNIQA	−16.8 [−18.3, −15.4]	−11.1 [−12.4, −9.7]	−11.4 [−12.8, −10.0]	−6.5 [−7.6, −5.4]	−3.9 [−4.9, −3.0]
FID↓	+174.4 [+150.2, +179.4]	+67.4 [+56.2, +80.3]	+78.9 [+66.7, +91.0]	+26.9 [+21.1, +39.1]	+7.7 [+2.6, +13.1]
KID↓	+158.0 [+140.3, +176.0]	+40.4 [+30.6, +52.9]	+50.4 [+39.1, +64.0]	+4.7 [+0.1, +10.5]	+1.0 [−1.5, +3.2]
FID_p ↓	+153.1 [+122.3, +149.6]	+72.8 [+58.7, +81.6]	+85.5 [+70.2, +93.7]	+26.7 [+18.8, +35.3]	+3.8 [−1.3, +7.5]
KID_p ↓	+161.9 [+146.6, +178.2]	+49.3 [+40.1, +60.0]	+62.4 [+52.0, +74.7]	+14.0 [+9.3, +20.0]	+2.3 [−0.1, +4.6]
Time (s)↓	+124.7 [+120.8, +129.4]	+124.4 [+120.5, +128.6]	+121.0 [+119.0, +123.0]	+118.9 [+118.0, +119.7]	+192.9 [+180.3, +206.4]

	Multi-stage, own recipe	
Metric	UltraImage	HRDiT
ImageReward	+0.055 [−0.011, +0.120]	**+0.114** [+0.049, +0.181]
Aesthetic	**+0.06** [+0.01, +0.10]	**+0.09** [+0.04, +0.13]
CLIP score	−0.31 [−0.61, −0.01]	−0.02 [−0.32, +0.27]
PickScore	**+0.24** [+0.14, +0.33]	**+0.37** [+0.27, +0.46]
HPSv2	**+0.21** [+0.06, +0.37]	+0.13 [−0.02, +0.28]
HPSv2.1	**+1.10** [+0.81, +1.39]	**+1.25** [+0.97, +1.53]
MUSIQ	−11.8 [−13.7, −9.8]	−5.1 [−6.7, −3.4]
MUSIQ_c	−19.5 [−20.8, −18.2]	−7.7 [−9.0, −6.4]
CLIP-IQA	−0.2 [−1.2, +0.9]	**+2.8** [+1.9, +3.8]
ARNIQA	**+1.0** [+0.3, +1.8]	+0.3 [−0.3, +0.9]
FID↓	+4.0 [+1.1, +10.2]	+0.4 [−2.3, +6.4]
KID↓	+0.2 [−1.8, +2.2]	−1.6 [−3.6, +0.3]
FID_p ↓	+10.4 [+6.2, +13.9]	+3.9 [+0.7, +8.0]
KID_p ↓	+7.3 [+5.1, +9.5]	+2.8 [+0.9, +4.8]
Time (s)↓	+237.0 [+233.5, +241.7]	+38.8 [+37.2, +40.5]

L SAMPLING-BUDGET, GUIDANCE, AND SCHEDULE CHECKS

This section expands the budget and schedule rows of Table 2: Figure L.1 plots quality against time, Table L.1 gives the budget and guidance checks, Table L.2 the progressive schedule against HRDiT, and Figure L.2 two random prompts under each schedule.

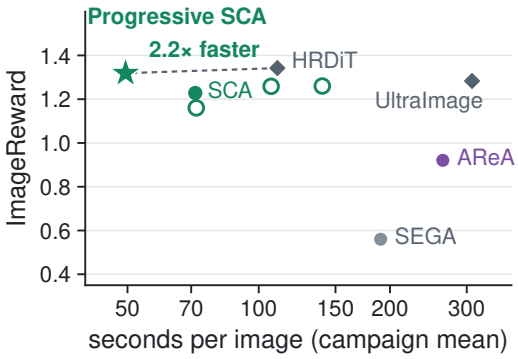

Figure L.1: **Quality against time.** 195 prompts, 4096^2. Open circles: SCA at guidance 7, 42 and 56 steps.

On the same 195 evaluation prompts, we compare the default 28-step SCA run with 56 steps at guidance 4.5 and with 28 steps at guidance 7 (Table L.1). The 56-step run keeps the noise threshold at 0.89; it therefore changes the discretization of both the opening and mixed phases. It improves ImageReward by 0.032 $[0.0002, 0.062]$, HPSv2.1 by 0.88, PickScore by 0.13, and MUSIQ$_c$ by 2.90, but increases corrected patch FID by 3.85 $[1.51, 6.33]$ with reference-image cluster resampling. Increasing guidance to 7 alone lowers seven per-image metrics, ImageReward by 0.068 $[0.025, 0.115]$ among them, and raises only MUSIQ. Thus the default schedule does not maximize every metric, and extra sampling can change the preference-quality balance.

Against UltraImage on these 195 prompts, 56-step SCA has ImageReward difference -0.023 $[-0.089, 0.040]$, HPSv2.1 difference -0.22 $[-0.51, 0.05]$, and PickScore difference -0.11 $[-0.21, -0.01]$. The first two intervals span zero and do not establish equivalence; PickScore favours UltraImage. Its MUSIQ advantage is 11.61 $[9.54, 13.60]$ and its MUSIQ$_c$ advantage 22.42 $[21.20, 23.59]$. Campaign means are 71.5 s for default SCA, 139.9 s for 56-step SCA, and 308.5 s for UltraImage on these 195 prompts. This is a sampling-budget sensitivity check, not an equal-FLOP or equal-wall-clock experiment: UltraImage has two resolutions, different guidance, and per-prompt preparation. Against HRDiT on the same prompts (Table 2), the 56-step run stays lower on ImageReward (1.26 against 1.34), HPSv2.1 (30.55 against 30.92), and PickScore (23.22 against 23.46) while taking longer (139.9 against 110.4 s); these are point estimates without paired intervals. Table L.2 adds a 42-step run sized to HRDiT's campaign time (106.8 against 110.4 s) and progressive SCA (Section 2.5). The extra steps raise PickScore, HPSv2, HPSv2.1, ARNIQA, and MUSIQ$_c$ over 28 steps but leave ImageReward unresolved, and HRDiT stays significantly ahead of the 42-step run on ImageReward, PickScore, HPSv2.1, aesthetic score, and CLIP-IQA while trailing it on both MUSIQ scores and ARNIQA. Progressive SCA is significantly worse than HRDiT on no per-image metric and better on HPSv2, MUSIQ, and ARNIQA; against single-pass SCA it gains ImageReward, PickScore, aesthetic score, both HPS scores, CLIP-IQA, and ARNIQA, and loses both MUSIQ scores. Figure L.1 places every setting of the panel by ImageReward and time, and Figure L.2 shows HRDiT, progressive SCA, and single-pass SCA on two random prompts.

Table L.1: **Doubling the steps raises eight metrics; guidance 7 lowers seven and raises MUSIQ.** On 195 prompts, the first two columns compare with default SCA; the last compares 56-step SCA with UltraImage. Per-image differences (first-named setting minus second) and 95% paired intervals, 2000 draws; **blue** / *orange*: the interval excludes zero in favour of / against the first-named setting.

Metric	56 vs. 28 steps	Guidance 7 vs. 4.5	56 steps vs. UltraImage
ImageReward	**+0.032** [+0.000, +0.062]	−0.068 [−0.115, −0.025]	−0.023 [−0.089, +0.040]
Aesthetic	**+0.03** [+0.01, +0.06]	−0.10 [−0.14, −0.07]	−0.03 [−0.07, +0.02]
CLIP score	−0.01 [−0.23, +0.22]	−0.21 [−0.45, +0.03]	+0.30 [−0.01, +0.62]
PickScore	**+0.13** [+0.07, +0.18]	−0.27 [−0.34, −0.20]	−0.11 [−0.21, −0.01]
HPSv2 (×100)	**+0.32** [+0.23, +0.40]	−0.25 [−0.35, −0.16]	+0.11 [−0.03, +0.26]
HPSv2.1 (×100)	**+0.88** [+0.71, +1.04]	−0.60 [−0.80, −0.41]	−0.22 [−0.51, +0.05]
MUSIQ	−0.2 [−1.5, +1.0]	**+1.4** [+0.3, +2.6]	**+11.6** [+9.5, +13.6]
MUSIQ$_c$	**+2.9** [+2.2, +3.6]	+0.2 [−0.4, +0.8]	**+22.4** [+21.2, +23.6]
CLIP-IQA (×100)	**+0.9** [+0.4, +1.4]	−1.0 [−1.7, −0.4]	**+1.1** [+0.0, +2.1]
ARNIQA (×100)	**+1.5** [+1.0, +1.9]	−1.2 [−1.5, −0.8]	+0.4 [−0.4, +1.2]

Distribution metrics of the progressive and 42-step runs. Against single-pass SCA at 28 steps (195 generated images, 168 reference identities, 1000 bootstrap draws; `results/`

Figure L.2: **Progressive SCA keeps one large subject, as HRDiT does; single-pass SCA can shrink or repeat it.** Two prompts drawn at random from the original 46-prompt panel; same 1:1 512×512 crop per row; times in Table 2.

`a4k195_panel/corrected_fidp_ci/paired_deltas.csv`), progressive SCA has significantly higher patch FID ($+7.2$ $[3.2, 11.0]$) and patch KID ($+0.005$ $[0.003, 0.007]$), while its FID ($+3.0$ $[-0.2, 8.3]$) and KID differences span zero. For 42-step SCA only FID is significantly higher ($+2.4$ $[0.1, 5.7]$); KID, patch FID ($+1.5$ $[-0.5, 3.6]$), and patch KID span zero.

Table L.2: **Against HRDiT, progressive SCA is not significantly worse on any per-image metric; single-pass SCA with 42 steps is.** 4096^2, 195 prompts, seed label 0. Per-image differences (first-named setting minus second) with 95% paired intervals over 2000 prompt draws; **blue** / *orange*: the interval excludes zero in favour of / against the first-named setting. Progressive: HRDiT's schedule with SCA attention (Section 2.5); 42 steps: single-pass SCA sized to HRDiT's campaign time. Intervals that include zero do not establish equivalence.

Metric	Progressive vs. HRDiT	Progressive vs. SCA	42 steps vs. SCA	HRDiT vs. 42 steps
ImageReward	−0.023 [−0.089, +0.034]	**+0.091** [+0.029, +0.156]	+0.030 [−0.001, +0.059]	**+0.083** [+0.023, +0.150]
Aesthetic	−0.028 [−0.069, +0.014]	**+0.063** [+0.017, +0.106]	+0.020 [−0.007, +0.047]	**+0.071** [+0.022, +0.119]
CLIP score	+0.24 [−0.03, +0.49]	+0.21 [−0.07, +0.49]	−0.01 [−0.19, +0.20]	−0.02 [−0.31, +0.27]
PickScore	−0.04 [−0.13, +0.05]	**+0.32** [+0.23, +0.41]	**+0.08** [+0.03, +0.13]	**+0.29** [+0.19, +0.39]
HPSv2 (×100)	**+0.39** [+0.25, +0.52]	**+0.52** [+0.36, +0.67]	**+0.23** [+0.15, +0.31]	−0.10 [−0.26, +0.07]
HPSv2.1 (×100)	+0.08 [−0.17, +0.32]	**+1.33** [+1.02, +1.62]	**+0.60** [+0.44, +0.75]	**+0.65** [+0.38, +0.93]
MUSIQ	**+1.3** [+0.2, +2.4]	*−3.7* [−5.5, −1.9]	−0.4 [−1.5, +0.7]	*−4.7* [−6.3, −2.9]
MUSIQ_c	+0.6 [−0.6, +1.8]	*−7.0* [−8.3, −5.7]	**+1.9** [+1.3, +2.5]	*−9.6* [−10.9, −8.4]
CLIP-IQA (×100)	−0.6 [−1.4, +0.2]	**+2.2** [+1.2, +3.2]	+0.4 [−0.1, +0.8]	**+2.4** [+1.5, +3.3]
ARNIQA (×100)	**+2.9** [+2.4, +3.5]	**+3.3** [+2.7, +3.8]	**+1.1** [+0.7, +1.4]	*−0.7* [−1.4, −0.1]

M EXPLORATORY 8192^2 COMPARISON

Table M.1 reports all 12 completed prompt pairs at 8192^2 under the shared FLUX.1-dev protocol. SCA has higher ImageReward, CLIP, PickScore, both HPS variants, and ARNIQA, with paired intervals excluding zero. Aesthetic score and the three remaining quality metrics have intervals spanning zero. In particular, MUSIQ, MUSIQ_c, and CLIP-IQA have point estimates favouring AReA. These mixed quality scores and the grid and stripe failures in Figure R.11 limit the evidence for reliable 8K generation. With only 12 prompts, this comparison is exploratory; we do not report distributional metrics for this set.

All 8K scores were computed on CPU. HPS uses float32 here, whereas the main GPU evaluation uses mixed precision, so absolute scores across resolutions should not be interpreted as a controlled scaling comparison. Both methods within this table use the same evaluator settings. For native-resolution MUSIQ, attention queries are processed in chunks of 256 while retaining every key and native image patch. This bounds intermediate memory without changing the attention context; pre-trained equivalence checks at three smaller image sizes reproduced the unchunked scores exactly.

Table M.1: **Exploratory 8192^2 comparison: SCA is significantly better than AReA on six of ten metrics.** FLUX.1-dev, 12 matched prompts, seed 0. All metrics are higher-is-better. Intervals are paired prompt-bootstrap 95% percentile intervals (2000 draws, bootstrap seed 0); **blue**: significantly better than AReA.

Metric	AReA	SCA	SCA − AReA [95% CI]
ImageReward	−0.376	0.675	**+1.051** [+0.386, +1.754]
Aesthetic	5.64	5.72	+0.08 [−0.26, +0.44]
CLIP score	22.38	25.52	**+3.14** [+0.63, +6.11]
PickScore	20.35	21.27	**+0.92** [+0.24, +1.75]
HPSv2 (×100)	24.60	26.13	**+1.53** [+0.56, +2.66]
HPSv2.1 (×100)	16.59	20.51	**+3.92** [+1.31, +6.85]
MUSIQ	28.6	25.7	−2.9 [−6.8, +0.8]
MUSIQ_c	44.0	40.1	−3.8 [−10.4, +1.6]
CLIP-IQA (×100)	57.1	52.0	−5.1 [−10.5, +0.6]
ARNIQA (×100)	37.3	54.6	**+17.3** [+11.9, +22.6]

N CALIBRATION TABLE AT NON-SQUARE SIZES

At $(s_H, s_W) = (2, 4)$ the two axes have different factors, and SCA uses the table of the emulated scale nearest the larger factor, scale 4. On the 46 ablation prompts at 2048×4096, the scale-2 table changes the paired metrics negligibly (Table J.4): HPSv2 +0.09, HPSv2.1 +0.17 (both ×100), and PickScore +0.05 are significant but small, and ImageReward (+0.01 $[-0.03, 0.05]$) and the other metrics are within noise. These small changes do not resolve the mixed SCA–AReA results at this size, and we keep the default rule.

O THE LOW-BAND DUPLICATION GUARD

Construction. Let LP_p average-pool a head's image output over $p \times p$ tokens and upsample it bilinearly, and $\text{HP}_p = \text{I} - \text{LP}_p$. The guard replaces the low band of the mix M by that of the global branch:

$$O \;=\; \text{LP}_p\big(O^{\text{glob}}\big) + \text{HP}_p\big(M\big) \;=\; O^{\text{glob}} + \pi_h(\sigma)\,\text{HP}_p\big(O^{\text{loc}} - O^{\text{glob}}\big). \tag{6}$$

The local branch then contributes only its difference from the global branch above the cutoff. The cutoff should lie below $1/(2s)$, where the global branch aliases, and above $1/w$, below which the window cannot resolve; with $w = 64$ and $s = 4$ this leaves periods of 8 to 64 tokens. We used $p = 16$; $p = 8$ and $p = 32$ were worse on four development prompts, and a noise-dependent cutoff behaved like $p = 16$. FreeScale (Qiu et al., 2024) also fuses two self-attention outputs by frequency band, in the opposite direction: it keeps the blurred low band of attention over overlapping crops and takes the high band from full self-attention at the target size. ResDiT also combines global low frequencies and local high frequencies, using patch-wise FFT masks (Ma et al., 2025). Our optional guard uses pooled global attention and preserves the fitted per-head weight on the local-minus-global high-band residual. Its global-low/local-high construction is therefore not itself novel.

Table O.1: **Ablations around the band-guard configuration at 4096^2**, 46 ablation prompts. The reference row gives absolute values for SCA + band guard (band 16); every other row changes one thing, keeps the guard unless stated, and reports its paired difference to the reference. Marks and units as in Table 3; corrected patch FID is shown without intervals.

Variant	Mix err.	Image Reward	Aesth.	HPSv2	HPSv2.1	Pick Score	CLIP	CLIP-IQA	ARNIQA	MUSIQ	MUSIQ_c	FID_p ↓
SCA + band guard (reference)	0.200	1.17	6.43	29.18	28.81	22.71	29.01	69.6	69.6	43.1	62.1	166.7
no band (**= SCA final**)	0.200	−0.04	+0.01	**+0.54**	**+1.17**	**+0.25**	+0.40	**+1.7**	+0.9	**+2.9**	+0.4	+1.3
no gate	0.200	−0.36	−0.23	−0.49	−1.39	−0.86	−1.15	−0.6	+1.6	−0.7	−1.0	+8.9
one shared $\pi(\sigma)$	0.230	−0.03	−0.06	−0.20	−0.33	+0.02	−0.09	−6.5	−4.6	+0.5	−5.7	−4.6
per-layer $\pi_\ell(\sigma)$	0.223	−0.02	+0.03	−0.08	−0.01	+0.01	−0.03	−1.2	−1.2	−0.3	−1.5	−1.3
local only ($\pi = 1$)	0.261	−0.01	+0.03	**+0.16**	**+0.55**	**+0.12**	+0.16	+0.5	+0.6	+1.0	**+0.9**	+1.6
hard routing, threshold 0.5	n/a	−0.05	−0.02	+0.02	+0.18	+0.04	−0.00	+0.3	+0.1	+1.2	+0.5	+1.5
hard routing, threshold 0.3	n/a	−0.03	+0.02	+0.11	**+0.34**	**+0.09**	+0.13	+0.6	+0.6	+0.8	**+0.8**	+1.1
mean pooling	n/a	−0.85	−0.39	−2.35	−5.81	−1.80	−2.96	−5.4	−1.1	−1.8	−0.8	+18.7

History: what it fixes and what it costs. On the development prompts, the band alone (no opening phase) fixed the boats but left extra statues and faces on the Moai and woman-and-bear prompts;

the opening phase alone got the main subject right, but small extra figures appeared late on some prompts; both together gave one subject on the boats, Moai, and woman-and-bear prompts. This was our first full configuration. On the 46 ablation prompts, however, removing the guard improves HPSv2 ($+0.54$), HPSv2.1 ($+1.17$), PickScore ($+0.25$), CLIP-IQA ($+1.7$), and MUSIQ ($+2.9$), all significant, leaves the other metrics within noise, and saves 17 s per image on an idle GPU. The guard also hid what the per-head mix contributes. With it (Table O.1), local only matched the per-head mix within 2% on preference metrics, because the guard replaces the low band of every head's output with that of the global branch whatever the head's weight, so it injects global low frequencies even at $\pi = 1$; without the guard local only loses on nearly every metric (Section 4.3). Likewise, with the guard one shared weight and per-layer weights matched the per-head table on preference metrics and lost only on no-reference quality, and hard per-head routing (10 to 17% of heads global at threshold 0.5, about 6% at 0.3) stayed within 2% of the mix. The guard is an option for prompts where late duplication is visible, not part of the default method.

P Compute and reproducibility

Compute. Generation and calibration used single NVIDIA GH200 GPUs on a shared cluster; the verified timing logs identify the GH200 120 GB model. Some metric evaluations ran on CPUs. The surviving logs do not establish the complete calibration runtime. Generation at 4096^2 takes about 72 s per image for SCA, 264 s for AReA, and 196 s for full attention on shared nodes (campaign means, Table 1); the main comparisons, ablations, and non-square sizes comprise more than 2000 generated images. Surviving worker logs establish a lower bound of 123.3 GH200 worker-hours through 2026-10-17 06:59 UTC, including model loading, calibration, evaluation, and failed attempts. The bound is the union of 581 closed task intervals per physical node, with timestamp precision treated conservatively; it measures occupied worker time, not GPU kernel activity. Missing historical logs, unmatched starts, and ongoing work prevent a complete campaign total. Overlapping scheduler allocation time and image-sidecar durations are not added to this bound.

Reproducibility. The code (attention processor, calibration, sampling, evaluation wrappers, AReA wrapper, and tests), the calibration table `k1_1024.npz`, the per-image metric files, and the per-comparison summary and paired-delta CSV files behind every table will be released. Main-text metric values and paired intervals are derived from those CSV files. Each SCA generation task ran from a frozen copy of the code named in the task. Per-image sidecar files record the sampling settings and, where available, a code fingerprint; the inherited Krea baseline records do not establish one source revision for every image.

Q Broader impact

SCA shortens generation at 4096^2 relative to full attention under our measured protocol, which helps applications such as print, design, and scientific visualization. Energy use remains unmeasured. The same reduction lowers the cost of producing high-resolution synthetic images that could be used to mislead. SCA adds no capability beyond the base model's and does not change its safety filters or watermarking; the risks and mitigations of the base model apply unchanged.

R QUALITATIVE GALLERIES

Figure R.1 shows curated SCA outputs. The other galleries, Figures R.2 to R.11, are uncurated unless stated: prompts are drawn at random with a fixed seed from pools fixed without looking at images, which exclude the development prompts (Figures R.2 to R.5 also exclude the 16 prompts then chosen for the curated and failure-case figures), and the selection seed, source hashes and crop coordinates of every gallery are archived in its `provenance.json`. Unless a caption says otherwise, images are FLUX.1-dev outputs at seed 0 with 28 steps and guidance 4.5. Every crop, here and in Figures 5 and L.2, shows native pixels of the outlined window, which is placed automatically at the highest edge energy averaged over the methods shown, never by hand. Figure R.6 uses FLUX.2-dev with guidance 4.0 and a FLUX.2 calibration table. Figure R.10 uses the FLUX.1-Krea-dev transformer with FLUX.1-dev text encoders and VAE and a Krea-calibrated SCA table; DyPE and SEGA come from the recorded Krea campaign, in which seven SEGA outputs were regenerated from an archived runtime because their original source snapshots could not be recovered, and the inherited baselines span several runtime revisions. In Figure R.8, UltraImage runs its two-stage defaults (30+30 steps, guidance 3.5 then 7), so that comparison is not equal-cost.

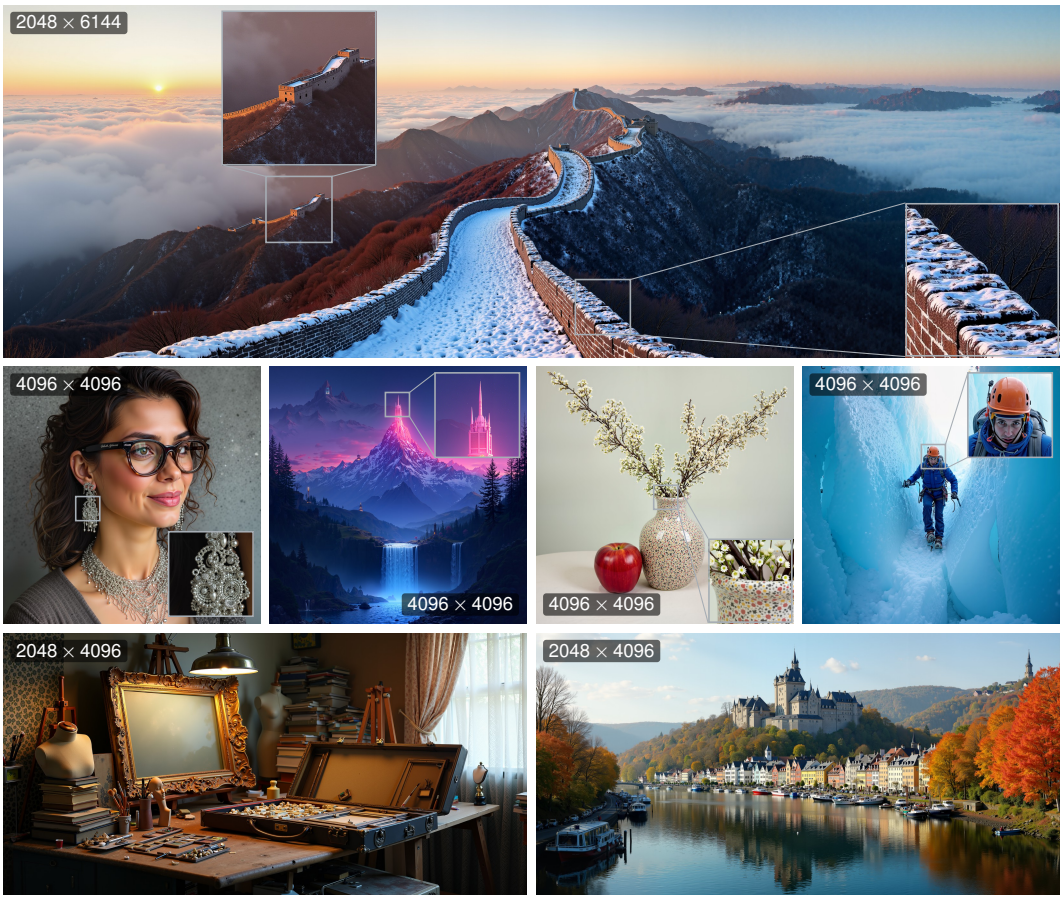

Figure R.1: **SCA keeps layout and fine detail across three aspect ratios without training.** Curated SCA outputs at 4096^2, 2048×4096 and 2048×6144, picked by eye for variety; insets are 1:1 crops of the outlined windows.

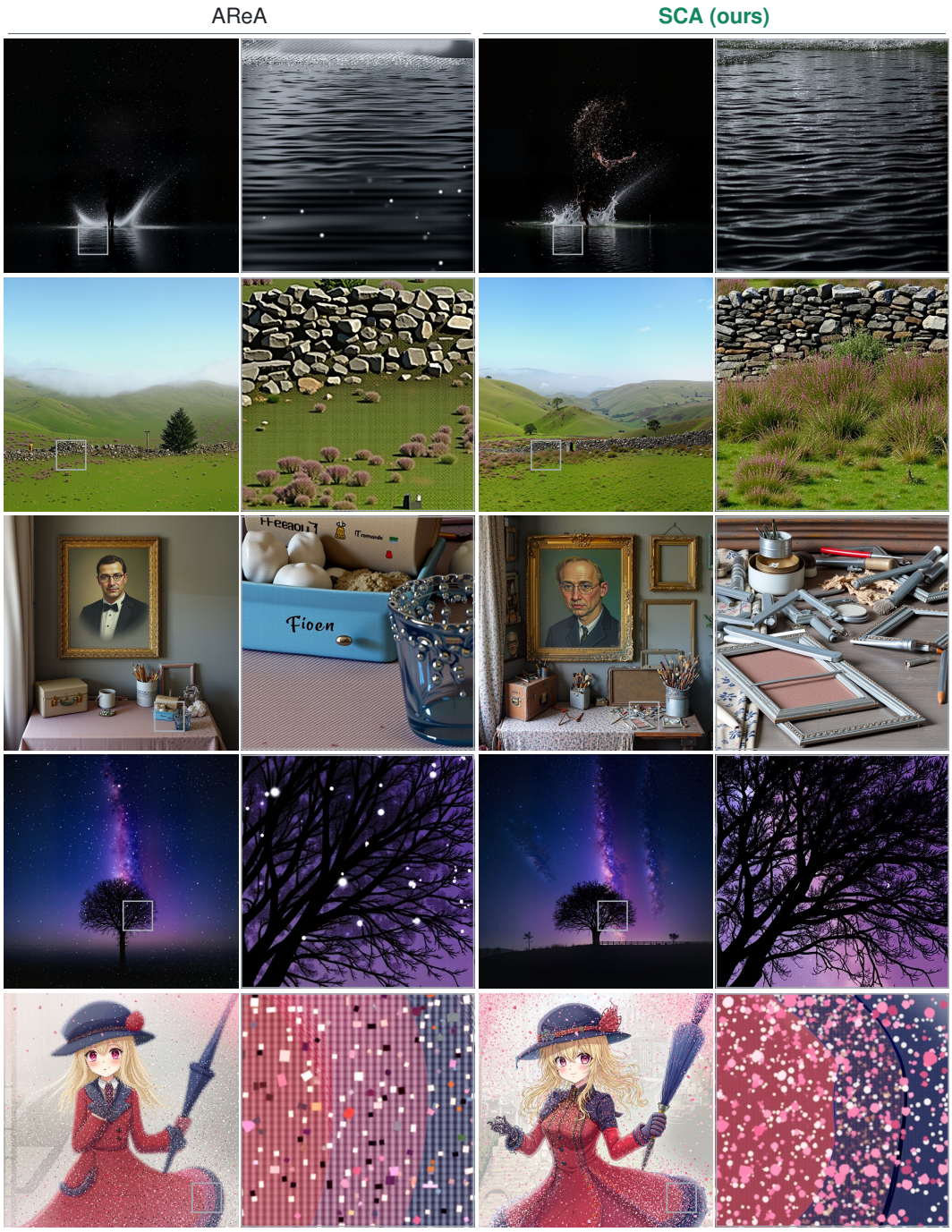

Figure R.2: **Uncurated, 4096^2: SCA keeps texture that AReA blurs or grids, but adds extra subjects in rows one and four.** Prompts one to five of 32 drawn at random (seed 0); after each image, a 1:1 crop of its outlined window.

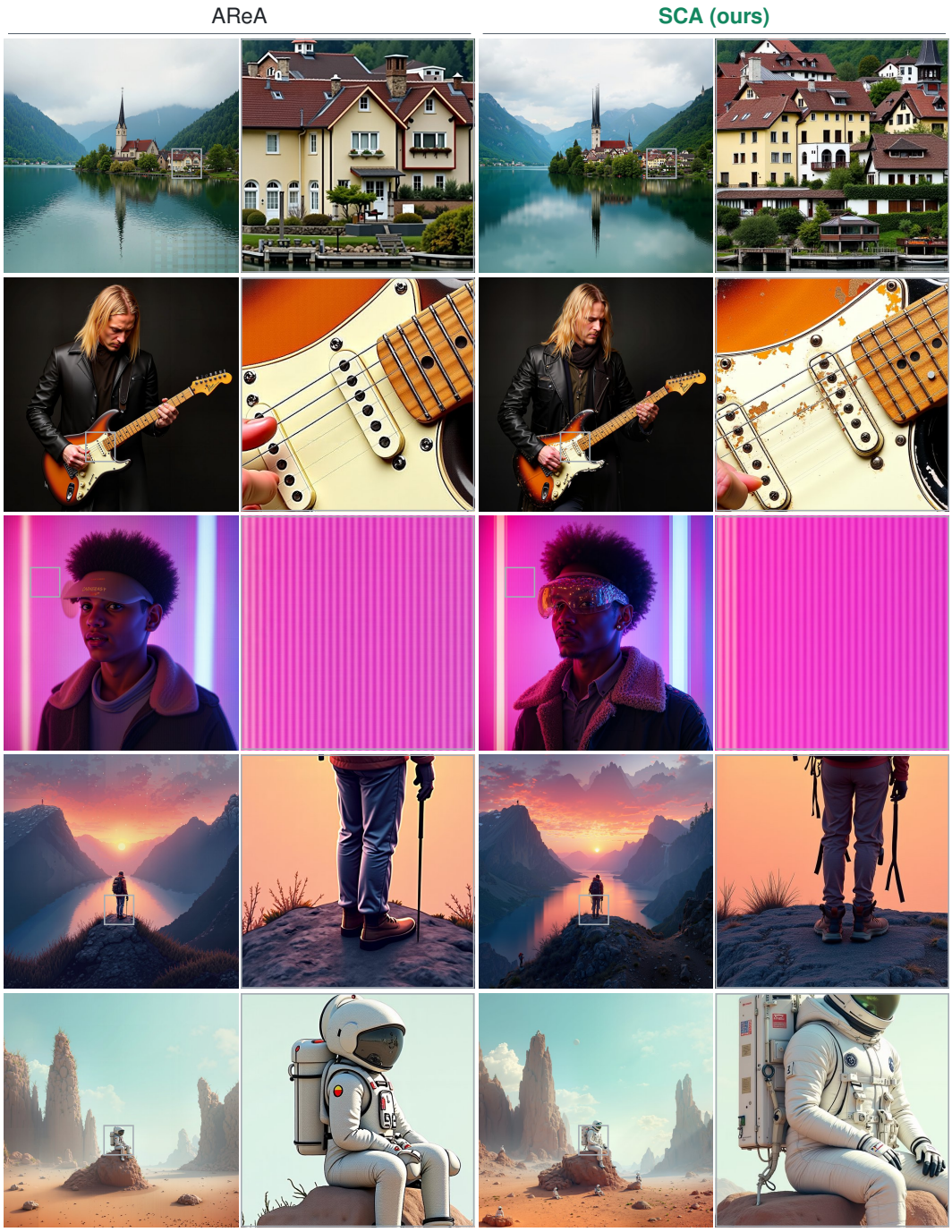

Figure R.3: **Uncurated samples at 4096^2, continued.** Prompts six to ten of the same draw; protocol as in Figure R.2. SCA adds extra spires, hikers, and astronauts in rows one, four, and five.

Figure R.4: **At 4096 × 2048 both methods keep the layout; SCA avoids AReA's weave pattern but is smoother.** The first two of six prompts drawn at random; AReA above, SCA below, each followed by a 1:1 crop of the outlined window.

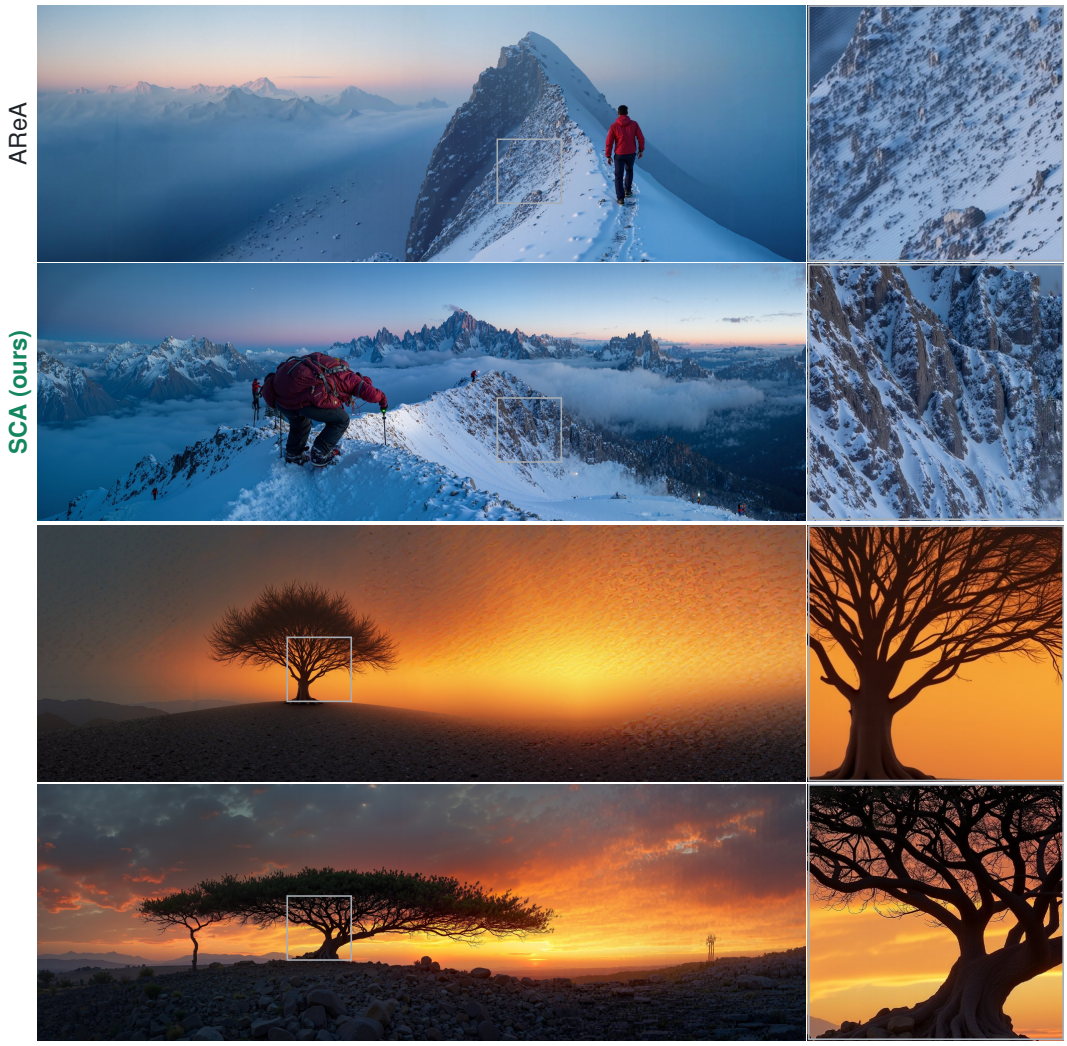

Figure R.5: **At 6144 × 2048 SCA resolves finer texture than AReA but adds extra climbers and trees.** The first two of six prompts drawn at random; AReA above, SCA below, each followed by a 1:1 crop of the outlined window.

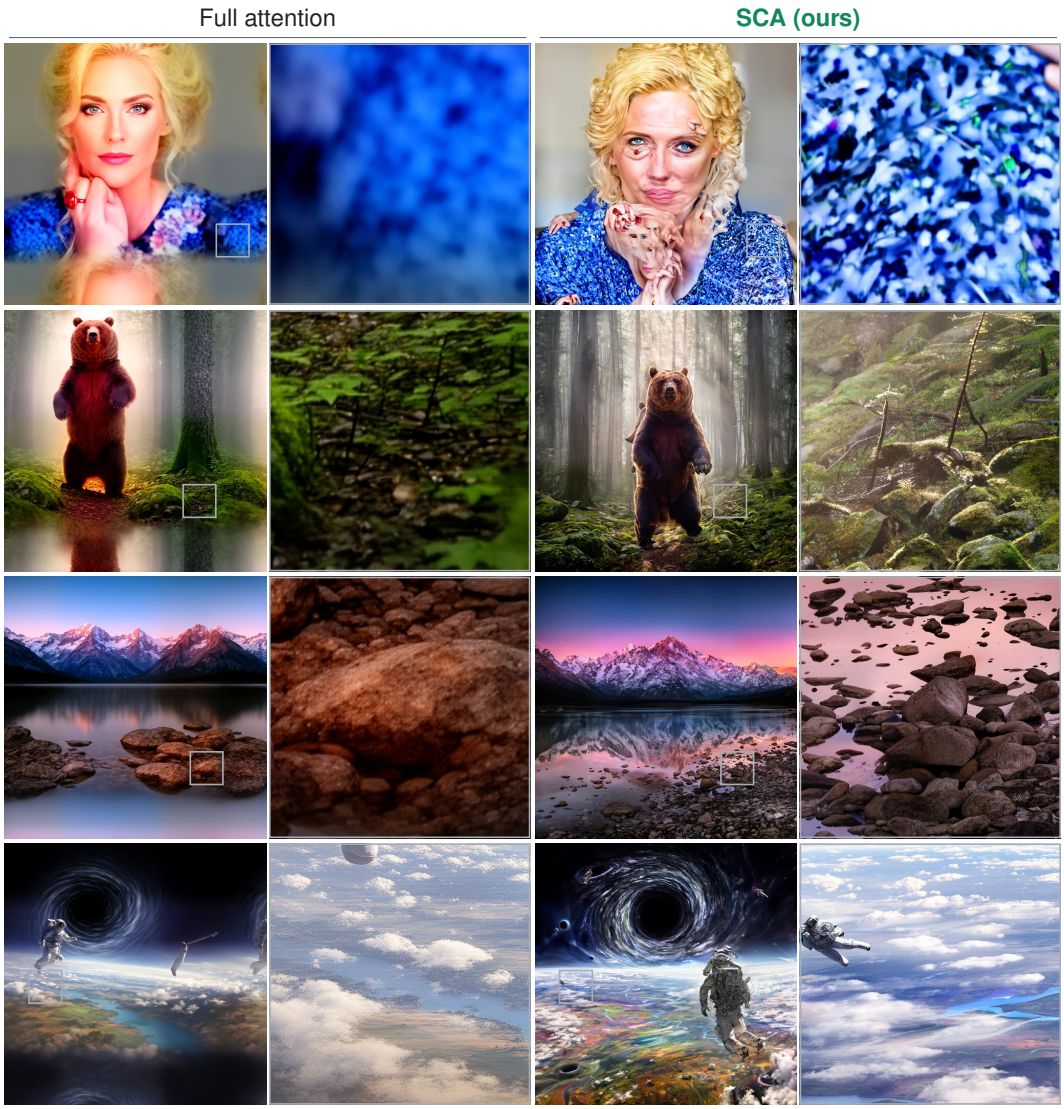

Figure R.6: **FLUX.2-dev at 4096^2: full attention blurs and repeats; SCA is sharp but duplicates subjects in three rows and distorts the portrait.** Four of 50 paired prompts drawn at random (seed 17); 1:1 crops follow each image.

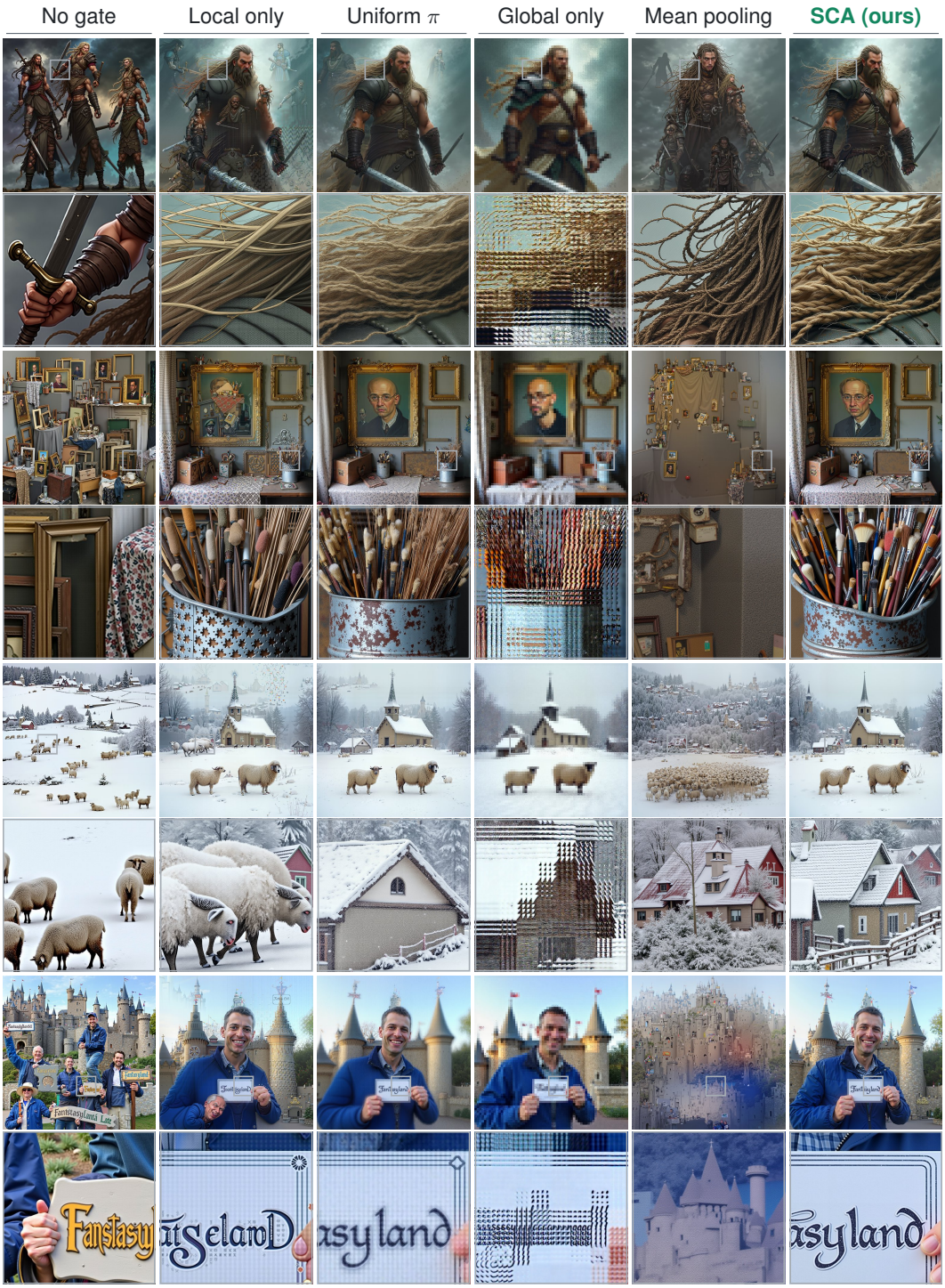

Figure R.7: **Without the gate the subject repeats; global only leaves the pooled branch's coarse grid; mean pooling loses the layout.** Four curated ablation prompts at 4096^2 (Table 3); full images above 1:1 crops of one window.

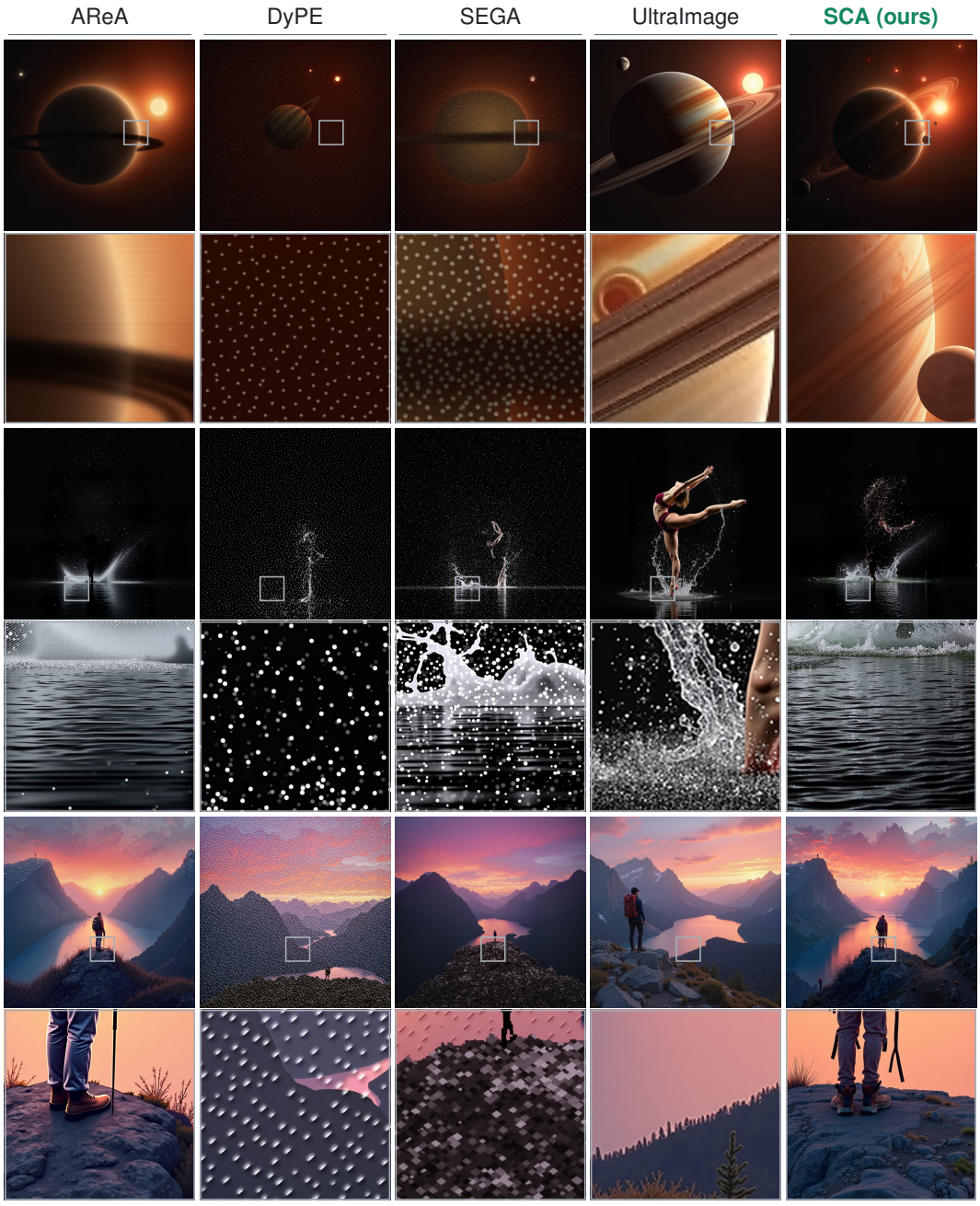

Figure R.8: **At 4096^2 DyPE and SEGA leave speckle, UltraImage softens the planet, and SCA is sharp but adds extra people in two prompts.** Three of 191 non-development prompts drawn at random (seed 17); full images above 1:1 crops of one window.

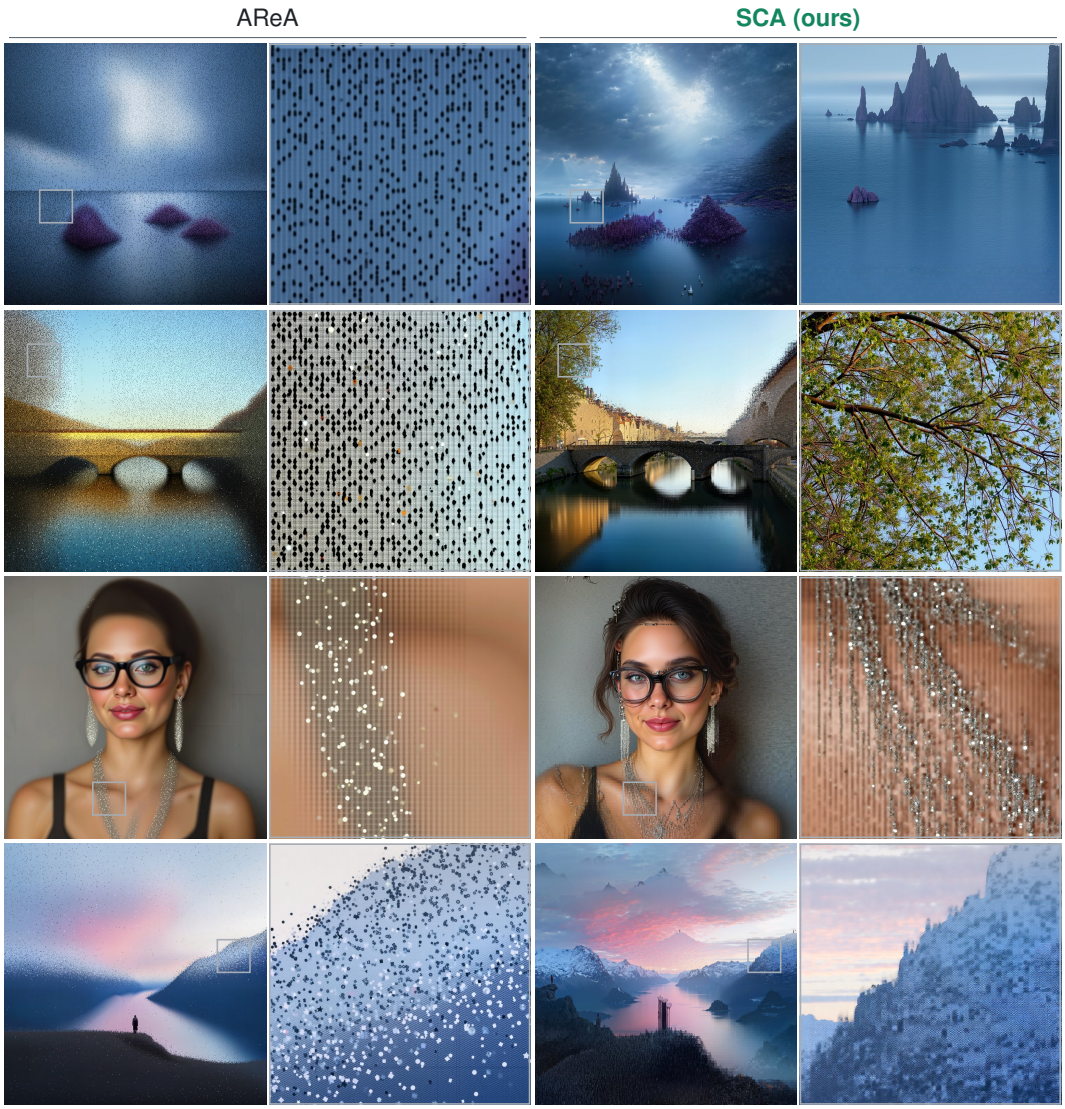

Figure R.9: **At 6144^2 AReA's crops dissolve into speckle or streaks; SCA keeps the layouts but shows stripes, block speckle, or extra figures.** Four of 100 prompts drawn at random; after each image, a 1:1 crop of its window.

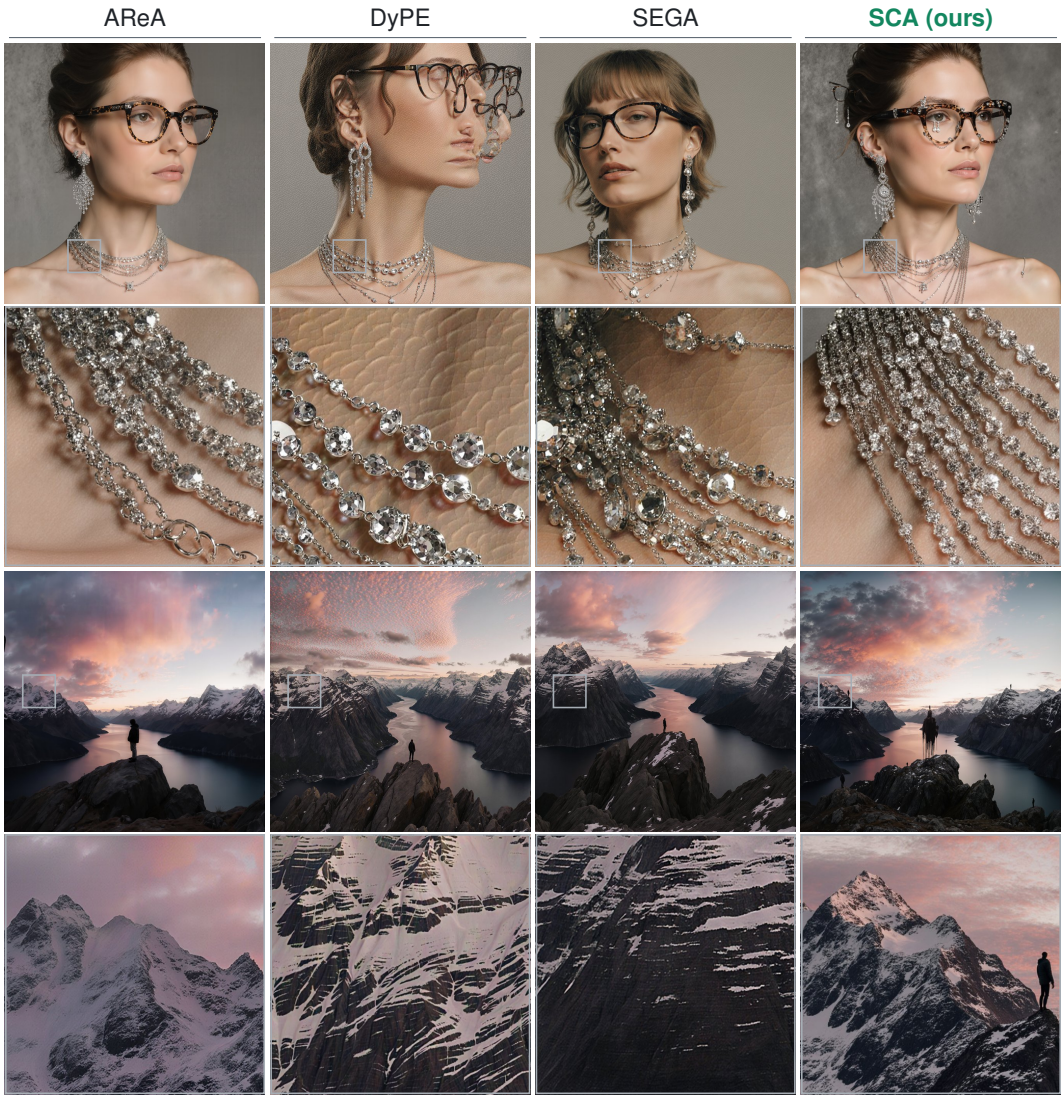

Figure R.10: **On FLUX.1-Krea-dev at 4096^2 the crops look alike, but DyPE ghosts the face and SCA melts the lone figure.** Two of 100 four-method prompts drawn at random; full images above 1:1 crops of one window.

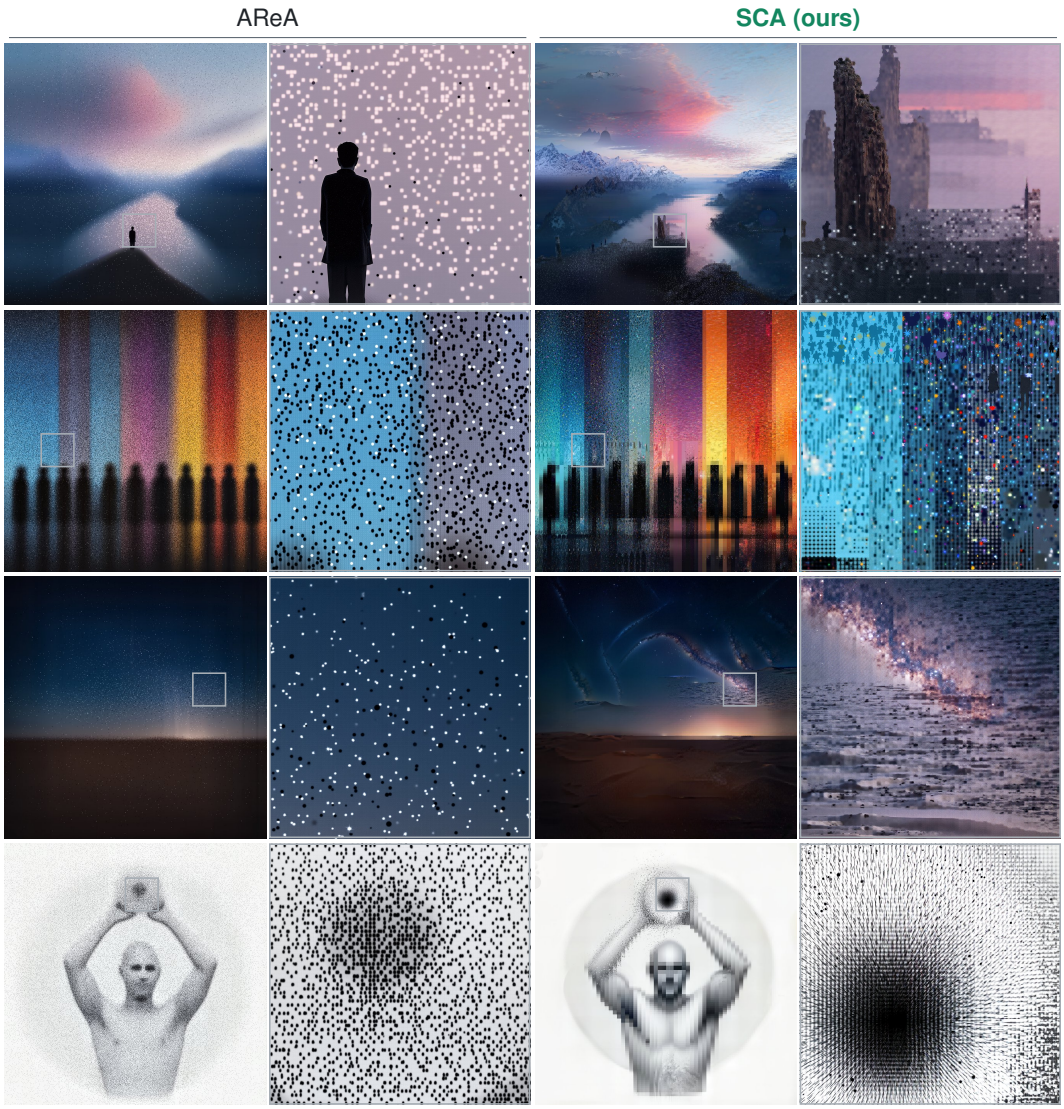

Figure R.11: **At 8192^2 both fail: AReA blurs and speckles, and SCA breaks into blocks and stripes.** Four of 12 paired prompts drawn at random; after each image, a 1:1 crop of its outlined window. Exploratory.

S FAILURE CASES

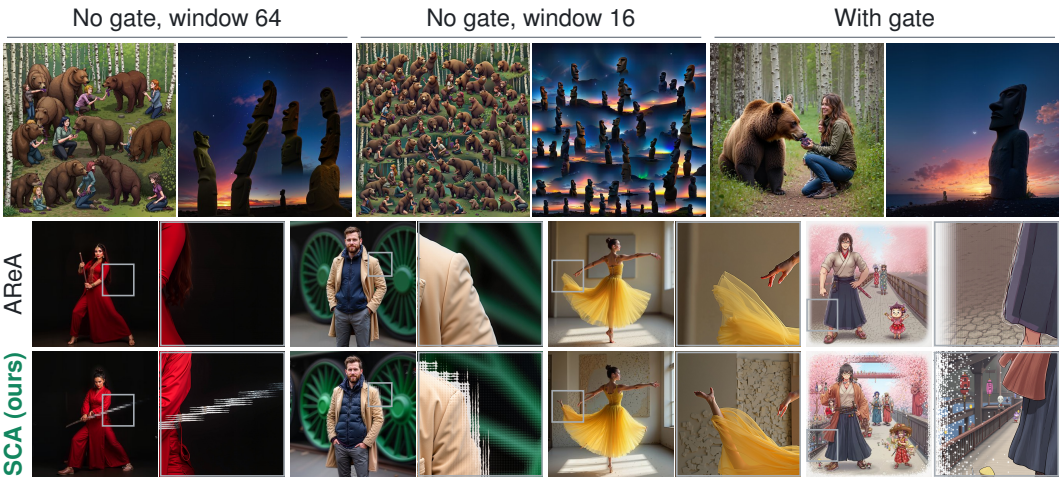

Figure S.1: **Curated failures: without the gate, windows duplicate subjects; SCA still makes local errors.** Top: development prompts; the gate leaves small duplicates. Below, 1:1 crops: staircase sword, blocky patch, extra foot, border speckle, none in AReA.

Figure S.1 shows the duplication that windows cause without the gate, and local errors that SCA still makes.

Late duplication. With the final method, small extra people, animals, or objects can appear in the last 20 steps, when the local branch dominates most heads. Among the 11 distinct uncurated FLUX.1-dev 4096^2 images in Figures R.2, R.3, and R.8, three add people and two repeat objects (`results/visual_audit_20261020`); AReA's images of these prompts blur or show lattice patterns instead. The band guard suppresses some of them but costs preference scores.

Articulated subjects. The underwater-surfer development prompt keeps doubled limbs in every variant; AReA's image shows a large translucent ghost.

Small factors. At 2048×4096 the window covers half the image height, so the local branch alone already sees most of the image along that axis, and the per-head mix gains little over AReA's shrinking windows; aesthetic score, CLIP-IQA, and corrected patch FID and KID are significantly worse.

Larger scales and another backbone. The gallery sampled from the completed 12-prompt 8192^2 run still shows grid and stripe artifacts in both methods (Figure R.11); this exploratory set does not establish general 8K quality. In FLUX.2-dev, improved aggregate metrics coexist with distorted facial geometry and fine texture in the randomly selected portrait example (Figure R.6).

Text-image offsets. The local branch keeps out-of-range text-image offsets (Remark 1).

Metrics. No-reference quality metrics reward sharp repeated subjects (v0 scores higher on CLIP-IQA and ARNIQA), and FID against 195 references at 299^2 resolution sees little of the 4K detail; we therefore report every metric and the qualitative galleries.